



MATERIAL SAFETY DATA SHEET (MSDS) COMPENDIUM FOR HPCL MUMBAI REFINERY

*Key for Safe Handling and Storage of
Hazardous Substances*

**MUMBAI REFINERY
HINDUSTAN PETROLEUM CORPORATION
LIMITED**

**FIRST EDITION
DATE: MARCH 2023**

MATERIAL SAFETY DATA SHEET (MSDS) COMPENDIUM FOR HPCL MUMBAI REFINERY

*Key for Safe Handling and Storage of Hazardous
Substances*

TECHNICAL DEPARTMENT

MUMBAI REFINERY

HINDUSTAN PETROLEUM CORPORATION LIMITED



FOREWORD

We at HPCL Mumbai Refinery have always been pro-active when it comes to safety. Mumbai Refinery produces and ships products which are considered hazardous and flammable in nature. Safe handling of these products is imperative for safety of all stakeholders. In line with regulatory requirements and our goal to ascertain safety of our employees and customers, a Material Safety Data Sheet (MSDS) of all products is required to be made available.

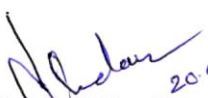
MSDS is a document that lists the hazardous nature of a product, its physical and chemical characteristics (e.g. flammability, explosive properties), its effect on human health, the chemicals with which it can adversely react, handling precautions, the types of measures that can be used to control exposure, emergency & first aid procedures, and methods to contain a spill. Effective use of MSDS will result in reduction in chemical source illness and injuries in the workplace.

The older MSDSs were prepared in non-standardized formats, which were subject to lack of some information required for safe handling of our products. The same has been now standardized internationally. In line with this requirement, **Bureau of Indian Standards (BIS)** has formulated and released **IS 17889_2022** standard on recommendation of the '*Chemical Hazards Sectional Committee and approval of the Chemical Division Council*' to harmonize with the internationally available format such as the one provided by **ISO 11014** and **Globally Harmonized System of Classification and Labelling of Chemicals (GHS)** for better hazard communication globally.

As and when new regulatory information is made available, the MSDS will be updated accordingly.

This compendium of MSDS has been prepared by PS&E division of Technical Department for all final products shipped from HPCL-Mumbai Refinery as per the standard IS 17889_2022, which will go a long way in achieving the objective of safe handling of Petroleum products at each stage.

Best Regards


Subhash S Nandanwar
CGM-Technical

Date:

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1. INTRODUCTION

In the Petroleum Industry, a number of hazardous chemicals are required to be stored, utilized and handled in appreciable quantities. In view of the stringent product quality requirement, various complex processes are introduced involving the handling of hazardous chemicals. Some additives are sometimes introduced to meet the quality requirement of various petroleum products. These chemicals pose various types of hazards like flammability, toxicity, explosivity, corrosivity etc.

Inadequate awareness about the hazardous properties of these chemicals and products may lead to serious accidents which will affect the human at work, the environment, This compendium will provide information which will help in facilitating the safe storage and handling of hazardous chemicals and in avoiding accidents.

2. SCOPE

This compendium has been prepared with the intent of providing information of Products being stored and handled in HPCL Mumbai Refinery. This compendium prepared to provide detailed information about MSDS for all final products shipped from HPCL-Mumbai Refinery as per the standard IS 17889_2022.

3. DEFINITION

- 3.1 SYNONYMS:** For common chemicals, several chemical names and numerous trade names may be applied to describe the chemical in question in synonyms field, many of these names are identified to aid users on the range of names which have been used to describe each substance.
- 3.2 CAS NO.:** Chemical Abstracts Service Registry Number is a numeric designation assigned by American Chemical Society's Chemical Abstracts Service and uniquely identifies a specific chemical compound. This entry allows one to conclusively identify a material regardless of the name or naming system used.
- 3.3 UN NUMBER:** The United Nations number is a four-figure code used to identify hazardous chemicals and is used for identification of chemicals transported internally by road, rail and by air.
- 3.4 Hazardous Waste ID. No.:** This means the categories of wastes specified in the schedule in The Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008.
- 3.5 Skin corrosion/irritation:** The production of irreversible damage of the skin
- 3.6 Specific target organ toxicity – single exposure (narcotic effects, drowsiness):** Specific non-lethal effects on organs or organ systems in the body following single exposure to a chemical.
- 3.7 Aspiration hazard:** An aspiration hazard is a hazardous substance that may be fatal if they are swallowed and enter the airways through inhalation.



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- 3.8 Hazardous to the aquatic environment - chronic hazard:** Chronic aquatic toxicity means the intrinsic property of a substance to cause adverse effects to aquatic organisms during aquatic exposures which are determined in relation to the life-cycle of the organism.
- 3.9 Acute toxicity:** Acute toxicity describes the adverse effects of a substance that result either from a single exposure or from multiple exposures in a short period of time (usually less than 24 hours).
- 3.10 Specific target organ toxicity - repeated exposure:** Specific toxic effects on target organs occurring after repeated exposure to a substance or mixture.
- 3.11 Serious Eye Damage/Irritation:** The production of changes in the eye following the application of a test substance to the anterior surface of the eye, which are fully reversible within 21 days of application.
- 3.12 Germ cell mutagenicity:** The classification system is primarily concerned with chemicals which may cause mutations in the germ cells of humans and these mutations can be transmitted to the progeny.
- 3.13 Acute toxicity, inhalation:** Acute inhalation toxicity is the total of adverse effects caused by a substance following a single uninterrupted exposure by inhalation over a short period of time (24 hours or less) to a substance capable of being inhaled.
- 3.14 DNEL:** The derived no-effect level (DNEL) is the level of exposure to a substance above which humans should not be exposed.
- 3.15 FLAMMABLE CHEMICALS**

- i. **Flammable Gases:** Gases which at 20°C and at standard pressure of 101.3 KPa are:
- ignitable when in a mixture of 13 percent or less by volume with air, or;
 - have a flammable range with air of at least 12 percentage points regardless of the lower flammable limits.

Note: The flammability shall be determined by tests or by calculation in accordance with methods adopted by International Standards Organisation ISO Number 10156 of 1990 or by Bureau of Indian Standard ISI Number 1446 of 1985.

- ii. **Extremely flammable liquids:** Chemicals which have been flash point lower than or equal to 23°C and boiling point less than 35°C.
- iii. **Very highly flammable liquids:** Chemicals which have a flash point lower than or equal to 23°C and initial boiling point higher than 35°C.
- iv. **Highly flammable liquids:** Chemicals which have a flash point lower than or equal to 60°C but higher than 23°C.
- v. **Flammable liquids:** Chemicals which have a flash point higher than 60°C but lower than 90°C.

3.16 FLAMMABLE (EXPLOSIVE) LIMIT AND RANGE

In the case of gases or vapor, which form flammable mixtures with air, there is a minimum



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concentration of vapor in air below which the propagation of flame does not occur in contact with a source of ignition. This is called Lower Explosive Limit (LEL). Similarly, there is a maximum concentration of vapor above which the propagation of flame does not occur on contact with a source of ignition. This is called Upper Explosive Limit (UEL). This range of minimum to maximum concentration is termed as Flammable (Explosive) range.

The flammable limits are not appreciably changed by normal variations in atmospheric pressure and temperature. However, at high temperatures, the upper limit is raised and the lower limit lowered, resulting in greater flammable or explosive range. The effect of high pressures on the limits is different for each gas or vapor depending upon its nature.

3.17 TDG FLAMMABILITY

Transport of dangerous goods flammability is used in Canada and is based on Classification of dangerous goods by UN Committee of experts on transport of danger goods. The classes indicated are 2, 3, 4 and 5 which are same as indicated in The Central Motor Vehicle Rules 1989.

3.18 FLASH POINT

The minimum temperature at which the liquid gives so much vapor that this vapor, when mixed with air, forms an ignitable mixture and gives a momentary flash on application of a small pilot flame under specified conditions of test.

3.19 AUTO IGNITION TEMPERATURE

The lowest temperature at which ignition occurs in a mixture of explosive gas and air when the method specified in IS: 7820-1975 is followed.

3.20 EXPLOSIVES

Explosives mean a solid or liquid or pyrotechnic substance (or a mixture of substances) or an article.

- a) Which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings.
- b) Which is designed to produce an effect by heat, light, sound, gas or smoke or a combination of these as the result of non-detonative self-sustaining exothermic chemical reaction.

3.21 CORROSIVE CHEMICALS

Chemicals, which will cause damage to the living tissue and/or chemically attack metallic



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containers and structures are corrosive chemicals.

3.22 OXIDISING SUBSTANCES (OXIDISER):

Oxidizing substances are those substances which readily give of oxygen or other oxidizing substances (such as bromine, chlorine or fluorine). They also include materials that react chemically to oxidize combustible materials; this means that oxygen combines chemically with the other material in a way that increases the chance of fire or explosion. This reaction may be spontaneous at either room temperature or may occur under slight heating. These materials can be severe fire and explosion hazards.

3.23 PYROPHORIC SUBSTANCES

Chemicals which are so reactive that on contact with air and its moisture, oxidation/ hydrolysis occurs at such a rate so as to cause ignition i.e., its auto ignition temperature is below the room temperature. Some of these reactions liberate flammable gases.

3.24 ORGANIC PEROXIDES

Organic peroxide means an organic compound that contains the bivalent –o-o- structure and which may be considered to be a structural derivative of hydrogen peroxide where one or both of hydrogen atoms has been replaced by an organic radical.

3.25 TOXIC CHEMICALS

Toxicity is the degree to which a substance is able to damage an exposed organism. Chemicals having the following values of acute toxicity and which owing to their physical & chemical properties, are capable of producing major accident hazards:

	ORAL TOXICITY Medium lethal dose by oral route (oral toxicity)	DERMAL TOXICITY Medium lethal dose by the dermal route (dermal toxicity)	INHALATION TOXICITY Medium lethal concentration by inhalation route (four hrs.)
	LD50 (mg/kg body weight of test animals)	LD50 (mg/kg body weight of test animals)	LC50 (mg/l inhalation in test animals)
1. Extremely toxic	<5	<40	<0.5
2. Highly toxic	5-50	>4-200	0.5-2.0
3. Toxic	>50-200	>200-1000	>2-10

3.26 INCOMPATIBLE CHEMICALS



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These are chemical substances which in contact with each other cause hazardous reactions which may be violent or produce toxic materials.

3.27 TLV (THRESHOLD LIMIT VALUE)

The term refers to concentrations of air contaminants in the working environment to which it is believed, nearly all workers may be exposed repeatedly day after day without adverse effect. Normally, it is the time weighted average concentration in ppm or gm/m³ for a normal 8-hour work day or 40-hour workweek. Excursions above this limit may occur if they are compensated during the workday, by equivalent excursion below the limit.

3.28 STEL (SHORT TERM EXPOSURE LIMIT)

It is defined as a 15 minutes time weighted average exposure, which should not be exceeded at any time during a workday even if the 8 hour's time weighted average is within the TLV. Exposure at the STEL should not be longer than 15 minutes and should not be repeated more than four times per day. There should be at least 60 minutes interval between successive exposures at the STEL. An averaging period other than 15 minutes may be recommended when this is warranted by observed biological effects.

3.29 REFERENCES:

- CLP Regulation (EC) No. 1272/2008 on the classification, labelling and packaging of substances and mixtures.
- Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
- OISD-STD-114-Safe Handling of Hazardous Chemicals

4. EXPLANATORY NOTES

4.1 NFPA (704M) PICTORIAL REPRESENTATION OF HAZARD

NFPA pictorial representations consist of a diamond as shown in the figure below, divided into 4 parts. These parts, starting from left block (in clockwise direction), represent Health Hazard, Flammability Hazard, Reactivity Hazard and Space for additional information such as water reactivity, oxidant and radiation hazard. The color codes for these hazards are: blue for health, red for flammability, yellow for reactivity and colorless for additional information.

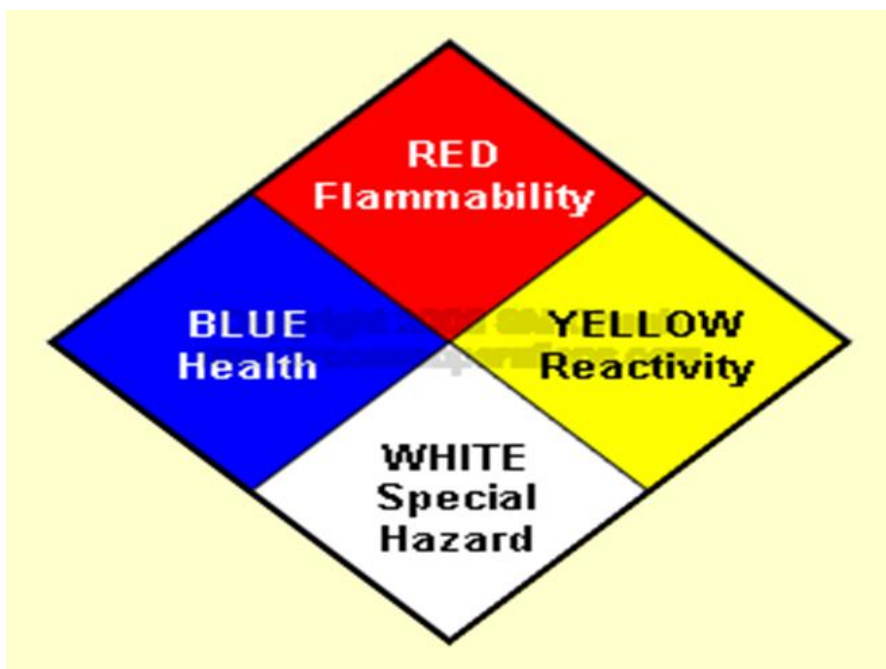
These are rated on a scale of 0 (no hazard) to 4 (extreme hazard). The ratings for individual substances can be found in the NFPA Fire Protection Guide to Hazardous Materials.



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4.1.1 HEALTH

Rating 4: A few whiffs of the gas or vapor could cause death or the gas, vapor or liquid could be fatal on penetrating the fire fighters normal full protective clothing which is designed for resistance to heat. For most chemicals having a Health 4 rating, the normal full protective clothing available to the average fire department will not provide adequate protection against skin contact with these materials. Only special protective clothing designed to protect against the specific hazard should be worn.

Rating 3: Materials extremely hazardous to health, but areas may be entered with extreme care. Full protective clothing, including self- contained breathing apparatus, rubber gloves, boots and bands around legs, arms and waist should be provided. No skin surface should be exposed.

Rating 2: Materials hazardous to health, but areas may be entered freely with self-contained breathing apparatus.

Rating 1: Materials only slightly hazardous to health. It may be desirable to wear self- contained breathing apparatus.

Rating 0: Materials which on exposure under fire conditions would offer no health hazard beyond that of ordinary combustible material.

4.1.2 FLAMMABILITY



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Rating 4: Very flammable gases, very volatile flammable liquids, and materials that in the form of dusts or mists readily form explosive mixtures when dispersed in air. Shut off flow of gas or liquid and keep cooling water streams on exposed tanks or containers. Use water spray carefully in the vicinity of dusts so as not to create dust clouds.

Rating 3: Liquids, which can be ignited under almost, all normal temperature conditions. Water may be ineffective on these liquids because of their low flash points.

Solids which form coarse dusts, solids in shredded or fibrous form that create flash fires, solids that burn rapidly, usually because they contain their own oxygen, and any material that ignites spontaneously at normal temperatures in air.

Rating 2: Liquids which must be moderately heated before ignition will occur and solids that readily give off flammable vapors. Water spray may be used to extinguish the fire because the material can be cooled to below its flash point.

Rating 1: Materials that must be pre-heated before ignition can occur. Water may cause frothing of liquids with this flammability rating number if it gets below the surface of the liquid and turns to steam.

However, water spray gently applied to the surface will cause a frothing which will extinguish the fire. Most combustible solids have flammability rating of 1.0 Materials that will not burn.

4.1.3 REACTIVITY

Rating 4: Materials which in themselves are readily capable of detonation or of explosive decomposition or explosive reaction at normal temperatures and pressures. Includes materials, which are sensitive to mechanical or localized thermal shock. If a chemical with this hazard rating is in an advanced or massive fire, the area should be evacuated.

Rating 3: Materials which in themselves are capable of detonation or of explosive decomposition or of explosive reaction but which require a strong initiating source or which must be heated under confinement before initiation. Includes materials which are sensitive to thermal or mechanical shock at elevated temperature and pressure or which react explosively with water without requiring of confinement. Firefighting should be done from an explosion-resistant location.

Rating 2: Materials which in themselves are normally unstable and readily undergo violent chemical change but do not detonate. Includes materials which can undergo chemical change with rapid release of energy at normal temperatures and pressures or which can undergo violent chemical change at elevated temperatures and pressures. Also includes those materials which may react violently with water or which may form potentially explosive mixtures with water. In advanced or massive fires, firefighting should be done from a protected location.



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Rating 1: Materials which in themselves are normally stable but which may become unstable at elevated temperature and pressure or which may react with water with some release of energy but not violently. Caution must be used in approaching the fire and applying water.

Rating 0: Materials which are normally stable even under fire exposure conditions and which are not reactive with water. Normal firefighting procedures may be used.

4.1.4 Explanation for bottom space (Colorless):

The bottom space is primarily used to identify unusual reactivity with water. With a line through its center alerts firefighting personnel to the possible hazard in use of water. This bottom space may also be used to identify a radiation hazard by the symbol. Oxidizing chemicals are identified in the bottom space, as OXY.

4.2 HAZCHEM CODE

Hazchem code provides vital information to fire brigade and other emergency services on the action to be taken to combat spillage, leakage or fire in an emergency involving hazardous substance.

The Central Motor Vehicle Rules requires that the vehicles transporting hazardous chemicals to display Emergency Information Panel (EIP). The EIP requires display of Hazchem code (also called the emergency action code).

The Hazchem code consists of a number from 1-4 (A) and any one of the letters – P, R, S, T, W, X, Y, Z (B) followed by at times by the letter E.

A	B	E
---	---	---

- The number (in place of A above) signify extinguishing medium to be employed as given below:

1. JETS	2. FOG	3. FOAM	4. DRY AGENT
---------	--------	---------	--------------

- The letters (in place of B above) signify three aspects viz.
 - Method of controlling spillage (a) By diluting or (b) By containing
 - Personal protective equipment to be worn by the fire fighter
 - Reactive nature of chemicals
- If letter E exists then, need for evacuation



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HAZCHEM Emergency Action Code

FOR FIRE OR SPILLAGE

1

COARSE SPRAY

2

FINE SPRAY

3

FOAM

4

DRY AGENT

•

ALCOHOL RESISTANT FOAM

P	V	LTS	DILUTE
R			
S	V	BA & FIRE KIT	
T			
W	V	LTS	CONTAIN
X			
Y	V	BA & FIRE KIT	
Z			

E

PUBLIC SAFETY HAZARD

Additional Information

DRY AGENT
Water **must not** be allowed to come into contact with the substance at risk.

ALCOHOL RESISTANT FOAM •2 or •3
Alcohol resistant foam is the preferred medium.
If not available:
- If •2 – use Fine Spray or Water Fog
- If •3 – use Normal Protein Foam

V
Substance can be violently or even explosively reactive, including combustion.

LTS
Liquid-Tight Chemical Protective Suit with BA.
Full **FIRE KIT** should also be worn for thermal protection if the substance is:
Liquid Oxygen
or Liquefied Toxic Gas (Division 2.3)
or Toxic Gas with sub-risk 2.1 or 5.1
or Class or sub-risk 3
or Division 5.1 PGI with sub-risk 6.1 or 8
or carried at temperature > 100 °C

DILUTE
May be washed to drain with large quantities of water.

CONTAIN
Prevent, by any means available, spillage from entering drains or water course.

E
People should be warned to stay indoors with all doors and windows closed, –but evacuation may need to be considered. Consult Control, Police and product expert.

5. CLASSIFICATION OF HAZARDOUS SUBSTANCES

It is imperative to summarize clearly and concisely the important hazards and adverse effects of the chemical product on human health and the environment, and physical and chemical hazards, including specific hazards, where appropriate. In order to facilitate this requirement, the United Nations (UN) has therefore developed a 'Globally Harmonized System' (GHS) of classification and labelling of chemicals. The GHS is a single worldwide system for classifying and communicating the hazardous properties of industrial and consumer chemicals. GHS sits alongside the UN 'Transport of Dangerous Goods' system. The GHS aims to ensure that information on the hazardous properties of chemicals is available throughout the world in order to enhance the protection of human health and the environment during the handling, transport and use of chemicals. GHS also provides the basis for harmonizing regulations on chemicals at national, regional and worldwide level.

5.1 GHS label elements: The standardized label elements included in the GHS are:



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- 5.1.1 Symbols (GHS hazard pictograms):** Convey health, physical and environmental hazard information, assigned to a GHS hazard class and category. Pictograms include the harmonized hazard symbols plus other graphic elements, such as borders, background patterns or covers and substances which have target organ toxicity.[8] Also, harmful chemicals and irritants are marked with an exclamation mark, replacing the European saltire. Pictograms will have a black symbol on a white background with a red diamond frame. For transport, pictograms will have the background, symbol and colors currently used in the UN Recommendations on the Transport of Dangerous Goods. Where a transport pictogram appears, the GHS pictogram for the same hazard should not appear.
- 5.1.2 Signal word:** "Danger" or "Warning" will be used to emphasize hazards and indicate the relative level of severity of the hazard, assigned to a GHS hazard class and category. Some lower level hazard categories do not use signal words. Only one signal word corresponding to the class of the most severe hazard should be used on a label.
- 5.1.3 GHS hazard statements:** Standard phrases assigned to a hazard class and category that describe the nature of the hazard. An appropriate statement for each GHS hazard should be included on the label for products possessing more than one hazard.

5.2 The additional label elements included in the GHS are:

- 5.2.1 GHS precautionary statements:** Measures to minimize or prevent adverse effects. There are four types of precautionary statements covering: prevention, response in cases of accidental spillage or exposure, storage, and disposal. The precautionary statements have been linked to each GHS hazard statement and type of hazard.
- 5.2.2 Product identifier (ingredient disclosure):** Name or number used for a hazardous product on a label or in the SDS. The GHS label for a substance should include the chemical identity of the substance. For mixtures, the label should include the chemical identities of all ingredients that contribute to acute toxicity, skin corrosion or serious eye damage, germ cell mutagenicity, carcinogenicity, reproductive toxicity, skin or respiratory sensitization, or Specific Target Organ Toxicity (STOT), when these hazards appear on the label.
- 5.2.3 Supplemental information:** Non-harmonized information on the container of a hazardous product that is not required or specified under the GHS. Supplemental information may be used to provide further detail that does not contradict or cast doubt on the validity of the standardized hazard information.

5.3 CLP Regulation (EC) No. 1272/2008 on the classification, labelling and packaging of substances and mixtures using GHS classification

The relationship of GHS hazard statements, pictograms, signal words, hazard classes, categories, and precautionary statements for Classification, Labelling and Packaging (CLP) of substances is summarized below:



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Classification			Labelling			
Hazard		Abbreviation of classification (without H set)	Pictogram, code*	Signal word	Code* Warning of danger	
Class	Category					Text
Explosives	Unstable explosive	Unst. Expl.		Danger	H200	Unstable explosive
	Division 1.1	Expl. 1.1			H201	Explosive; mass explosion hazard
	Division 1.2	Expl. 1.2			H202	Explosive; severe projection hazard
	Division 1.3	Expl. 1.3			H203	Explosive; fire, blast or projection hazard
	Division 1.4	Expl. 1.4	GHS01	Warning	H204	Fire or projection hazard
	Division 1.5	Expl. 1.5	No Pictogram	Danger	H205	May mass explode in fire
	Division 1.6	Expl. 1.6	No Pictogram	-	-	No hazard statement
Desensitised Explosives	Category 1	Desen. Expl. 1		Danger	H206	Fire, blast or projection hazard; increased risk of explosion if desensitising agent is reduced
	Category 2	Desen. Expl. 2			H207	Fire or projection hazard; increased risk of
	Category 3	Desen. Expl. 3	GHS02	Warning		



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						explosion if desensitising agent is reduced
	Category 4	Desen. Expl. 4			H208	Fire hazard; increased risk of explosion if desensitising agent is reduced
Flammable Gases (including chemically unstable gases)	Category 1A flammable gas and gases categorised as 1A meeting pyrophoric or unstable gas A/B criteria	Flam. Gas 1A		Danger	H220	Extremely flammable gas
		Pyr. Gas			H220	Extremely flammable gas
					H232	May ignite spontaneously if exposed to air
		Chem. Unst. Gas A			H220	Extremely flammable gas
		Chem. Unst. Gas B			H230	May react explosively even in the absence of air
					H220	Extremely flammable gas
					H231	May react explosively even in the absence of air at elevated pressure and/or temperature
			GHS02			



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	Category 1B	Flam. Gas 1B	 GHS02	Danger	H221	Flammable gas
	Category 2	Flam. Gas 2	No Pictogram	Warning	H221	Flammable gas
Aerosol	Category 1	Aerosol 1	 GHS02	Danger	H222	Extremely flammable aerosol
					H229	Pressurised container: May burst if heated
	Category 2	Aerosol 2	GHS02	Warning	H223	Flammable aerosol
					H229	Pressurised container: May burst if heated
	Category 3	Aerosol 3	No Pictogram	Warning	H229	Pressurised container: May burst if heated
Oxidising Gases	Category 1	Ox. Gas 1	 GHS03	Danger	H270	May cause or intensify fire; oxidiser
Gases under Pressure (1)	Compressed gas	Press. Gas	 GHS04	Warning	H280	Contains gas under pressure; may explode if heated
	Liquefied gas					
	Refrigerated liquefied gas				H281	Contains refrigerated gas; may cause cryogenic burns or injury



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	Dissolved gas				H280	Contains gas under pressure; may explode if heated
(1) = The hazard class “Gases under Pressure” is subdivided into ‘Groups’ (not ‘Categories’)						
Flammable Liquids	Category 1	Flam. Liq. 1		Danger	H224	Extremely flammable liquid and vapour
	Category 2	Flam. Liq. 2			H225	Highly flammable liquid and vapour
	Category 3	Flam. Liq. 3		Warning	H226	Flammable liquid and vapour
Flammable Solids	Category 1	Flam. Sol. 1	GHS02	Danger	H228	Flammable solid
	Category 2	Flam. Sol. 2		Warning		
Self-reactive substances and mixtures (2) Organic Peroxides (2)	Type A	Self-react. A	 GHS01	Danger	H240	Heating may cause an explosion
		Org. Perox. A				
	Type B	Self-react. B	 GHS01+GHS02	Danger	H241	Heating may cause a fire or explosion
	Type C and D	Self-react. C & D	 GHS02			
		Org. Perox. C & D				



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	Type E and F	Self-react. E & F		Warning		
		Org. Perox. E & F				
	Type G	Self-react. G	No Pictogram	No Signal word	-	No hazard statement
		Org. Perox. G				
	(2) = Two separate hazard classes have the same categories (and are therefore grouped).					
Pyrophoric Liquids	Category 1	Pyr. Liq. 1		Danger	H250	Catches fire spontaneously if exposed to air
Pyrophoric Solids	Category 1	Pyr. Sol. 1				
Self-heating substances and mixtures	Category 1	Self-heat. 1		Danger	H251	Self-heating; may catch fire
	Category 2	Self-heat. 2		Warning	H252	Self-heating in large quantities; may catch fire
Substances or mixtures which in contact with water emit flammable gases	Category 1	Water-react. 1		Danger	H260	In contact with water releases flammable gases which may ignite spontaneously
	Category 2	Water-react. 2		Danger	H261	In contact with water releases flammable gases
	Category 3	Water-react. 3		Warning		
Oxidising Liquids (2)	Category 1	Ox. Liq. 1		Danger	H271	May cause fire or explosion; strong oxidiser
		Ox. Sol. 1				
-			GHS03			



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Oxidising solids (2)	Category 2	Ox. Liq. 2		Danger	H272	May intensify fire; oxidiser		
		Ox. Sol. 2						
	Category 3	Ox. Liq. 3						
		Ox. Sol. 3						
	(2) = Two separate hazard classes have the same categories (and therefore grouped).							
Corrosive to metals	Category 1	Met. Corr. 1	 GHS05	Warning	H290	May be corrosive to metals		
Acute Toxicity	Category 1	Acute Tox. 1	 GHS06	Danger	H300	Fatal if swallowed		
					H310	Fatal in contact with skin		
					H330	Fatal if inhaled		
	Category 2	Acute Tox. 2			H300	Fatal if swallowed		
					H310	Fatal in contact with skin		
					H330	Fatal if inhaled		
	Category 3	Acute Tox. 3	 GHS06		H301	Toxic if swallowed		
					H311	Toxic in contact with skin		
					H331	Toxic if inhaled		
	Category 4	Acute Tox. 4	 GHS07	Warning	H302	Harmful if swallowed		
					H312	Harmful in contact with skin		
					H332	Harmful if inhaled		
Skin corrosi	Category 1(3)	Skin Corr. 1		Danger	H314	Causes severe skin		



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on / irritatio n	Category 1A	Skin Corr. 1A	 GHS05			burns and eye damage
	Category 1B	Skin Corr. 1B				
	Category 1C	Skin Corr. 1C				
	Category 2	Skin Irr. 2	 GHS07	Warning	H315	Causes skin irritation
(3) = Conditions in place for the use of Category 1, please see Annex I to CLP						
Serious eye damage / eye irritation	Category 1	Eye Dam. 1	 GHS05	Danger	H318	Causes serious eye damage
	Category 2	Eye Irr. 2	 GHS07	Warning	H319	Causes serious eye irritation
Sensitisation of the respiratory tract or the skin	Respiratory Sensitisers Category 1(3) and Sub-Categories 1A and 1B	Resp. Sens. 1 1A or 1B	 GHS08	Danger	H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled
	Skin Sensitisers Category 1(3) and Sub-Categories 1A and 1B	Skin. Sens. 1 1A or 1B	 GHS07	Warning	H317	May cause an allergic skin reaction
	(3) = Conditions in place for the use of Category 1, please see Annex I to CLP					
Germ cell mutagenicity	Category 1 and Sub-Categories 1A and 1B	Muta. 1, 1A or 1B		Danger	H340	May cause genetic defects (4)



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	Category 2	Muta. 2	 GHS08	Warning	H341	Suspected of causing genetic defects (4)
Carcinogenicity	Category 1 and Sub-Categories 1A and 1B	Carc. 1, 1A or 1B		Danger	H350	May cause cancer (4)
					H350i	May cause cancer when inhaled
	Category 2	Carc. 2		Warning	H351	Suspected of causing cancer (4)
(4) = State route of exposure if it is conclusively proven that no other routes of exposure cause the hazard.						
Reproductive toxicity	Category 1 and Sub-Categories 1A and 1B	Repr. 1, 1A or 1B	 GHS08	Danger	H360 (5)	May damage fertility or the unborn child.
					H360 F (6)	May damage fertility.
					H360 D(6)	May damage the unborn child
					H360 FD(6)	May damage fertility. May damage the unborn child.
					H360 Fd(6)	May damage fertility. Suspected of damaging the unborn child.



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					H360 Df(6)	May damage the unborn child. Suspected of damaging fertility.
	Category 2	Repr. 2		Warning	H361 (5)	Suspected of damaging fertility or the unborn child.
					H361f (6)	Suspected of damaging fertility.
					H361 d (6)	Suspected of damaging the unborn child
					H361f d (6)	Suspected of damaging fertility. Suspected of damaging the unborn child.
	Additional category for effects on or via lactation	Lact.	No Pictogram	No Signal Word	H362	May cause harm to breast-fed children.
(5) = (state specific effect if known)(state route of exposure if it is conclusively proven that no other routes of exposure cause the hazard) (6) F = Fertility, D= Development (lowercase f, d = suspected effect)						



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Specific target organ toxicity (single exposure)	Category 1	STOT SE 1	 GHS08	Danger	H370	Causes damage to organs (7)
	Category 2	STOT SE 2		Warning	H371	May cause damage to organs (7)
	Category 3	STOT SE 3	 GHS07	Warning	H335	May cause respiratory irritation
					H336	May cause drowsiness or dizziness
(7) = (or state all organs affected, if known)(state route of exposure if it is conclusively proven that no other routes of exposure cause the hazard)						
Specific target organ toxicity (repeat exposure)	Category 1	STOT RE 1	 GHS08	Danger	H372	Causes damage to organs (8) through prolonged or repeated exposure (9)
	Category 2	STOT RE 2		Warning	H373	May cause damage to organs (8) through prolonged or repeated exposure (9)
	(8) = (state all organs affected, if known) (9) = (state route of exposure if it is conclusively proven that no other routes of exposure cause the hazard)					
Aspiration Toxicity	Category 1	Asp. Tox. 1	 GHS08	Danger	H304	May be fatal if swallowed and enters airways



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Hazardous to the aquatic environment	Acute Category 1	Aquatic Acute 1	 GHS09	Warning	H400	Very toxic to aquatic life
	Chronic Category 1	Aquatic Chronic 1			H410	Very toxic to aquatic life with long lasting effects
	Chronic Category 2	Aquatic Chronic 2		No Signal Word	H411	Toxic to aquatic life with long lasting effects
	Chronic Category 3	Aquatic Chronic 3	No Pictogram	No Signal Word	H412	Harmful to aquatic life with long lasting effects
	Chronic Category 4	Aquatic Chronic 4			H413	May cause long lasting harmful effects to aquatic life
Hazardous to the ozone layer	Category 1	Ozone 1	 GHS07	Warning	H420	Harms public health and the environment by destroying ozone in the upper atmosphere
* = The Code for the Pictogram and the H-statement do not need to be included on the label.						



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6.1 MSDS of LPG

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Liquefied Petroleum Gas
1.1.1.1 Trade Name	LPG
1.1.2 CAS No.	68476-85-7

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuels Propellant Welding
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Extremely flammable gas	1	Flam. Liq. 2	H220
Gases under pressure: Liquefied gas		Press. Gas (Liq.)	H280
Germ cell mutagenicity	1B	Muta. 1B	H340
Carcinogenicity	1B	Carc. 1B	H350

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):



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H220	Extremely flammable gas
H280	Contains gas under pressure; may explode if heated
H340	May cause genetic defects.
H350	May cause cancer

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P201	Obtain special instructions before use
P210	Keep away from heat / sparks / open flames / hot surfaces. No smoking
P280	Wear eye protection, face protection, protective clothing, protective gloves

Precautionary statements – response

P377	Leaking gas fire: Do not extinguish, unless leak can be stopped safely
P308+P313	IF exposed or concerned: Get medical advice
P381	Eliminate all ignition sources if safe to do so
P403	Store in a well-ventilated place

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

This substance is not considered to be very persistent nor very bioaccumulating (vPvB). This substance is not considered to be persistent, bioaccumulating nor toxic (PBT).

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Liquefied Petroleum Gas

3.1.1 Systematic Chemical Name or Trivial Name: LPG

3.1.2 CAS No.: 68476-85-7

3.1.3 Other Identifiers

Molar mass 44.09 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Provide fresh air. Put victim at rest, cover with a blanket and keep warm. Keep at rest. In case of shortness of breath, give oxygen. Get medical advice/attention



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4.1.2 Skin Contact

Treat frozen body-parts appropriately. In case of frostbite, wash with plenty of water; do not remove clothing. Call a physician immediately

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Call a physician immediately.

4.1.4 Ingestion

No hazards which require special first aid measures.

4.2 Most important symptoms and anticipated effects both acute and delayed
headache, nausea, dizziness, frostbite, vomiting and diarrhea.

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder
- **Unsuitable extinguishing media**
None known

5.2 Special hazards arising from the chemical product:

Evacuate personnel to a safe area. Heating causes rise in pressure with risk of bursting. Use water spray or fog for cooling exposed containers. Vapours may form explosive mixture with air. Vapours are heavier than air and may spread along floors. Vapours are heavier than air and may travel considerable distance to an ignition source and flash back to source of vapours. Do not allow run-off from fire-fighting to enter drains or water courses. Leaking gas fire: Do not extinguish, unless leak can be stopped safely.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Special protective equipment for firefighters. In case of fire: Wear self-contained breathing apparatus. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Ensure procedures and training for emergency decontamination and disposal are in place.

6.2 Environmental precautions:

Do not allow to enter into surface water or drains.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Remove all sources of ignition. Use only explosion-proof equipment. Do not smoke. Provide for sufficient ventilation, particularly in closed rooms. Stop leak if safe to do so.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a dry, cool and well-ventilated area.

Incompatible substances or mixtures

Avoid heat, sparks, open flames and other ignition sources. Incompatible with oxidizing agents (e.g. hypochlorite), acids, (e.g. Nitric acid), heat and ignition sources.

Ventilation requirements

Use local and general ventilation.



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Specific designs for storage rooms or vessels
Recommended storage temperature: 15 – 50 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values: Contains no substances with occupational exposure limit values.

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



a. Hand protection

Wear insulated or leather gloves.

b. Other protection measures

Wear non-static long sleeved shirts and trousers, or coveralls.

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Use self-contained respiratory apparatus for rescue and maintenance work in storage vessels.



8.2.3 Environmental Exposure Controls:

Keep away from drains, surface and ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state

Gas



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Colour	colourless
Odour	Normally an odourless
Melting point/freezing point	-187.6 to -138.3 °C
Boiling point or initial boiling point and boiling range	-161.48 to -0.5 °C
Flammability	Extremely flammable
Lower and upper explosion limit	2.1 vol% (LEL) – 9.6 vol% (UEL)
Flash point	-104 to -60 °C
Auto-ignition temperature	287 - 537 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	not determined
<u>Solubility(ies)</u>	
Water solubility	24.4 – 60.4 mg/l @20°C
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	1.09 – 2.8
Vapour pressure	520-1050 kPa at 40 °C
<u>Density and/or relative density</u>	
Specific Gravity	0.51 to 0.58 (water = 1)
Relative vapour density	1.55 to 2.01 (Air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	None
% Volatiles	100%

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Flammable gases.

10.2 Chemical stability

Stable at ambient temperature and under normal conditions of use.

10.3 Possibility of hazardous reactions

In use may form flammable/explosive vapour-air mixture

10.4 Conditions to avoid



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Keep away from sources of heat (e.g. hot surfaces), sparks and open flames. Temperatures above 38 °C. Exposure to sunlight.

10.5 Incompatible materials

Incompatible with oxidizing agents (e.g. hypochlorite), acids, (e.g. Nitric acid), heat and ignition sources. Do not use natural rubber flexible hoses. Also, incompatible (potentially violently) with oxygen, halogens and metal halides.

10.6 Hazardous decomposition products

Burning produces noxious and toxic fumes. Possible decomposition products are: Carbon oxides.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects:

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	658 mg/l/4h	rat

11.1.2 Skin Corrosion/ Irritation: Not classified

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified

11.1.4 Respiratory or Skin Sensitization: Not classified

11.1.5 Mutagenicity: May cause genetic defects.

11.1.6 Carcinogenicity: May cause cancer.

11.1.7 Reproductive Toxicity: Not classified

11.1.8 Specific target organ toxicity-single exposure: Not classified.

11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: Not classified

11.1.11 Additional Information: Symptoms related to the physical, chemical and toxicological characteristics. Reference to other sections: 4.2.

Section 12: Ecological Information

12.1 Toxicity: No information provided.

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	1.09 – 2.8
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12.4 Mobility in Soil: Immiscible.

12.5 Other adverse effects: Data are not available



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Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**

Do not burn, or use a cutting torch on the empty drum. Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material.

13.1.1 Contaminated Packaging: Dispose of spent product packaging responsibly and lawfully with due consideration for health, safety and the environment.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1965

14.2 UN Proper Shipping Name: LPG

14.3 Transport Class(es): 2

14.3.1 Hazard Labels: 2

14.3.2 Symbols:



14.4 Packaging Group: Not applicable

14.5 Environmental Hazards: No information provided

14.6 Special Precautions for User: No data available

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

The storage of significant quantities of gas cylinders must comply with AS4332 The storage and handling of gases in cylinders, and AS/NZS 1596 The storage and handling of LP Gas.

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE



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- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.2 MSDS of MS

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Motor Spirit
1.1.1.1 Trade Name	Gasoline
1.1.2 CAS No.	86290-81-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuel
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Extremely flammable liquid and vapour	1	Flam. Liq. 1	H224
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411
Germ cell mutagenicity	1B	Muta. 1B	H340
Carcinogenicity	1B	Carc. 1B	H350

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS02



GHS07



GHS08



GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H225	Extremely flammable liquid and vapour.
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H361	Suspected of damaging fertility or the unborn child.
H340	May cause genetic defects.
H350	May cause cancer.
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P201	Obtain special instructions before use.
P280	Wear protective gloves/eye protection
P210	Keep away from heat/sparks/open flames/hot surfaces. No smoking.
P273	Avoid release to the environment.

Precautionary statements – response

P331	Do NOT induce vomiting.
P301+P310	IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician.
P403+P233	Store in a well-ventilated place. Keep container tightly closed.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Unleaded Motor Gasoline is extremely flammable liquid which may generate explosive mixtures of hydrocarbon vapours and air at ambient temperatures.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Motor Spirit

3.1.1 Systematic Chemical Name or Trivial Name: Gasoline

3.1.2 CAS No.: 86290-81-5



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3.1.3 Other Identifiers

Molar mass

150 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Remove person to fresh air and keep comfortable for breathing. If breathing is irregular or stopped, administer artificial respiration. Get medical advice/attention.

4.1.2 Skin Contact

Wash with plenty of soap and water. When in doubt or if symptoms are observed, get medical advice. Remove contaminated clothing and wash it before reuse.

4.1.3 Eye Contact

Rinse immediately carefully and thoroughly with eye-bath or water. When in doubt or if symptoms are observed, get medical advice.

4.1.4 Ingestion

Rinse mouth thoroughly with water. Do NOT induce vomiting. Get immediate medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Dizziness, Drowsiness, Cough, Mental Confusion Headache, Erythema, Central nervous system depression.

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, alcohol resistant foam, Dry extinguishing powder, Carbon dioxide

- **Unsuitable extinguishing media**

Strong water jet for safety reasons

5.2 Special hazards arising from the chemical product:



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Extremely flammable liquid and vapour. Heating causes rise in pressure with risk of bursting. Vapours can form explosive mixtures with air. Vapours are heavier than air, spread along floors and form explosive mixtures with air. Vapours can travel considerable distances to a source of ignition where they can ignite, flash back, or explode.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides (COx), Organic compounds, Hydrogen sulphide (H₂S) Sulphur oxides, Sulphuric acid

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire: Wear self-contained breathing apparatus. Use water spray jet to protect personnel and to cool endangered containers. Evacuate area. Do not allow run-off from fire-fighting to enter drains or water courses. Dispose according to legislation.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Evacuate area. Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Personal protection equipment: see section 8. Avoid contact with skin, eyes and clothes. Do not breathe vapour/spray. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Ensure that the equipment is adequately grounded. Use explosion-proof machinery, apparatus, ventilation facilities, tools etc. Use only non-sparking tools. As appropriate: Product may release Hydrogen Sulphide: A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if safe to do so. Dam up.

Advice on how to clean up a spill

Use foam on spills to minimise vapours.

Other information relating to spills and releases

Dispose of waste product or used containers according to local regulations.

6.4 Reference to other sections:



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Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Risk of explosive mixtures of vapour and air. Ensure that all relevant regulations regarding explosive atmospheres, and handling and storage facilities of flammable products, are followed.

Measures to prevent fire as well as aerosol and dust generation

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.



Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep in a dry, cool and well-ventilated place.

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

Bund storage facilities to prevent soil and water pollution in the event of spillage.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STE L [ppm]	STEL [mg/m ³]	Ceilin g- C [ppm]	Ceilin g-C [mg/m ³]	Notation
USA	Gasoline	86290-81-5	ACGIH	300	72	500				



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- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	1300 mg/m ³	Human, inhalatory	worker (industry)	Acute - systemic effects
DNEL	1100 mg/m ³	Human, inhalatory	worker (industry)	Acute -local effects
DNEL	840 mg/m ³	Human, inhalatory	worker (industry)	Long-term - local effects
DNEL	1200 mg/m ³	Human, inhalatory	general population	Acute - systemic effects
DNEL	640 mg/m ³	Human, inhalatory	general population	Acute - local effects
DNEL	180 mg/m ³	Human, inhalatory	general population	Long-term - local effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.



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- **Skin protection**



- a) **Hand protection**

- Wear petroleum product resistant gloves

- b) **Type of material**

- NBR (Nitrile rubber)

- c) **Material thickness**

- ≥0.3 mm

- d) **Breakthrough times of the glove material**

- >480 minutes (permeation: level 6)

- e) **Other protection measures**

- The quality of the protective gloves resistant to chemicals must be chosen as a function of the specific working place concentration and quantity of hazardous substances.

- **Respiratory protection:** Respiratory protection necessary at: Aerosol or mist formation. Type: A2 The filter class must be suitable for the maximum contaminant concentration



(gas/vapour/aerosol/particulates) that may arise when handling the product. If the concentration is exceeded, self-contained breathing apparatus must be used.

8.2.3 Environmental Exposure Controls:

Do not allow to enter into surface water or drains. Comply with applicable Community environmental protection legislation. Do not allow contact with soil, surface or ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Orange
Odour	like: - petroleum hydrocarbon odour
Melting point/freezing point	< -60 °C
Boiling point or initial boiling point and boiling range	210 °C
Flammability	Not applicable, liquid
Lower and upper explosion limit	No data available
Flash point	< -40 °C (closed cup)
Research Octane Number (RON)	91



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Motor Ocatne Number (MON)	81
Auto-ignition temperature	> 290 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	< 1 mm ² /s at 40 °C
<u>Solubility(ies)</u>	
Water solubility	30 - 100 mg/l (at 20 °C)
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	60 kPa @ 38 °C
<u>Density and/or relative density</u>	
Density	720-775 kg/m ³ at 15 °C
Relative vapour density	3 – 4 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising Stability	360 minutes

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Extremely flammable liquid and vapour.

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Violent reaction with: Vapours can form explosive mixtures with air.

10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Oxidising substances

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.



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Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	> 5.2 mg/l/4h	rat
Oral	LD50	14000 mg/kg	rat
Dermal	LD50	> 2000 mg/kg	rabbit
Dermal	LD50	> 2000 mg/kg	rat

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: May cause genetic defects

11.1.6 Carcinogenicity: May cause cancer.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: Symptoms related to the physical, chemical and toxicological characteristics, For further information see section 4

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	8.2 mg/l	fish	96 h
EC50	4.5 mg/l	aquatic invertebrates	48 h
EC50	3.1 mg/l	Algae	

12.2 Persistence and Degradability: Not available.

12.3 Bio Accumulative Potential: Heavier hydrocarbons of the product are possibly accumulative in aquatic organisms.



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n-octanol/water (log KOW)

1 - 8

12.4 Mobility in Soil: Spillage, depending on the ambient temperatures, may evaporate in significant quantities, and the rest of the spilled product may penetrate the soil and contaminate ground waters.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Waste is disposed of by de-harming it in accordance with national requirements and local regulations or via a licensed waste disposal contractor. Identify the hazards of waste handling and undertake required safety measures. Personal protective equipment is necessary for waste managing personnel.



- Sewage disposal-relevant information**

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

13.1.1 Contaminated Packaging: Do not burn, or use a cutting torch on, the empty drum. Do not puncture or incinerate.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1268

14.2 UN Proper Shipping Name: Petroleum

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3

14.3.2 Symbols:



14.4 Packaging Group: I

14.5 Environmental Hazards: yes (sea pollutant)

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.



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Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.3 MSDS of FOOD GRADE HEXANE

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	n-Hexane
1.1.1.1 Trade Name	Hexane
1.1.2 Chemical Formula	C ₆ H ₁₄
1.1.3 CAS No.	110-54-3

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Solvent Laboratory chemical Laboratory and analytical use
Uses advised against	Do not use for products which come into contact with foodstuffs. Do not use for private purposes (household)

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	2	Flam. Liq. 2	H225
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361f
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411



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For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H225	Highly flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H361f	Suspected of damaging fertility
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P202	Do not handle until all safety precautions have been read and understood
P280	Wear protective gloves/eye protection

Precautionary statements – response

P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]
P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
P308+P313	IF exposed or concerned: Get medical advice/attention



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2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: n-Hexane

3.1.1 Systematic Chemical Name or Trivial Name: Hexane

3.1.2 IUPAC Name: Hexane

3.1.3 CAS No.: 110-54-3

3.1.4 Other Identifiers

Molecular formula

C₆H₁₄

Molar mass

86.18 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Provide fresh air. In all cases of doubt, or when symptoms persist, seek medical advice.

4.1.2 Skin Contact

Rinse skin with water/shower. In case of skin irritation, consult a physician.

4.1.3 Eye Contact

Rinse cautiously with water for several minutes. In all cases of doubt, or when symptoms persist, seek medical advice.

4.1.4 Ingestion

Call a physician immediately. In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Observe aspiration hazard if vomiting occurs.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



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- **Suitable extinguishing media**

Co-ordinate firefighting measures to the fire surroundings water spray, dry extinguishing powder, BC-powder, carbon dioxide (CO₂)

- **Unsuitable extinguishing media**

Water jet

5.2 Special hazards arising from the chemical product:

Combustible. In case of insufficient ventilation and/or in use, may form flammable/explosive vapour- air mixture. Solvent vapours are heavier than air and may spread along floors. Places which are not ventilated, e.g., unventilated below ground level areas such as trenches, conduits and shafts, are particularly prone to the presence of flammable substances or mixtures. Vapours are heavier than air, spread along floors and form explosive mixtures with air. Vapours may form explosive mixtures with air.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire and/or explosion do not breathe fumes. Do not allow firefighting water to enter drains or water courses. Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Do not breathe vapours, aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert.

6.2 Environmental precautions:

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. Danger of explosion.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Covering of drains.



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Advice on how to clean up a spill

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge. Due to danger of explosion, prevent leakage of vapours into cellars, flues and ditches.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a well-ventilated place. Keep container tightly closed.

Incompatible substances or mixtures

Observe hints for combined storage.

Consideration of other advice:

Ground/bond container and receiving equipment.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 15 – 25 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters



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National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
EU	n-hexane	110-54-3	IOELV	20	72						2006/15/EC
IE	n-hexane	110-54-3	OELV	20	72						S.I. No. 619 of 2001

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	75 mg/m ³	human, inhalatory	worker (industry)	chronic - systemic effects
DNEL	11 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.



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- **Skin protection**



- a) **Hand protection**

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves. The times are approximate values from measurements at 22 ° C and permanent contact. Increased temperatures due to heated substances, body heat etc. and a reduction of the effective layer thickness by stretching can lead to a considerable reduction of the breakthrough time. If in doubt, contact manufacturer. At an approx. 1.5 times larger / smaller layer thickness, the respective breakthrough time is doubled / halved. The data apply only to the pure substance. When transferred to substance mixtures, they may only be considered as a guide.

- b) **Type of material**

NBR (Nitrile rubber)

- c) **Material thickness**

≥0,4 mm

- d) **Breakthrough times of the glove material**

>480 minutes (permeation: level 6)

- e) **Other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

Flame-retardant protective clothing.

- **Respiratory protection:** Respiratory protection necessary at: Aerosol or mist formation. Type: A (against organic gases and vapours with a boiling point of > 65 °C , colour code: Brown).



8.2.3 Environmental Exposure Controls:

Keep away from drains, surface and ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	colourless
Odour	like: - Gasoline



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Melting point/freezing point	-95 °C at 1.013 hPa (ECHA)
Boiling point or initial boiling point and boiling range	63 – 69 °C at 1.013 hPa
Flammability	flammable liquid in accordance with GHS criteria
Lower and upper explosion limit	1.1 vol% (LEL) – 7.5 vol% (UEL)
Flash point	-22 °C at 1.013 hPa (ECHA)
Auto-ignition temperature	225 °C (ECHA)
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	0.5 mm ² /s at 20 °C
<u>Solubility(ies)</u>	
Water solubility	<0.1 g/l at 20 °C
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 hPa at 20 °C
<u>Density and/or relative density</u>	
Density	0.66-0.687 g/ml at 20 °C
Relative vapour density	2.79 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

It's a reactive substance. Risk of ignition. Vapours may form explosive mixtures with air.

If heated

Risk of ignition.

10.2 Chemical stability

The material is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

10.3 Possibility of hazardous reactions

Violent reaction with: strong oxidiser, Chlorine, Iodine, Peroxides, Nitrogen oxides (NO_x),
=> Explosive properties



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10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Rubber articles, different plastics

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
nausea, vomiting, aspiration hazard
- **If in eyes**
slightly irritant but not relevant for classification
- **If inhaled**
irritant effects, headache, vertigo, fatigue, dizziness, narcosis
- **If on skin**
causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	185 mg/l/4h	rat	TOXNET
Oral	LD50	25.000 mg/kg	rat	TOXNET
Dermal	LD50	>2.000 mg/kg	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.



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11.1.9 Specific target organ toxicity - repeated exposure

May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled).

Hazard category	Target organ	Exposure route
2	nervous system	if inhaled

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Source	Exposure time
LL50	12.51 mg/l	fish	ECHA	96 h
EL50	21.85 mg/l	aquatic invertebrates	ECHA	48 h

12.2 Persistence and Degradability:

Theoretical Oxygen Demand: 3,527 mg/mg

Theoretical Carbon Dioxide: 3,064 mg/mg

Process of degradability

Process	Degradation rate	Time
oxygen depletion	83 %	10 d

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	4 (pH value: 7, 20 °C) (ECHA)
BCF	501.2 (ECHA)

12.4 Mobility in Soil:

The Organic Carbon normalized adsorption coefficient	3.34 (ECHA)
--	-------------

12.5 Other adverse effects: Data are not available



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Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

13.1.1 Contaminated Packaging: It is a dangerous waste; only packagings which are approved may be used.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1208

14.2 UN Proper Shipping Name: Hexanes

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3, "Fish and tree"

14.3.2 Symbols:



14.4 Packaging Group: II

14.5 Environmental Hazards: yes (hazardous to the aquatic environment)

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable



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Section 16: Other Information

Can cause motor neuropathy in exposed workers. May be irritating to respiratory tract and narcotic in high concentrations. Inhalation of 5000 ppm for 1/6 hours produces marked vertigo.

25000—1000 ppm for 12 hours produces drowsiness, fatigue, loss of appetite, paresthesia in distal extremities. 2000 ppm for 1/6 hours produces no symptoms. Dangerous if abused. A solvent, permitted in food industry for extraction of oil. A very dangerous fire and explosion hazard when exposed to heat or flame

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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6.4 MSDS of Solvent-1425

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Solvent-1425
1.1.1.1 Trade Name	Solvent-1425
1.1.2 CAS No.	64742-49-0

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Solvent Laboratory chemical Laboratory and analytical use
Uses advised against	Do not use for products which come into contact with foodstuffs. Do not use for private purposes (household)

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.: 022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	2	Flam. Liq. 2	H225
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361f
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects



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Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02

GHS08

GHS09

2.2.2 Signal Word(s): **Danger**

2.2.3 Hazard Statements(s):

H225	Highly flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H361f	Suspected of damaging fertility
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P202	Do not handle until all safety precautions have been read and understood
P280	Wear protective gloves/eye protection

Precautionary statements – response

P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]
P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.



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Section 3: Composition/Information on Ingredients

3.1 Name of Product: Solvent-1425

3.1.1 Systematic Chemical Name or Trivial Name: Solvent-1425

3.1.2 CAS No.: 64742-49-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Provide fresh air. In all cases of doubt, or when symptoms persist, seek medical advice.

4.1.2 Skin Contact

Rinse skin with water/shower. In case of skin irritation, consult a physician.

4.1.3 Eye Contact

Rinse cautiously with water for several minutes. In all cases of doubt, or when symptoms persist, seek medical advice.

4.1.4 Ingestion

Call a physician immediately. In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). Observe aspiration hazard if vomiting occurs.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Co-ordinate firefighting measures to the fire surroundings
water spray, dry extinguishing powder, BC-powder, carbon dioxide (CO₂)

- **Unsuitable extinguishing media**

Water jet

5.2 Special hazards arising from the chemical product:



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Combustible. In case of insufficient ventilation and/or in use, may form flammable/explosive vapour- air mixture. Solvent vapours are heavier than air and may spread along floors. Places which are not ventilated, e.g., unventilated below ground level areas such as trenches, conduits and shafts, are particularly prone to the presence of flammable substances or mixtures. Vapours are heavier than air, spread along floors and form explosive mixtures with air. Vapours may form explosive mixtures with air.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire and/or explosion do not breathe fumes. Do not allow firefighting water to enter drains or water courses. Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Do not breathe vapours, aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert.

6.2 Environmental precautions:

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. Danger of explosion.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Covering of drains.

Advice on how to clean up a spill

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents).

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.



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Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge. Due to danger of explosion, prevent leakage of vapours into cellars, flues and ditches.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a well-ventilated place. Keep container tightly closed.

Incompatible substances or mixtures

Observe hints for combined storage.

Consideration of other advice:

Ground/bond container and receiving equipment.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 15 – 25 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling g-C [ppm]	Ceiling g-C [mg/m ³]	Notation	Source
EU	Solvent-1425	64742-49-0	IOE LV	20	72						2006/15/EC
IE	Solvent-1425	64742-49-0	OEL V	20	72						S.I. No. 619 of 2001



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Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	75 mg/m ³	human, inhalatory	worker (industry)	chronic - systemic effects
DNEL	11 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.

- Skin protection



a. Hand protection

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves. The times are approximate values from measurements at 22 ° C and



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permanent contact. Increased temperatures due to heated substances, body heat etc. and a reduction of the effective layer thickness by stretching can lead to a considerable reduction of the breakthrough time. If in doubt, contact manufacturer. At an approx. 1.5 times larger / smaller layer thickness, the respective breakthrough time is doubled / halved. The data apply only to the pure substance. When transferred to substance mixtures, they may only be considered as a guide.

b. Type of material

NBR (Nitrile rubber)

c. Material thickness

≥0,4 mm

d. Breakthrough times of the glove material

>480 minutes (permeation: level 6)

e. Other protection measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

Flame-retardant protective clothing.

- **Respiratory protection:** Respiratory protection necessary at: Aerosol or mist formation. Type: A (against organic gases and vapours with a boiling point of > 65 °C, colour code: Brown).



8.2.3 Environmental Exposure Controls:

Keep away from drains, surface and ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Clear Colourless Liquid
Odour	Solvent odour
Melting point/freezing point	-95 °C at 1.013 hPa (ECHA)
Boiling point or initial boiling point and boiling range	68 – 69 °C at 1.013 hPa
Flammability	flammable liquid in accordance with GHS criteria
Lower and upper explosion limit	1 vol% (LEL) – 6 vol% (UEL)
Flash point	-22 °C
Auto-ignition temperature	280 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	0.5 mm ² /s at 20 °C
<u>Solubility(ies)</u>	



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Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 hPa at 20 °C
<u>Density and/or relative density</u>	
Density	0.66 g/cm ³ at 20 °C
Relative vapour density	4.8
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

It's a reactive substance. Risk of ignition. Vapours may form explosive mixtures with air.

If heated

Risk of ignition.

10.2 Chemical stability

The material is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

10.3 Possibility of hazardous reactions

Violent reaction with: strong oxidiser, Chlorine, Iodine, Peroxides, Nitrogen oxides (NO_x),
=> Explosive properties

10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Rubber articles, different plastics

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.



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Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
nausea, vomiting, aspiration hazard
- **If in eyes**
slightly irritant but not relevant for classification
- **If inhaled**
irritant effects, headache, vertigo, fatigue, dizziness, narcosis
- **If on skin**
causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	18 mg/l/4h	rat	TOXNET
Oral	LD50	3280 mg/kg	rat	TOXNET

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure

May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled).

Hazard category	Target organ	Exposure route
2	nervous system	if inhaled

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.



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Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	7.72 mg/l	fish	ECHA	96 h
LC50	17 mg/l	aquatic invertebrates	ECHA	48 h

12.2 Persistence and Degradability: Not Classified

12.3 Bio Accumulative Potential: Not Classified

12.4 Mobility in Soil: Not Classified

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- Sewage disposal-relevant information**

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

13.1.1 Contaminated Packaging: It is a dangerous waste; only packaging which are approved may be used.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3295

14.2 UN Proper Shipping Name: Flammable Liquid N.O.S

14.3 Transport Class(es): 3.1B

14.3.1 Hazard Labels: 3

14.3.2 Symbols:





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14.4 Packaging Group: II

14.5 Environmental Hazards: yes (hazardous to the aquatic environment)

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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6.5 MSDS of Naphtha

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Naphtha
1.1.1.1 Trade Name	Naphtha
1.1.2 CAS No.	64741-46-4

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Raw material for industry Petroleum Petroleum products
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	1	Flam. Liq. 1	H224
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361fd
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411
Germ cell mutagenicity	1B	Muta. 1B	H340
Carcinogenicity	1B	Carc. 1B	H350

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H225	Highly flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H361fd	Suspected of damaging fertility
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H340	May cause genetic defects
H350	May cause cancer
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention	
P201	Obtain special instructions before use
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P273	Avoid release to the environment
P280	Wear eye protection, face protection, protective clothing, protective gloves
Precautionary statements – response	
P301+P310+P331	IF SWALLOWED: Immediately call a doctor, a POISON CENTER. Do NOT induce vomiting
P403+P233	Store in a well-ventilated place. Keep container tightly closed

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Naphtha



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3.1.1 Systematic Chemical Name or Trivial Name: Naphtha

3.1.2 CAS No.: 64741-46-4

3.1.3 Other Identifiers

Molar mass 101-215 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Keep at rest. Provide fresh air. Give oxygen or artificial respiration if necessary. Call a physician immediately.

4.1.2 Skin Contact

Take off immediately all contaminated clothing. Wash with plenty of water/. In case of doubt or persistent symptoms, consult always a physician. Wash contaminated clothing before reuse.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If eye irritation persists: Get medical advice/attention.

4.1.4 Ingestion

Do Not induce vomiting. Rinse mouth immediately and drink plenty of water. Never give anything by mouth to an unconscious person. Get medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Mental confusion, Irritation, Dizziness, Drowsiness, Cough, Headache, Erythema

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder

- **Unsuitable extinguishing media**

Strong water jet



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5.2 Special hazards arising from the chemical product:

Heating causes rise in pressure with risk of bursting. Vapours may form explosive mixture with air. Vapours are heavier than air and may spread along floors. Vapours are heavier than air and may travel considerable distance to an ignition source and flash back to source of vapours. As appropriate. Hydrogen sulphide (H₂S). Sulphur oxides. Sulphuric acid. Do not allow run-off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Special protective equipment for firefighters. In case of fire: Wear self-contained breathing apparatus. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Evacuate personnel to a safe area. Stay upwind/keep distance from source. Provide adequate ventilation.

6.2 Environmental precautions:

Do not allow to enter into surface water or drains

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if safe to do so. Dam up

Advice on how to clean up a spill

Take up liquid spill into absorbent material, e.g.: sand, earth, vermiculite or powdered limestone. Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of as special waste in compliance with local and national regulations

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.



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Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge. Due to danger of explosion, prevent leakage of vapours into cellars, flues and ditches.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Storage of flammable liquids. Keep container tightly closed in a cool, well-ventilated place

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling g-C [ppm]	Ceiling g-C [mg/m ³]	Notation	Source
EU	Naptha	64741-46-4	IOELV	20	72						
IE	Naptha	64741-46-4	OELV	20	72						



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Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



a. Hand protection

Wear chemically resistant gloves.

b. Type of material

NBR (Nitrile rubber)

c. Material thickness

≥ 0.28 – 0.54 mm

d. Breakthrough times of the glove material

>480 minutes (permeation: level 6)

e. Other protection measures

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space, such as (but not limited to): other chemicals that are possibly used, physical requirements (protection against cutting/drilling, skill, thermal protection), and the instructions/specification of the supplier of gloves.



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- **Respiratory protection:** Respiratory protection necessary at: Aerosol or mist formation.
Type: A



Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	colourless
Odour	like: - petroleum hydrocarbon odour
Melting point/freezing point	< -20 °C
Boiling point or initial boiling point and boiling range	30 - 190 °C
Flammability	Not applicable, liquid
Lower and upper explosion limit	0.8 vol% (LEL) – 5.0 vol% (UEL)
Flash point	≤ 0 °C (closed cup)
Auto-ignition temperature	200 - 230 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	0.5 – 0.6 mm ² /s (@ 20°C)
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	2.7
Vapour pressure	12 PSI 37.8°C
<u>Density and/or relative density</u>	
Density	0.74 g/ml (@ 15 °C)
Relative vapour density	3.5 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.



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Section 10: Stability and Reactivity

10.1 Reactivity

Extremely flammable liquid and vapour.

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.

10.5 Incompatible materials

Oxidising substances.

10.6 Hazardous decomposition products

Carbon oxides. As appropriate: Hydrogen sulphide (H₂S). SO_x. Sulphuric acid: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	> 5610 mg/m ³	rat
Oral	LD50	> 5000 mg/kg	rat
Dermal	LD50	> 2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: May cause genetic defects.

11.1.6 Carcinogenicity: May cause cancer.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.



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Section 12: Ecological Information

12.1 Toxicity: Not Classified.

12.2 Persistence and Degradability: No data available

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	2.1
BCF	3.5 – 4.4

12.4 Mobility in Soil: No data available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**
Do not allow to enter into surface water or drains.

13.1.1 Contaminated Packaging: Handle with care.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1255

14.2 UN Proper Shipping Name: Naphtha

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3

14.3.2 Symbols:



14.4 Packaging Group: I

14.5 Environmental Hazards: No



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14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.6 MSDS of Sp. Cut Naphtha

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Sp. Cut Naphtha
1.1.1.1 Trade Name	Sp. Cut Naphtha
1.1.2 CAS No.	64741-46-4

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Raw material for industry Petroleum Petroleum products
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	1	Flam. Liq. 1	H224
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361fd
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411
Germ cell mutagenicity	1B	Muta. 1B	H340
Carcinogenicity	1B	Carc. 1B	H350

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H225	Highly flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H361fd	Suspected of damaging fertility
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H340	May cause genetic defects
H350	May cause cancer
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention	
P201	Obtain special instructions before use
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P273	Avoid release to the environment
P280	Wear eye protection, face protection, protective clothing, protective gloves
Precautionary statements – response	
P301+P310+P331	IF SWALLOWED: Immediately call a doctor, a POISON CENTER. Do NOT induce vomiting
P403+P233	Store in a well-ventilated place. Keep container tightly closed

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Sp. Cut Naphtha



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3.1.1 Systematic Chemical Name or Trivial Name: Sp. Cut Naphtha

3.1.2 CAS No.: 64741-46-4

3.1.3 Other Identifiers

Molar mass 101-215 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Keep at rest. Provide fresh air. Give oxygen or artificial respiration if necessary. Call a physician immediately.

4.1.2 Skin Contact

Take off immediately all contaminated clothing. Wash with plenty of water/. In case of doubt or persistent symptoms, consult always a physician. Wash contaminated clothing before reuse.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If eye irritation persists: Get medical advice/attention.

4.1.4 Ingestion

Do Not induce vomiting. Rinse mouth immediately and drink plenty of water. Never give anything by mouth to an unconscious person. Get medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Mental confusion, Irritation, Dizziness, Drowsiness, Cough, Headache, Erythema

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder

- **Unsuitable extinguishing media**

Strong water jet



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5.2 Special hazards arising from the chemical product:

Heating causes rise in pressure with risk of bursting. Vapours may form explosive mixture with air. Vapours are heavier than air and may spread along floors. Vapours are heavier than air and may travel considerable distance to an ignition source and flash back to source of vapours. As appropriate. Hydrogen sulfide (H₂S). Sulphur oxides. Sulphuric acid. Do not allow run-off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Special protective equipment for firefighters. In case of fire: Wear self-contained breathing apparatus. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Evacuate personnel to a safe area. Stay upwind/keep distance from source. Provide adequate ventilation.

6.2 Environmental precautions:

Do not allow to enter into surface water or drains

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if safe to do so. Dam up

Advice on how to clean up a spill

Take up liquid spill into absorbent material, e.g.: sand, earth, vermiculite or powdered limestone. Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of as special waste in compliance with local and national regulations

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.



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Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge. Due to danger of explosion, prevent leakage of vapours into cellars, flues and ditches.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Storage of flammable liquids. Keep container tightly closed in a cool, well-ventilated place

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling g-C [ppm]	Ceiling g-C [mg/m ³]	Notation	Source
EU	Sp. Cut Naptha	64741-46-4	IOEL V	20	72						
IE	Sp. Cut Napht	64741-46-4	OEL V	20	72						



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Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.

- Skin protection



a. Hand protection

Wear chemically resistant gloves.

b. Type of material

NBR (Nitrile rubber)

c. Material thickness

≥ 0.28 – 0.54 mm

d. Breakthrough times of the glove material

>480 minutes (permeation: level 6)

e. Other protection measures

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space, such as (but not limited to): other chemicals that are possibly used, physical requirements (protection against cutting/drilling, skill, thermal protection), and the instructions/specification of the supplier of gloves.



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- **Respiratory protection:** Respiratory protection necessary at: Aerosol or mist formation.
Type: A



Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	colourless
Odour	like: - petroleum hydrocarbon odour
Melting point/freezing point	< -20 °C
Boiling point or initial boiling point and boiling range	30 - 105 °C
Flammability	Not applicable, liquid
Lower and upper explosion limit	0.8 vol% (LEL) – 5.0 vol% (UEL)
Flash point	≤ 0 °C (closed cup)
Auto-ignition temperature	200 - 230 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	0.5 – 0.6 mm ² /s (@ 20°C)
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	2.7
Vapour pressure	66.7 – 85.3 kPa
<u>Density and/or relative density</u>	
Density	0.75-0.785 g/ml @ 15 °C
Relative vapour density	3.5 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.



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Section 10: Stability and Reactivity

10.1 Reactivity

Extremely flammable liquid and vapour.

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.

10.5 Incompatible materials

Oxidising substances.

10.6 Hazardous decomposition products

Carbon oxides. As appropriate: Hydrogen sulfide (H₂S). SO_x. Sulphuric acid: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	> 5610 mg/m ³	rat
Oral	LD50	> 5000 mg/kg	rat
Dermal	LD50	> 2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: May cause genetic defects.

11.1.6 Carcinogenicity: May cause cancer.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.



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Section 12: Ecological Information

12.1 Toxicity: Not Classified.

12.2 Persistence and Degradability: No data available

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	2.1
BCF	3.5 – 4.4

12.4 Mobility in Soil:

The Organic Carbon normalized adsorption coefficient	No data available.
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12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**
Do not allow to enter into surface water or drains.

13.1.1 Contaminated Packaging: Handle with care.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1255

14.2 UN Proper Shipping Name: Sp. Cut Naphtha

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3

14.3.2 Symbols:





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14.4 Packaging Group: I

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- v. Chemical hazard awareness
- vi. Safety control measures including PPE
- vii. Housekeeping, hygiene and chemical storage
- viii. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.7 MSDS of MTO

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	MTO
1.1.1.1 Trade Name	MTO
1.1.2 CAS No.	8006-64-2

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuel
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid and vapor	3	Flam. Liq. 3	H226
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger



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2.2.3 Hazard Statements(s):

H226	Flammable liquid and vapor
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P210	Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
P102	Keep out of reach of children.
P273	Avoid release to the environment.
P280	Wear eye protection, face protection, protective clothing, protective gloves.

Precautionary statements – response

P301+P310+P331	IF SWALLOWED: Immediately call a doctor, a POISON CENTER. Do NOT induce vomiting.
P501	Dispose of contents and container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Mineral Turpentine Oil

3.1.1 Systematic Chemical Name or Trivial Name: MTO

3.1.2 CAS No.: 8008-64-2

3.1.3 Other Identifiers

Molar mass	170 g/mol
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Section 4: First-Aid Measures

4.1 Description of first aid measures





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4.1.1 Inhalation

Keep at rest. Provide fresh air. Give oxygen or artificial respiration if necessary. Call a physician immediately.

4.1.2 Skin Contact

Take off immediately all contaminated clothing. Wash with plenty of water/. In case of doubt or persistent symptoms, consult always a physician. Wash contaminated clothing before reuse.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If eye irritation persists: Get medical advice/attention.

4.1.4 Ingestion

Do Not induce vomiting. Rinse mouth immediately and drink plenty of water. Get medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Cough, Nausea, Erythema

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder.
- **Unsuitable extinguishing media**
Strong water jet.

5.2 Special hazards arising from the chemical product:

Vapours may form explosive mixture with air. Vapours are heavier than air and may spread along floors. Vapours are heavier than air and may travel considerable distance to an ignition source and flash back to source of vapours. On heating there is a risk of a build-up of pressure in hermetically sealed containers or tanks. Do not allow run-off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide, carbon dioxide and unburned hydrocarbons (smoke). Organic compounds. nitrogen oxides (NOx) and sulphur oxides.

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters



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Special protective equipment for firefighters. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Evacuate personnel to a safe area. Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Concerning personal protective equipment to use.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if safe to do so. Dam up.

Advice on how to clean up a spill

Take up liquid spill into absorbent material, e.g.: sand, earth, vermiculite or powdered limestone. Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of as special waste in compliance with local and national regulations.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provide adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take any precaution to avoid mixing with combustibles.

Measures to protect the environment



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Do not allow to enter into surface water or drains.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed in a cool, well-ventilated place.

Incompatible substances or mixtures

Oxidising substances. Strong acids

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling - C [ppm]	Ceiling -C [mg/m ³]	Notation	Source
USA	MTO	8006-64-2	ACGIH		200						

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	19 mg/kg bw/day	human, oral	general population	Long-term - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:



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Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- **Hand protection**

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

- a) **Type of material**

NBR (Nitrile rubber)

- b) **Material thickness**

≥0.54 mm

- c) **Breakthrough times of the glove material**

>360 minutes

- d) **Other protection measures**

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Half-face mask (DIN EN 140). Full face mask (EN 136). Filter type: AP (EN 141). Use self-contained respiratory apparatus for rescue and maintenance work in



storage vessels. (EN 137)

8.2.3 Environmental Exposure Controls:

Avoid release to the environment. Comply with applicable Community environmental protection legislation.



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Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Water white liquid with kerosene like odour.
Odour	like: - petroleum hydrocarbon odour.
Melting point/freezing point	-47 °C
Boiling point or initial boiling point and boiling range	145°C to 205°C
Flammability	Not applicable, liquid
Lower and upper explosion limit	0.6 vol% (LEL) – 4.9 vol% (UEL)
Flash point	35°C
Auto-ignition temperature	5%
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	1 - 2 cPs (@ 40°C)
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	> 3
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	4 mm of Hg at 20°C
<u>Density and/or relative density</u>	
Density	0.775 – 0.84 (@ 15°C)
Relative vapour density	4.8(Air=1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Flammable liquid.

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.



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10.4 Conditions to avoid

Can undergo auto-oxidation in air & generate heat which can build up in a confined space to cause spontaneous combustion

10.5 Incompatible materials

Incompatible with oxidizing agents & chlorine. Reacts vigorously with oxidising materials. Can react violently with Ca(OCl), CrO, Cr(OCl), SnCl, chloromelamines .

10.6 Hazardous decomposition products

Hazardous combustion products: Carbon di oxide, carbon mono oxide

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Oral	LD50	5760 mg/kg	rat	OECD

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure
Not classified

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Source
LC50	2 - 5 mg/l	Fish 1	OECD
EC50	1.4 mg/l	Daphnia 1	OECD



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ErC50	1 - 3 mg/l	algae	OECD
Aquatic toxicity (chronic)			
NOEC	0.48 mg/l	daphnia	NOEL
NOEC	0.98 mg/l	fish	NOEL

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	> 3
BCF	Low potential.

12.4 Mobility in Soil:

The Organic Carbon normalized adsorption coefficient	Low potential
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12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**
Do not allow to enter into surface water or drains.

13.1.1 Contaminated Packaging: Never use pressure to empty container.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1223

14.2 UN Proper Shipping Name: Kerosene



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14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.8 MSDS of ATF

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Kerosene(petroleum)
1.1.1.1 Trade Name	Aviation Turbine Fuel
1.1.2 CAS No.	8008-20-6

1.2 Relevant identified uses of the chemical product and uses advised against

Uses advised against	No data available
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1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	3	Flam. Liq. 3	H226
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411

2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):



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H226	Flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P102	Keep out of reach of children
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P273	Avoid release to the environment

Precautionary statements – response

P301+P310	IF SWALLOWED: Immediately call a POISON CENTER/ doctor
P331	Do NOT induce vomiting
P261	Avoid breathing vapours
P280	Wear protective gloves
P501	Dispose of contents/container to an approved waste disposal plant

2.3 Other hazards except those mentioned in DGCL Rules, 201:

Evaporates slowly. May cause eye and respiratory system irritation. Risk of soil and ground water contamination

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Aviation Turbine Fuel

3.1.1 Systematic Chemical Name or Trivial Name: Kerosene(petroleum)

3.1.2 CAS No.: 8008-20-6

3.1.4 Other Identifiers

Molar mass	185 g/mol
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Section 4: First-Aid Measures

4.1 Description of first aid measures





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4.1.1 Inhalation

Remove person to fresh air and keep comfortable for breathing. For breathing difficulties, oxygen may be necessary. If breathing stops, provide artificial respiration. Get medical attention if symptoms are severe or persist.

4.1.2 Skin Contact

Remove contaminated clothing immediately and wash skin with soap and water. Get medical attention if irritation persists after washing

4.1.3 Eye Contact

Rinse immediately with plenty of water. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation persists after washing

4.1.4 Ingestion

Do not induce vomiting. Get medical attention immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritating to skin. May irritate eyes. Vapours in high concentrations are narcotic. May cause nausea, headache, dizziness and intoxication. Entry into the lungs following ingestion or vomiting may cause chemical pneumonitis.

4.3 Advice for immediate medical attention and special treatment needed

Notes for the doctor

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray, foam, dry powder or carbon dioxide
- **Unsuitable extinguishing media**
Do not use water jet as an extinguisher, as this will spread the fire

5.2 Special hazards arising from the chemical product:

Flammable liquid and vapour. Containers can burst violently or explode when heated, due to excessive pressure build-up

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool containers exposed to heat with water spray and remove them from the fire area if it can be done without risk. Prevent fire extinguishing water from contaminating surface water or the ground water system.

5.4.2 Special protective equipment for fire-fighters:



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Wear positive-pressure self-contained breathing apparatus (SCBA) and appropriate protective clothing

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Avoid inhalation of vapours and contact with skin and eyes. Wear adequate protective equipment at all operations. Prevent unauthorized access. Vapours are heavier than air and may spread near ground and travel a considerable distance to a source of ignition and flash back. Eliminate all ignition sources if safe to do so. Take precautionary measures against static discharge

6.2 Environmental precautions:

Avoid release to the environment. Stop leak if safe to do so. Avoid the spillage or runoff entering drains, sewers or watercourses. Contain spillage with sand, earth or other suitable non-combustible material. Inform the relevant authorities if environmental pollution occurs (sewers, waterways, soil or air). Risk of soil and ground water contamination.

6.3 Methods and materials for containment and cleaning up:

Advice on how to clean up a spill

Immediately start clean-up of the liquid and contaminated soil. Small Spillages: Absorb spillage with sand or other inert absorbent. Pay attention to the fire and health hazards caused by the product. Take care as floors and other surfaces may become slippery

6.4 Reference to other sections: For personal protection, see Section 8.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge Use only non-sparking tools

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:



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Store in accordance with local regulations. Store in a demarcated bunded area to prevent release to drains and/or watercourses.

Incompatible substances or mixtures

Observe hints for combined storage.

Consideration of other advice:

Ground/bond container and receiving equipment.

Ventilation requirements

Protect from sunlight.

Specific designs for storage rooms or vessels

Mild steel. Stainless steel

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
US	Kerosene	8008-20-6		0.2							

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Composition	Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
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Renewable hydrocarbons	DNEL	147 mg/m ³	human, inhalatory	worker (industry)	Long term systemic effects
		42 mg/kg/day	human, dermal	worker (industry)	Long term systemic effects
Kerosene	DNEL	18.5 mg/kg bw/day	human, oral	worker (industry)	Long term systemic effects
Naphthalene	DNEL	3.57 mg/kg	human, dermal	worker (industry)	Long term systemic effects
		25 mg/m ³	human, inhalatory	worker (industry)	Long term local effects
		25 mg/m ³		worker (industry)	Long term systemic effects

DNEL: Derived No-Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. **Hand protection**

- Wear suitable gloves.

- b. **Type of material**

- Nitrile rubber, Neoprene, Polyvinyl chloride (PVC).

- c. **Material thickness**

- ≥0.54 mm



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d. Breakthrough times of the glove material

>360 minutes (permeation: level 6)

e. Other protection measures

Protective clothing when needed. Wear anti-static protective clothing if there is a risk of ignition from static electricity

- **Respiratory protection:** Respiratory protection necessary at: airborne contamination exceeds the recommended occupational exposure limit. Type: A2



8.2.3 Environmental Exposure Controls:

Store in a demarcated bunded area to prevent release to drains and/or watercourses.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Clear light yellow
Odour	like: - Hydrocarbons
Melting point/freezing point	≤ -47°C (ASTM D2386, D5972, IP 529)
Boiling point or initial boiling point and boiling range	300°C (ASTM D 86)
Flammability	flammable liquid in accordance with GHS criteria
Lower and upper explosion limit	0.6 vol% (LEL) – 6 vol% (UEL)
Flash point	38°C (IP 170)
Smoke point	25 mm
Freezing point	-47 °C
Auto-ignition temperature	~ 250°C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	8 mm ² /s @ -20°C
Solubility(ies)	
Water solubility	< 50 mg/l @ 20°C
Partition coefficient	
Partition coefficient n-octanol/water (log value):	log Kow: > 3
Vapour pressure	~ 2 kPa @ 38°C
Density and/or relative density	



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Density	775-840 kg/m ³ @ 15°C (ASTM D4052)
Relative vapour density	> 3 (Air = 1.0)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

There are no known reactivity hazards associated with this product

10.2 Chemical stability

Stable at normal ambient temperatures and when used as recommended.

10.3 Possibility of hazardous reactions

No potentially hazardous reactions known

10.4 Conditions to avoid

Keep away from heat, sparks and open flame.

10.5 Incompatible materials

Oxidising agents

10.6 Hazardous decomposition products

Does not decompose when used and stored as recommended.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If in eyes**
None
- **If on skin**
causes skin irritation

11.1.1 Acute Toxicity:

Composition	Exposure route	Endpoint	Value	Species	Source
Renewable hydrocarbons	oral	LD50	> 2000 mg/kg	rat	EC B1 tris
	dermal	LD50	>2000 mg/kg	rat	EC B3



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Kerosene	inhalation	LC50	> 5.28 mg/l(4h)	rat	OECD 403
	oral	LD50	> 5000 mg/kg	rat	OECD 420, EPA OTS 798.1175
	dermal	LD50	>2000 mg/kg	rabbit	EC B3OECD 402, EPA OTS 798.1100

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause nausea, headache, dizziness and intoxication.

11.1.9 Specific target organ toxicity - repeated exposure
Shall not be classified.

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Composition	Endpoint	Value	Species	Source	Exposure time
Renewable hydrocarbons	LL50	> 1000 mg/l	Fish	WAF	96 hours
	EL50	> 100 mg/l	Aquatic invertebrates	WAF	48 hours
	EL50	> 100 mg/l	Aquatic plants	WAF	72 hours
	EC ₅₀	> 1000 mg/l,	Microorganisms		3 hours
Kerosene	LL50	5-17 mg/l	Fish	WAF	24 hours
		2-5 mg/l			48 hours
	EL50	4.6 mg/l	Aquatic invertebrates	WAF	24 hours
		1.4 mg/l			48 hours
	NOEL	0.3 mg/l	Aquatic plants	WAF	24 hours
	EL50	1-3 mg/l			
	NOEL	1 mg/l			



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Aquatic toxicity(chronic)

Renewable hydrocarbons	NOEC	1 mg/l,	Aquatic invertebrates	WAF	21 days
	LOEC	3.2 mg/l,		WAF	
	NOEC	373 mg/kg			10 days
	LC50	1200 mg/kg			
Kerosene	NOEL	0.1 mg/l	Fish early life stage	QSAR	28 days
	EL50	0.81 mg/l	Aquatic invertebrates	WAF	21 days
	NOEL	0.48 mg/l			

12.2 Persistence and Degradability:

The product contains volatile substances which may spread in the atmosphere. Can be photodegraded in the atmosphere.

12.3 Bio Accumulative Potential: The substance fulfils the Possibly bioaccumulative criterion.

12.4 Mobility in Soil:

Evaporates slowly. The product has poor water-solubility. Product can penetrate soil until reaching the surface of ground water. The product contains substances which are bound to particulate matter and are retained in soil.

12.5 Other adverse effects: No data available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

13.1.1 Contaminated Packaging

Care should be taken when handling emptied containers that have not been thoroughly cleaned or rinsed out. Empty containers or liners may retain some product residues and hence be potentially hazardous.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1863



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14.2 UN Proper Shipping Name: UN 1863 FUEL, AVIATION

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3, "Fish and tree"

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: yes (hazardous to the aquatic environment)

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.9 MSDS of SKO

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Superior Kerosene Oil
1.1.1.1 Trade Name	SKO
1.1.2 CAS No.	8008-20-6

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuel
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid and vapor	3	Flam. Liq. 3	H226
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger



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2.2.3 Hazard Statements(s):

H226	Flammable liquid and vapor
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H336	May cause drowsiness or dizziness
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P210	Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
P102	Keep out of reach of children.
P273	Avoid release to the environment.
P280	Wear eye protection, face protection, protective clothing, protective gloves.

Precautionary statements – response

P301+P310+P331	IF SWALLOWED: Immediately call a doctor, a POISON CENTER. Do NOT induce vomiting.
P501	Dispose of contents and container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: SKO

3.1.1 Systematic Chemical Name or Trivial Name: Superior Kerosene Oil

3.1.2 CAS No.: 8008-20-6

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation



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Keep at rest. Provide fresh air. Give oxygen or artificial respiration if necessary. Call a physician immediately.

4.1.2 Skin Contact

Take off immediately all contaminated clothing. Wash with plenty of water/. In case of doubt or persistent symptoms, consult always a physician. Wash contaminated clothing before reuse.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If eye irritation persists: Get medical advice/attention.

4.1.4 Ingestion

Do Not induce vomiting. Rinse mouth immediately and drink plenty of water. Get medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Cough, Nausea, Erythema

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder.
- **Unsuitable extinguishing media**
Strong water jet.

5.2 Special hazards arising from the chemical product:

Vapours may form explosive mixture with air. Vapours are heavier than air and may spread along floors. Vapours are heavier than air and may travel considerable distance to an ignition source and flash back to source of vapours. On heating there is a risk of a build-up of pressure in hermetically sealed containers or tanks. Do not allow run-off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide, carbon dioxide and unburned hydrocarbons (smoke). Organic compounds. nitrogen oxides (NO_x) and sulphur oxides.

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters



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Special protective equipment for firefighters. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Evacuate personnel to a safe area. Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Concerning personal protective equipment to use.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if safe to do so. Dam up.

Advice on how to clean up a spill

Take up liquid spill into absorbent material, e.g.: sand, earth, vermiculite or powdered limestone. Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of as special waste in compliance with local and national regulations.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provide adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take any precaution to avoid mixing with combustibles.

Measures to protect the environment



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Do not allow to enter into surface water or drains.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed in a cool, well-ventilated place.

Incompatible substances or mixtures

Oxidising substances. Strong acids

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/mS]	STEL [ppm]	STEL [mg/mS]	Ceiling - C [ppm]	Ceiling -C [mg/mS]	Notation	Source
USA	SKO	8008-20-6	ACGIH		200						

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	19 mg/kg bw/day	human, oral	general population	Long-term - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:



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Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. Hand protection**

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

- b. Type of material**

NBR (Nitrile rubber)

- c. Material thickness**

≥0.54 mm

- d. Breakthrough times of the glove material**

>360 minutes

- e. Other protection measures**

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Half-face mask (DIN EN 140). Full face mask (EN 136). Filter type: AP (EN 141). Use self-contained respiratory apparatus for rescue and maintenance work in



storage vessels. (EN 137)

8.2.3 Environmental Exposure Controls:

Avoid release to the environment. Comply with applicable Community environmental protection legislation.



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Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Clear to pale yellow liquid.
Odour	like: - petroleum hydrocarbon odour.
Melting point/freezing point	-47 °C
Boiling point or initial boiling point and boiling range	300 °C
Flammability	Not applicable, liquid
Lower and upper explosion limit	0.6 vol% (LEL) – 4.9 vol% (UEL)
Flash point	38 °C
Smoke point	18 mm
Auto-ignition temperature	227 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	1 - 2 cPs (@ 40°C)
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	> 3
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	< 0.48 mm Hg @20°C
<u>Density and/or relative density</u>	
Density	785-825 kg/m ³ (@ 15°C)
Relative vapour density	4.5 (Air=1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Flammable liquid.

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.



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10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.

10.5 Incompatible materials

Oxidising substances. Strong acids.

10.6 Hazardous decomposition products

Hazardous combustion products: Burning produces noxious and toxic fumes. Hazardous decomposition products COx. see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	> 5200 mg/m ³ /4h	rat	OECD
Oral	LD50	> 5000 mg/kg	rat	OECD
Dermal	LD50	>2000 mg/kg	rabbit	OECD

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Suspected of damaging fertility.

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure
Not classified

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.



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Endpoint	Value	Species	Source
LC50	2 - 5 mg/l	Fish 1	OECD
EC50	1.4 mg/l	Daphnia 1	OECD
ErC50	1 - 3 mg/l	algae	OECD
Aquatic toxicity (chronic)			
NOEC	0.48 mg/l	daphnia	NOEL
NOEC	0.98 mg/l	fish	NOEL

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	> 3
BCF	Low potential.

12.4 Mobility in Soil:

The Organic Carbon normalized adsorption coefficient	Low potential.
--	----------------

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**
Do not allow to enter into surface water or drains.

13.1.1 Contaminated Packaging: Never use pressure to empty container.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**



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Section 14: Transport Information

14.1 UN Number: UN 1223

14.2 UN Proper Shipping Name: Kerosene

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 3

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.10 MSDS of HFHSD

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	HFHSD
1.1.1.1 Trade Name	HFHSD
1.1.2 CAS No.	68334-30-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuels
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid and vapour	3	Flam. Liq. 3	H226
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Acute toxicity	4	Acute Tox.4	H332
Carcinogenicity	2	Carc. 2	H351
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411
Serious Eye Damage/Irritation	2	Eye Irr. 2	H319

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS02

GHS07

GHS08

GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H226	Flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H332	Harmful if inhaled
H336	May cause drowsiness or dizziness
H351	Suspected of causing cancer
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H319	May cause eye damage/irritation
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P210	Keep away from heat/sparks/open flames/hot surfaces. - No smoking
P260	Do not breathe mist, vapours, spray, gas, fume
P273	Avoid release to the environment
P280	Wear protective gloves/eye protection

Precautionary statements – response

P301+P310+P331	IF SWALLOWED: Immediately call a POISON CENTER, a doctor. Do NOT induce vomiting
P303+361+353	If on skin (or hair): Take off immediately all contaminated clothing. Rinse with water/shower
P391	Collect spillage
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.



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Section 3: Composition/Information on Ingredients

3.1 Name of Product: HFHSD

3.1.1 Systematic Chemical Name or Trivial Name: HFHSD

3.1.2 CAS No.: 68334-30-5

3.1.3 Other Identifiers

Molar mass 198-202 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Remove person to fresh air. If person is not breathing, ensure an open airway and provide artificial respiration. If necessary, provide additional oxygen once breathing is restored if trained to do so. Seek medical attention immediately

4.1.2 Skin Contact

Remove contaminated clothing and shoes. After contact with skin, wash immediately with plenty of water and soap. In case of doubt or persistent symptoms, consult always a physician. In the event of a high pressure injection injury, worker should obtain immediate medical assistance

4.1.3 Eye Contact

In case of contact with eyes, immediately flush with clean, low-pressure water for at least 15 min. Hold eyelids open to ensure adequate flushing. Seek medical attention. In case of contact lenses, remove immediately

4.1.4 Ingestion

Do not induce vomiting. Rinse mouth immediately and drink plenty of water. Get immediate medical advice/attention

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Skin contact, Eyes contact

4.3 Advice for immediate medical attention and special treatment needed

No data available

Section 5: Firefighting Measures

5.1 Extinguishing Media





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- **Suitable extinguishing media**
Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder
- **Unsuitable extinguishing media**
Strong water jet

5.2 Special hazards arising from the chemical product:

Vapours are heavier than air and may spread along floors. On heating there is a risk of a build-up of pressure in hermetically sealed containers or tanks. Do not allow run off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides (CO, CO₂), Hydrogen sulphide, Sulphur oxides

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Special protective equipment for firefighters. In case of fire: Wear self-contained breathing apparatus. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: Isolate area around container involved in fire

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Do not breathe vapours/aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert.

6.2 Environmental precautions:

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. Danger of explosion.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop the source of the spill.

Advice on how to clean up a spill

Absorb with liquid-binding material (sand, non-combustible, inert oil absorbing materials).

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections:



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Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a well-ventilated place. Keep container tightly closed in a cool

Incompatible substances or mixtures

A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances

Consideration of other advice:

Ensure proper process control to avoid excess waste discharge.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Keep only in the original container. Steel, Stainless steel

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA	TWA	STEL	STEL	Ceiling - C	Ceiling -C	Notation
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				[ppm] 1	[mg/mS] 1	[ppm] 1	[mg/mS] 1	[ppm] 1	[mg/mS] 1	
USA	HFHS D	68334 -30-5	ACGIH	0. 1	100					

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	4300 mg/m ³	human, inhalatory	worker (industry)	Acute - systemic effects
DNEL	2.9 mg/kg bw/day	human, dermal	worker (industry)	Long-term - systemic effects
DNEL	68 mg/m ³	human, inhalatory	worker (industry)	Long-term - systemic effects
DNEL	2600 mg/kg bw/day	human, inhalatory	general population	Acute - systemic effects
DNEL	20 mg/m ³	human, inhalatory	general population	Long-term - systemic effects
DNEL	1.3 mg/kg bw/day	human, dermal	general population	Long-term - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.



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- **Skin protection**



- a. Hand protection**

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space, such as (but not limited to): other chemicals that are possibly used, physical requirements (protection against cutting/drilling, skill, thermal protection), and the instructions/specification of the supplier of gloves.

- b. Type of material**

NBR (Nitrile rubber)

- c. Material thickness**

≥0.54 mm

- d. Breakthrough times of the glove material**

>240 minutes

- e. Other protection measures**

Wear suitable coveralls to prevent exposure to the skin, Protective helmets, antistatic boots

- **Respiratory protection:** Respiratory protection necessary at: Type: A



8.2.3 Environmental Exposure Controls:

Avoid release to the environment. Comply with applicable Community environmental protection legislation.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	yellow
Odour	Petroleum hydrocarbon odour.
Melting point/freezing point	-22 to -0.4 °F (-30 to -18 °C)
Boiling point or initial boiling point and boiling range	320 to 690 °F (160 to 366 °C)
Flammability	Flammable liquid
Lower and upper explosion limit	0.6 vol% (LEL) – 6.5vol% (UEL)
Flash point	66 °C PMCC



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Pour point	+3(Winter) +15(Summer)
Auto-ignition temperature	494 °F (257 °C)
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	2-5 cst @ 40°C
Cetane Index	45
<u>Solubility(ies)</u>	
Water solubility	Insoluble in water
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Log Kow range of 3.3 to >.6.0
Vapour pressure	0.009 psia @ 70 °F
<u>Density and/or relative density</u>	
Density	810 - 845 kg/m ³
Relative vapour density	> 1
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	25 g/m ³

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Material is not self-reacting. Flammable concentrations may be present in air. Compound can react with oxidizing materials

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Avoid high temperatures, open flames, sparks, static electricity, welding, smoking and other ignition sources.

10.5 Incompatible materials

Keep away from strong oxidizers such as nitric and sulphuric acids.



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10.6 Hazardous decomposition products

Carbon monoxide, carbon dioxide and non-combusted hydrocarbons (smoke).

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If in eyes**
Causes serious eye irritation
- **If on skin**
Causes skin irritation.
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	6 mg/l/4h	rat
Oral	LD50	>5000 mg/kg	rabbit
Dermal	LD50	>5000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Causes serious eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Suspected of causing cancer.

11.1.7 Reproductive Toxicity: Not classified

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure
May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled).

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: Symptoms related to the physical, chemical and toxicological characteristics. Reference to other sections: 4.2.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	21 mg/l	fish	96 h



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EL50	68 mg/l	aquatic invertebrates	48 h
IL50	22 mg/l	aquatic algae	72 h
NOEL	0.21 mg/l	aquatic invertebrates	14 days
NOEL	0.083 mg/l	Fish	

12.2 Persistence and Degradability: Not applicable

12.3 Bio Accumulative Potential: Not available

12.4 Mobility in Soil: Not available

12.5 Other adverse effects: Do not allow to enter into surface water or drains

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- Sewage disposal-relevant information**

Do not burn, or use a cutting torch on the empty drum. Do not puncture or incinerate

13.1.1 Contaminated Packaging: Never use pressure to empty container.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1202

14.2 UN Proper Shipping Name: Diesel Fuel

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 9

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.



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Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.11 MSDS of HSD BS VI

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	HSD BS VI
1.1.1.1 Trade Name	Diesel
1.1.2 CAS No.	68334-30-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Fuels
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid and vapour	3	Flam. Liq. 3	H226
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Acute toxicity	4	Acute Tox.4	H332
Carcinogenicity	2	Carc. 2	H351
Specific target organ toxicity - single exposure (narcotic effects, drowsiness)	3	STOT SE 3	H336
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	2	Aquatic Chronic 2	H411
Serious Eye Damage/Irritation	2	Eye Irr. 2	H319

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS02



GHS07



GHS08



GHS09

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H226	Flammable liquid and vapour
H304	May be fatal if swallowed and enters airways
H315	Causes skin irritation
H332	Harmful if inhaled
H336	May cause drowsiness or dizziness
H351	Suspected of causing cancer
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H319	May cause eye damage/irritation
H411	Toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P210	Keep away from heat/sparks/open flames/hot surfaces. - No smoking
P260	Do not breathe mist, vapours, spray, gas, fume
P273	Avoid release to the environment
P280	Wear protective gloves/eye protection

Precautionary statements – response

P301+P310+P331	IF SWALLOWED: Immediately call a POISON CENTER, a doctor. Do NOT induce vomiting
P303+361+353	If on skin (or hair): Take off immediately all contaminated clothing. Rinse with water/shower
P391	Collect spillage
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.



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Section 3: Composition/Information on Ingredients

3.1 Name of Product: HSD BS VI

3.1.1 Systematic Chemical Name or Trivial Name: Diesel

3.1.2 CAS No.: 68334-30-5

3.1.3 Other Identifiers

Molar mass

198-202 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Remove person to fresh air. If person is not breathing, ensure an open airway and provide artificial respiration. If necessary, provide additional oxygen once breathing is restored if trained to do so. Seek medical attention immediately

4.1.2 Skin Contact

Remove contaminated clothing and shoes. After contact with skin, wash immediately with plenty of water and soap. In case of doubt or persistent symptoms, consult always a physician. In the event of a high pressure injection injury, worker should obtain immediate medical assistance

4.1.3 Eye Contact

In case of contact with eyes, immediately flush with clean, low-pressure water for at least 15 min. Hold eyelids open to ensure adequate flushing. Seek medical attention. In case of contact lenses, remove immediately

4.1.4 Ingestion

Do NOT induce vomiting. Rinse mouth immediately and drink plenty of water. Get immediate medical advice/attention

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Skin contact, Eyes contact

4.3 Advice for immediate medical attention and special treatment needed

No data available

Section 5: Firefighting Measures

5.1 Extinguishing Media



- Suitable extinguishing media



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- Water spray, Alcohol resistant foam, Carbon dioxide, Dry extinguishing powder
- **Unsuitable extinguishing media**
Strong water jet

5.2 Special hazards arising from the chemical product:

Vapours are heavier than air and may spread along floors. On heating there is a risk of a build-up of pressure in hermetically sealed containers or tanks. Do not allow run off from fire-fighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides (CO, CO₂), Hydrogen sulphide, Sulphur oxides

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Special protective equipment for firefighters. In case of fire: Wear self-contained breathing apparatus. Use water spray or fog for cooling exposed containers. Evacuate personnel to a safe area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

5.4.2 Special protective equipment for fire-fighters: Isolate area around container involved in fire

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Do not breathe vapours/aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert.

6.2 Environmental precautions:

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. Danger of explosion.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop the source of the spill.

Advice on how to clean up a spill

Absorb with liquid-binding material (sand, non-combustible, inert oil absorbing materials).

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.



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Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provision of sufficient ventilation. Avoid exposure.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking



Take precautionary measures against static discharge.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a well-ventilated place. Keep container tightly closed in a cool

Incompatible substances or mixtures

A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances

Consideration of other advice:

Ensure proper process control to avoid excess waste discharge.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Keep only in the original container. Steel, Stainless steel

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Countr y	Name of agent	CAS No	Identifie r	TWA [ppm]	TWA [mg/mS]	STE L [ppm]	STEL [mg/mS]	Ceiling - C [ppm]	Ceiling -C [mg/ mS]	Notatio n
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Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	4300 mg/m ³	human, inhalatory	worker (industry)	Acute - systemic effects
DNEL	2.9 mg/kg bw/ day	human, dermal	worker (industry)	Long-term - systemic effects
DNEL	68 mg/m ³	human, inhalatory	worker (industry)	Long-term - systemic effects
DNEL	2600 mg/kg bw/day	human, inhalatory	general population	Acute - systemic effects
DNEL	20 mg/m ³	human, inhalatory	general population	Long-term - systemic effects
DNEL	1.3 mg/kg bw/ day	human, dermal	general population	Long-term - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Use safety goggle with side protection.



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- **Skin protection**



- a. Hand protection**

The selection of specific gloves for a specific application and time of use in a working area, should also take into account other factors on the working space, such as (but not limited to): other chemicals that are possibly used, physical requirements (protection against cutting/drilling, skill, thermal protection), and the instructions/specification of the supplier of gloves.

- b. Type of material**

NBR (Nitrile rubber)

- c. Material thickness**

≥0.54 mm

- d. Breakthrough times of the glove material**

>240 minutes

- e. Other protection measures**

Wear suitable coveralls to prevent exposure to the skin, Protective helmets, antistatic boots

- **Respiratory protection:** Respiratory protection necessary at: Type: A



8.2.3 Environmental Exposure Controls:

Avoid release to the environment. Comply with applicable Community environmental protection legislation.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	yellow
Odour	Petroleum hydrocarbon odour.
Melting point/freezing point	-22 to -0.4 °F (-30 to -18 °C)
Boiling point or initial boiling point and boiling range	320 to 690 °F (160 to 366 °C)
Flammability	Flammable liquid
Lower and upper explosion limit	0.6 vol% (LEL) – 6.5vol% (UEL)
Flash point	35 °C
Pour point	+3(Winter)



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	+15(Summer)
Auto-ignition temperature	494 °F (257 °C)
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	2-4.5 cst @ 40°C
Cetane Index	46
<u>Solubility(ies)</u>	
Water solubility	Insoluble in water
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Log Kow range of 3.3 to >.6.0
Vapour pressure	0.009 psia @ 70 °F
<u>Density and/or relative density</u>	
Density	810 - 845 kg/m ³
Relative vapour density	> 1
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	25 g/m ³

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Material is not self-reacting. Flammable concentrations may be present in air. Compound can react with oxidizing materials

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Avoid high temperatures, open flames, sparks, static electricity, welding, smoking and other ignition sources.

10.5 Incompatible materials

Keep away from strong oxidizers such as nitric and sulphuric acids.

10.6 Hazardous decomposition products



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Carbon monoxide, carbon dioxide and non-combusted hydrocarbons (smoke).

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If in eyes**
Causes serious eye irritation
- **If on skin**
Causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	6 mg/l/4h	rat
Oral	LD50	>5000 mg/kg	rabbit
Dermal	LD50	>5000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Causes serious eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Suspected of causing cancer.

11.1.7 Reproductive Toxicity: Not classified

11.1.8 Specific target organ toxicity-single exposure: May cause drowsiness or dizziness.

11.1.9 Specific target organ toxicity - repeated exposure
May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled).

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: Symptoms related to the physical, chemical and toxicological characteristics. Reference to other sections: 4.2.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	21 mg/l	fish	96 h



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EL50	68 mg/l	aquatic invertebrates	48 h
IL50	22 mg/l	aquatic algae	72 h
NOEL	0.21 mg/l	aquatic invertebrates	14 days
NOEL	0.083 mg/l	Fish	

12.2 Persistence and Degradability: Not applicable

12.3 Bio Accumulative Potential: Not available

12.4 Mobility in Soil: Not available

12.5 Other adverse effects: Do not allow to enter into surface water or drains

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- Sewage disposal-relevant information**

Do not burn, or use a cutting torch on the empty drum. Do not puncture or incinerate.

13.1.1 Contaminated Packaging: Never use pressure to empty container.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 1202

14.2 UN Proper Shipping Name: Diesel Fuel

14.3 Transport Class(es): 3

14.3.1 Hazard Labels: 9

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.



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Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.12 MSDS of LDO

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Light Distillate Oil
1.1.1.1 Trade Name	LDO
1.1.2 CAS No.	68476-30-2

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Not Available
Uses advised against	Not Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product: Not Classified

2.2 Label Elements: Not Classified

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: LDO

3.1.1 Systematic Chemical Name or Trivial Name: Light Distillate Oil

3.1.2 CAS No.: 68476-30-2

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air in case of accidental inhalation of vapours. In case of shortness of breath, give oxygen. If breathing is irregular or stopped, administer artificial respiration. Call a physician immediately.

4.1.2 Skin Contact



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Wash off immediately with plenty of water for at least 15 minutes. If symptoms persist, call a physician.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses. If eye irritation persists, consult a specialist.

4.1.4 Ingestion

If swallowed, seek medical advice immediately and show this container or label. Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person.

4.2 Most important symptoms and anticipated effects both acute and delayed

Carcinogenic effects

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray, Dry powder, Foam

5.2 Special hazards arising from the chemical product:

Flash back possible over considerable distance. Do not allow run-off from fire fighting to enter drains or water courses.

5.3 Hazardous Combustion Products: Not available

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Standard procedure for chemical fires. In the event of fire, cool tanks with water spray.

5.4.2 Special protective equipment for fire-fighters: Wear self-contained breathing apparatus and protective suit.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:





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Use personal protective equipment. Do not breathe vapours or spray mist. Ensure adequate ventilation.

6.2 Environmental precautions:

Avoid subsoil penetration. Do not flush into surface water or sanitary sewer system.

6.3 Methods and materials for containment and cleaning up:

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). The material taken up must be disposed of in accordance with regulations.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Wear personal protective equipment. Avoid contact with skin and eyes.

Measures to prevent fire as well as aerosol and dust generation

Keep away from sources of ignition - No smoking. Provide sufficient air exchange and/or exhaust in work rooms.



7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep containers tightly closed in a cool, well-ventilated place.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters: Contains no substances with occupational exposure limit values.

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.



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• **Skin protection**



a. Hand protection

Wear suitable gloves.

b. Type of material

Butyl-rubber

c. Material thickness

0.5 mm

d. Breakthrough times of the glove material

4 h

e. Other protection measures

Wash hands before breaks and immediately after handling the product.

8.2.3 Environmental Exposure Controls: Not Available

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	light brown
Odour	Not available
Melting point/freezing point	Not available
Boiling point or initial boiling point and boiling range	Not available
Flammability	Not available
Lower and upper explosion limit	Not available
Flash point	66 °C
Pour point	12 °C(Winter) 21 °C(Summer)
Auto-ignition temperature	Not available
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	2.5 – 16 cst @ 40°C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not available
Soil organic carbon/water (log KOC)	Not available



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Vapour pressure	Not available
Density and/or relative density	
Density	Not available
Relative vapour density	Not available
Particle characteristics	not relevant (liquid)
Other safety parameters	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Incompatible materials

Oxidizing agents, Reducing agents, Acids and bases

10.2 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	4.6 mg/l/4h	rat	TOXNET
Oral	LD50	> 5000 mg/kg	rat	TOXNET
Dermal	LD50	>2000 mg/kg	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Moderately irritating.

11.1.3 Serious Eye Damage/ Eye Irritation: Slightly irritating.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer.

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Not Classified.

11.1.8 Specific target organ toxicity-single exposure: Not Classified.

11.1.9 Specific target organ toxicity - repeated exposure: Not Classified.

11.1.10 Aspiration Hazard: Not Classified.

11.1.11 Additional Information: Not Classified.



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Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LC50	54 mg/l	fish	ECHA	96 h

12.2 Persistence and Degradability: Not Available

12.3 Bio Accumulative Potential: Not Available

12.4 Mobility in Soil: Not Classified

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: In accordance with local and national regulations.



13.1.1 Contaminated Packaging: Packaging that cannot be cleaned must be disposed of in the same way as the material itself.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified

14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects



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- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.13 MSDS of SPO-90N

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Spindle Oil
1.1.1.1 Trade Name	SPO-90N
1.1.2 CAS No.	64742-46-7

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Hydraulic fluid.
Uses advised against	No Data Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Skin corrosion/irritation	2	Skin Irrit. 2	H315
ACUTE TOXICITY (inhalation)	4	Acute Tox. 4	H332
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS07 GHS08



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2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways
H315 Causes skin irritation
H332 Harmful if inhaled.

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves/eye protection
P261	Avoid breathing vapour.
P264	Wash hands thoroughly after handling.

Precautionary statements – response

P304 + P340 + P312	IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or physician if you feel unwell.
P301 + P310 + P331	IF SWALLOWED: Immediately call a POISON CENTER or physician. Do NOT induce vomiting.
P302 + P352 + P362 + P364	IF ON SKIN: Wash with plenty of soap and water. Take off contaminated clothing and wash it before reuse.
P332 + P313	If skin irritation occurs: Get medical attention.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Defatting to the skin. Note: High Pressure Applications Injections through the skin resulting from contact with the product at high pressure constitute a major medical emergency. See 'Notes to physician' under First-Aid Measures, Section 4 of this Safety Data Sheet

Section 3: Composition/Information on Ingredients

3.1 Name of Product: SPO-90N

3.1.1 Systematic Chemical Name or Trivial Name: Spindle Oil

3.1.2 CAS No.: 64742-46-7

Section 4: First-Aid Measures

4.1 Description of first aid measures





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4.1.1 Inhalation

If inhaled, remove to fresh air. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. Get medical attention if symptoms occur.

4.1.2 Skin Contact

In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention.

4.1.3 Eye Contact

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.

4.1.4 Ingestion

Do not induce vomiting. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Aspiration hazard if swallowed. Can enter lungs and cause damage. Get medical attention immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Mild irritation of the nose, throat, and respiratory tract

4.3 Advice for immediate medical attention and special treatment needed

No specific treatment.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
In case of fire, use foam, dry chemical or carbon dioxide extinguisher or spray.
- **Unsuitable extinguishing media**
Do not use water jet.

5.2 Special hazards arising from the chemical product:

In a fire or if heated, a pressure increase will occur and the container may burst.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training



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5.4.2 Special protective equipment for fire-fighters: Fire-fighters should wear positive pressure self-contained breathing apparatus (SCBA) and full turnout gear.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Avoid breathing vapour or mist. Provide adequate ventilation. Put on appropriate personal protective equipment. Floors may be slippery; use care to avoid falling. Contact emergency personnel.

For emergency responders: Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in “For nonemergency personnel”.

6.2 Environmental precautions:

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop leak if without risk. Move containers from spill area

Advice on how to clean up a spill

Absorb with an inert material and place in an appropriate waste disposal container.

Other information relating to spills and releases

Dispose of via a licensed waste disposal contractor.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Use explosion proof ventilation to prevent vapour accumulation.

Measures to prevent fire as well as aerosol and dust generation

Avoid heat and open flames, including pilot lights and strong oxidizing agents.



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Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials and food and drink.

Incompatible substances or mixtures

Oxidizing materials.

Consideration of other advice:

Do not store in open or unlabelled containers. Store in a cool, dry place with adequate ventilation. Keep away from open flames and high temperatures.

Ventilation requirements

Keep containers closed when not in use.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
US	SP O-90 N	64742-46-7	ACGIH	0.005	5						NOM-010-STPS-2014

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)



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- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



a. Hand protection

Wear protective gloves if prolonged or repeated contact is likely. Wear chemical resistant gloves. Recommended: Nitrile gloves. The correct choice of protective gloves depends upon the chemicals being handled, the conditions of work and use, and the condition of the gloves (even the best chemically resistant glove will break down after repeated chemical exposures). Most gloves provide only a short time of protection before they must be discarded and replaced. Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. Gloves should therefore be chosen in consultation with the supplier/ manufacturer and with a full assessment of the working conditions.

b. Type of material

NBR (Nitrile rubber)

- **Respiratory protection:** Use with adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use,



and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be



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chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Light yellow Liquid
Odour	Mild hydrocarbon odour
Melting point/freezing point	Not available
Boiling point or initial boiling point and boiling range	Not available
Flammability	Not available
Lower and upper explosion limit	Not available
Flash point	170 °C
Pour point	-12 °C
Auto-ignition temperature	>100°C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	16 - 19 cSt at 40°C 3 – 4.5 cSt at 100°C
Viscosity Index	97
<u>Solubility(ies)</u>	
Water solubility	insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not available
Soil organic carbon/water (log KOC)	Not available
Vapour pressure	<0.033 kPa (<0.25 mm Hg) [20°C (68°F)]
<u>Density and/or relative density</u>	
Density	870 kg/m ³ at 15°C
Relative vapour density	>1 [Air = 1]
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity



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No specific test data available for this product. Refer to Conditions to avoid and Incompatible materials for additional information.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur. Under normal conditions of storage and use, hazardous polymerization will not occur.

10.4 Conditions to avoid

Avoid all possible sources of ignition (spark or flame).

10.5 Incompatible materials

Oxidizing materials.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Nausea, vomiting, aspiration hazard
- **If in eyes**
Pain or irritation, watering, Redness
- **If inhaled**
Nausea or vomiting, headache, drowsiness/fatigue, dizziness/vertigo, Unconsciousness
- **If on skin**
Irritation, redness, dryness, cracking
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Oral	LD50	> 5000 mg/kg	rat	TOXNET

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.



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11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: No known significant effects or critical hazards.

11.1.8 Specific target organ toxicity-single exposure: No known significant effects or critical hazards.

11.1.9 Specific target organ toxicity - repeated exposure
No known significant effects or critical hazards.

11.1.10 Aspiration Hazard: No known significant effects or critical hazards.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	54 mg/l	fish	96 h
EC50	68 mg/l	aquatic invertebrates	48 h

12.2 Persistence and Degradability: There is no data available on the preparation itself.

12.3 Bio Accumulative Potential: Not Measured

12.4 Mobility in Soil: Spillages may penetrate the soil causing ground water contamination.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: The generation of waste should be avoided or minimized wherever possible. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant. The generation of waste should be avoided or minimized wherever possible. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant.



- Sewage disposal-relevant information**

Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Waste packaging should be recycled. Incineration



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or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way.

13.1.1 Contaminated Packaging: Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3802

14.2 UN Proper Shipping Name: Spindle Oil

14.3 Transport Class(es): 9

14.3.1 Hazard Labels: 3

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: yes

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet



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6.14 MSDS of 150 N-I

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	150 NEUTRAL SOLVENT
1.1.1.1 Trade Name	150 N-I
1.1.2 CAS No.	64742-65-0

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Base oil.
Uses advised against	No Data Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product: The product is not classified as dangerous according to Regulation (EC) No. 1272/2008

2.2 Label Elements: Not classified/No labelling required

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Contaminated surfaces will be extremely slippery. Should not be released into the environment.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: 150 N-I

3.1.1 Systematic Chemical Name or Trivial Name: 150 NEUTRAL SOLVENT

3.1.2 CAS No.: 64742-65-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air.

4.1.2 Skin Contact



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Remove contaminated clothing and shoes, wash off with soap and water, Wash contaminated clothing before reuse

4.1.3 Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids.

4.1.4 Ingestion

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Call a physician or poison control centre immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed
gastrointestinal irritation, nausea, vomiting, Irritation, diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Carbon dioxide (CO₂). ABC powder. Foam. Water spray or fog.
- **Unsuitable extinguishing media**
Do not use a solid water stream as it may scatter and spread fire

5.2 Special hazards arising from the chemical product:

Incomplete combustion and thermolysis may produce gases of varying toxicity such as carbon monoxide, carbon dioxide, various hydrocarbons, aldehydes and soot. These may be highly dangerous if inhaled in confined spaces or at high concentration.

5.3 Hazardous Combustion Products:

Not Classified

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool containers / tanks with water spray, Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

5.4.2 Special protective equipment for fire-fighters: Wear self-contained breathing apparatus and protective suit.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Do not touch or walk through spilled material. Contaminated surfaces will be extremely slippery. Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition.

6.2 Environmental precautions:

Do not allow material to contaminate ground water system. Try to prevent the material from entering drains or water courses. Local authorities should be advised if significant spillages cannot be contained.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Collect with non-combustible absorbent material (sand, diatomaceous earth, earth, vermiculite).

Other information relating to spills and releases

Place in appropriate containers for disposal.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Use only in well-ventilated areas.

Measures to prevent fire as well as aerosol and dust generation

When using, do not eat, drink or smoke.



Take precautionary measures against static discharges. Ground/bond containers, tanks and transfer/receiving equipment.

Measures to protect the environment

Avoid release to the environment.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values



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Occupational exposure limit values (Workplace Exposure Limits)

Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling - C [ppm]	Ceiling - C [mg/m ³]	Notation	Source
150 N-I	64742-65-0		0.001	10						

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	5.4 mg/m ³ /8h	human, inhalatory	worker (industry)	Long term, local effects
DNEL	1.2 mg/m ³ /24h	human, inhalatory	Consumer	Long term, local effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses with side-shields.



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- **Skin protection**



- a. Hand protection**

Hydrocarbon-proof gloves. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the EC approved gloves.

- **Respiratory protection:** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Respirator with combination filter for vapour/particulate (EN 141). The use of breathing apparatus must comply strictly



with the manufacturer's instructions and the regulations governing their choices and uses.

8.2.3 Environmental Exposure Controls:

The product should not be allowed to enter drains, water courses or the soil.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	yellow
Odour	like: - characteristic
Melting point/freezing point	No information available
Boiling point or initial boiling point and boiling range	No information available
Flammability	No information available
Lower and upper explosion limit	Not explosive
Flash point	200 °C
Pour point	-1 °C
Auto-ignition temperature	> 250 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	28-32 cst @ 40 °C
Viscosity Index	95
Solubility(ies)	



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Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 kPa at 20 °C
<u>Density and/or relative density</u>	
Density	872 kg/m ³ @ 15 °C
Relative vapour density	No information available
Particle characteristics	No information available
<u>Other safety parameters</u>	
Oxidising properties	not applicable

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Chemical stability

Stable under recommended storage conditions.

10.2 Possibility of hazardous reactions

None under normal processing

10.3 Conditions to avoid

Heat (temperatures above flash point), sparks, ignition points, flames, static electricity.

10.4 Incompatible materials

Strong oxidising agents.

10.5 Hazardous decomposition products

None under normal use

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not classified.
- **If in eyes**
Not classified.
- **If inhaled**
Not classified.



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- **If on skin**
Not classified.
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	5 mg/l/4h	rat	OECD 403
Oral	LD50	> 5000 mg/kg bw	rat	OECD 420
Dermal	LD50	> 5000 mg/kg bw	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Not classified.

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified.

11.1.4 Respiratory or Skin Sensitization: None known.

11.1.5 Mutagenicity: This product is not classified as mutagenic.

11.1.6 Carcinogenicity: This product is not classified carcinogenic.

11.1.7 Reproductive Toxicity: This product does not contain any known or suspected reproductive hazards.

11.1.8 Specific target organ toxicity-single exposure: No information available.

11.1.9 Specific target organ toxicity - repeated exposure: No information available.

11.1.10 Aspiration Hazard: No information available.

11.1.11 Additional Information: Characteristic skin lesions (oil blisters) may develop following prolonged and repeated exposures (contact with contaminated clothing).

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 100 mg/l	fish	OECD 203	96 h
EL50	> 10000 mg/l	aquatic invertebrates	OECD 202	48 h

Aquatic toxicity (Chronic)

NOEL	10 mg/l	fish	QSAR Petrotox	14/28 d
NOEL	> 1000 mg/l	aquatic invertebrates	OECD 211	21 d



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12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: No information available.

n-octanol/water (log KOW)

> 6 @ 20 °C

12.4 Mobility in Soil: Given its physical and chemical characteristics, the product generally shows low soil mobility.

12.5 Other adverse effects: No information available.

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of in accordance with the European Directives on waste and hazardous waste. Where possible recycling is preferred to disposal or incineration.



13.1.1 Contaminated Packaging: Empty containers should be taken to an approved waste handling site for recycling or disposal

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified

14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency



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6.15 MSDS of 500 N-I

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	500 Neutral
1.1.1.1 Trade Name	500 N-I
1.1.2 CAS No.	64742-54-7

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Lubricating Oil Base Stock
Uses advised against	Not Classified

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product: Not classified

2.2 Label Elements: Not classified

2.2.1 Precautionary Statement(s):

Precautionary statements – response

P301+310	IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician
P405	Store locked up.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: 500 N-I

3.1.1 Systematic Chemical Name or Trivial Name: 500 Neutral

3.1.2 CAS No.: 64742-54-7

Section 4: First-Aid Measures

4.1 Description of first aid measures



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4.1.1 Inhalation

Move the exposed person to fresh air at once. If respiratory problems, artificial respiration/oxygen.

4.1.2 Skin Contact

Remove contaminated clothing immediately and wash skin with soap and water. Get medical attention promptly if symptoms occur after washing.

4.1.3 Eye Contact

Immediately flush with plenty of water for up to 15 minutes. Remove any contact lenses and open eyes wide apart. Get medical attention.

4.1.4 Ingestion

Get medical attention immediately! Immediately rinse mouth and provide fresh air.

4.2 Most important symptoms and anticipated effects both acute and delayed

Not Classified.

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Extinguish with foam, carbon dioxide or dry powder.
- **Unsuitable extinguishing media**
Water jet

5.2 Special hazards arising from the chemical product:

When heated and in case of fire, corrosive vapours/gases may be formed.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂), Aldehydes, Ketones, Nitrous gases (NO_x)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

NOTE! Use air-supplied respirators to protect against gases/fumes. Use water SPRAY only to cool containers! Do not put water on leaked material.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Wear protective clothing as described in Section 8 of this safety data sheet.

6.2 Environmental precautions:

Do not allow ANY environmental contamination. Do not discharge into drains, water courses or onto the ground.

6.3 Methods and materials for containment and cleaning up:

Advice on how to clean up a spill

Collect with absorbent, non-combustible material into suitable containers.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Avoid spilling, skin and eye contact.



Wear full protective clothing for prolonged exposure and/or high concentrations. Always remove oil with soap and water or skin cleaning agent, never use organic solvents. Do not use oil-contaminated clothing or shoes, and do not put rags moistened with oil into pockets. Ventilate well, avoid breathing vapours. Use approved respirator if air contamination is above accepted level.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in tightly closed original container in a dry, cool and well-ventilated place.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values: Not Classified

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:



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Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. **Hand protection**

Chemical resistant gloves required for prolonged or repeated contact.

- **Respiratory protection:** In case of inadequate ventilation use suitable respirator. Wear mask supplied with: Gas cartridge suitable for organic substances.



Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Viscous liquid
Colour	Bright and Clear
Odour	like: - Gasoline
Melting point/freezing point	Not Classified
Boiling point or initial boiling point and boiling range	N.A.
Flammability	Not Classified
Lower and upper explosion limit	Not Classified
Flash point	224 °C (ASTM D 92)
Pour point	-1 °C
Auto-ignition temperature	Not Classified
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	9.5 cSt 11.5 cst @ 100 °C (ASTM D 445)



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Viscosity	95
<u>Solubility(ies)</u>	
Water solubility	Not Classified
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not Classified
Soil organic carbon/water (log KOC)	Not Classified
Vapour pressure	N.A.
<u>Density and/or relative density</u>	
Density	0.8811 @ 15°C
Relative vapour density	Not Classified
Particle characteristics	Not Classified
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Not Classified.

10.2 Chemical stability

Stable under normal temperature conditions and recommended use.

10.3 Possibility of hazardous reactions

Not Classified.

10.4 Conditions to avoid

Not Classified.

10.5 Incompatible materials

Strong oxidising substances.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If inhaled**



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Not relevant at normal room temperatures. When heated, irritating vapours may be formed.

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Oral	LD50	>5000 mg/kg	rat
Dermal	LD50	>5000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: No specific health warnings noted.

11.1.3 Serious Eye Damage/ Eye Irritation: No specific health warnings noted.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC 50	>1000 mg/l	fish	96 h
EC 50	>1000 mg/l	aquatic invertebrates	48 h
IC 50	>1000 mg/l	ALGAE	72 h

12.2 Persistence and Degradability: The product is not readily biodegradable.

12.3 Bio Accumulative Potential: Will not bio-accumulate.

12.4 Mobility in Soil: The product is insoluble in water and will spread on the water surface.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of waste and residues in accordance with local authority requirements.



13.1.1 Contaminated Packaging:

Incinerate with provision for removal of effluent gases by scrubber.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information



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The product is not covered by international regulation on the transport of dangerous goods (IMDG, IATA, ADR/RID).

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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6.16 MSDS of 150 N-II

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	150 NEUTRAL SOLVENT
1.1.1.1 Trade Name	150 N-II
1.1.2 CAS No.	64742-65-0

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Base oil.
Uses advised against	No Data Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product: The product is not classified as dangerous according to Regulation (EC) No. 1272/2008

2.2 Label Elements: Not classified/No labelling required

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Contaminated surfaces will be extremely slippery. Should not be released into the environment.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: 150 N-II

3.1.1 Systematic Chemical Name or Trivial Name: 150 NEUTRAL SOLVENT

3.1.2 CAS No.: 64742-65-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air.

4.1.2 Skin Contact



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Remove contaminated clothing and shoes, Wash off with soap and water, Wash contaminated clothing before reuse

4.1.3 Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids.

4.1.4 Ingestion

Do not induce vomiting. Never give anything by mouth to an unconscious person. Call a physician or poison control centre immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed
gastrointestinal irritation, nausea, vomiting, Irritation, diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Carbon dioxide (CO₂). ABC powder. Foam. Water spray or fog.
- **Unsuitable extinguishing media**
Do not use a solid water stream as it may scatter and spread fire

5.2 Special hazards arising from the chemical product:

Incomplete combustion and thermolysis may produce gases of varying toxicity such as carbon monoxide, carbon dioxide, various hydrocarbons, aldehydes and soot. These may be highly dangerous if inhaled in confined spaces or at high concentration.

5.3 Hazardous Combustion Products:

Not Classified

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool containers / tanks with water spray, Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

5.4.2 Special protective equipment for fire-fighters: Wear self-contained breathing apparatus and protective suit.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Do not touch or walk through spilled material. Contaminated surfaces will be extremely slippery. Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition.

6.2 Environmental precautions:

Do not allow material to contaminate ground water system. Try to prevent the material from entering drains or water courses. Local authorities should be advised if significant spillages cannot be contained.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Collect with non-combustible absorbent material (sand, diatomaceous earth, earth, vermiculite).

Other information relating to spills and releases

Place in appropriate containers for disposal.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Use only in well-ventilated areas.

Measures to prevent fire as well as aerosol and dust generation

When using, do not eat, drink or smoke.



Take precautionary measures against static discharges. Ground/bond containers, tanks and transfer/receiving equipment.

Measures to protect the environment

Avoid release to the environment.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values



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Occupational exposure limit values (Workplace Exposure Limits)

Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling - C [ppm]	Ceiling - C [mg/m ³]	Notation	Source
150 N-II	64742-65-0		0.001	10						

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	5.4 mg/m ³ /8h	human, inhalatory	worker (industry)	Long term, local effects
DNEL	1.2 mg/m ³ /24h	human, inhalatory	Consumer	Long term, local effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses with side-shields.



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- **Skin protection**



- a. Hand protection**

Hydrocarbon-proof gloves. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the EC approved gloves.

- **Respiratory protection:** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Respirator with combination filter for vapour/particulate (EN 141). The use of breathing apparatus must comply strictly



with the manufacturer's instructions and the regulations governing their choices and uses.

8.2.3 Environmental Exposure Controls:

The product should not be allowed to enter drains, water courses or the soil.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	yellow
Odour	like: - characteristic
Melting point/freezing point	No information available
Boiling point or initial boiling point and boiling range	No information available
Flammability	No information available
Lower and upper explosion limit	Not explosive
Flash point	210 °C
Pour point	-15 °C
Auto-ignition temperature	> 250 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	28-33 cst @ 40 °C
Viscosity Index	105
Solubility(ies)	



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Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 hPa at 20 °C
<u>Density and/or relative density</u>	
Density	872 kg/m ³ @ 15 °C
Relative vapour density	No information available
Particle characteristics	No information available
<u>Other safety parameters</u>	
Oxidising properties	not applicable

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Chemical stability

Stable under recommended storage conditions.

10.2 Possibility of hazardous reactions

None under normal processing

10.3 Conditions to avoid

Heat (temperatures above flash point), sparks, ignition points, flames, static electricity.

10.4 Incompatible materials

Strong oxidising agents.

10.5 Hazardous decomposition products

None under normal use

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not classified.
- **If in eyes**
Not classified.
- **If inhaled**
Not classified.



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- **If on skin**
Not classified.
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	5 mg/l/4h	rat	OECD 403
Oral	LD50	> 5000 mg/kg bw	rat	OECD 420
Dermal	LD50	> 5000 mg/kg bw	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Not classified.

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified.

11.1.4 Respiratory or Skin Sensitization: None known.

11.1.5 Mutagenicity: This product is not classified as mutagenic.

11.1.6 Carcinogenicity: This product is not classified carcinogenic.

11.1.7 Reproductive Toxicity: This product does not contain any known or suspected reproductive hazards.

11.1.8 Specific target organ toxicity-single exposure: No information available.

11.1.9 Specific target organ toxicity - repeated exposure: No information available.

11.1.10 Aspiration Hazard: No information available.

11.1.11 Additional Information: Characteristic skin lesions (oil blisters) may develop following prolonged and repeated exposures (contact with contaminated clothing).

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 100 mg/l	fish	OECD 203	96 h
EL50	> 10000 mg/l	aquatic invertebrates	OECD 202	48 h

Aquatic toxicity (Chronic)

NOEL	10 mg/l	fish	QSAR Petrotox	14/28 d
NOEL	> 1000 mg/l	aquatic invertebrates	OECD 211	21 d



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12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: No information available.

n-octanol/water (log KOW)

> 6 @ 20 °C

12.4 Mobility in Soil: Given its physical and chemical characteristics, the product generally shows low soil mobility.

12.5 Other adverse effects: No information available.

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of in accordance with the European Directives on waste and hazardous waste. Where possible recycling is preferred to disposal or incineration.



13.1.1 Contaminated Packaging: Empty containers should be taken to an approved waste handling site for recycling or disposal

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified

14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency



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6.17 MSDS of 500 N-II

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name 500 Neutral Solvent

1.1.1.1 Trade Name 500 N-II

1.1.2 CAS No. 64742-54-7

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses Lubricating Oil Base Stock

Uses advised against Not Classified

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name Hindustan Petroleum Corporation Limited

1.3.2 Address HPCL Mumbai Refinery, B D Patil Marg
Chembur, Mumbai-400074

1.3.3 Contact No. 022-25076605

1.3.4 Contact person in emergency Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.: 022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product: Not classified

2.2 Label Elements: Not classified

2.2.1 Precautionary Statement(s):

Precautionary statements – response

P301+310 IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician

P405 Store locked up.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: 500 N-II

3.1.1 Systematic Chemical Name or Trivial Name: 500 Neutral Solvent

3.1.2 CAS No.: 64742-54-7

Section 4: First-Aid Measures

4.1 Description of first aid measures



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4.1.1 Inhalation

Move the exposed person to fresh air at once. If respiratory problems, artificial respiration/oxygen.

4.1.2 Skin Contact

Remove contaminated clothing immediately and wash skin with soap and water. Get medical attention promptly if symptoms occur after washing.

4.1.3 Eye Contact

Immediately flush with plenty of water for up to 15 minutes. Remove any contact lenses and open eyes wide apart. Get medical attention.

4.1.4 Ingestion

Get medical attention immediately! Immediately rinse mouth and provide fresh air.

4.2 Most important symptoms and anticipated effects both acute and delayed

Not Classified.

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Extinguish with foam, carbon dioxide or dry powder.
- **Unsuitable extinguishing media**
Water jet

5.2 Special hazards arising from the chemical product:

When heated and in case of fire, corrosive vapours/gases may be formed.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon monoxide (CO), Carbon dioxide (CO₂), Aldehydes, Ketones, Nitrous gases (NO_x)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

NOTE! Use air-supplied respirators to protect against gases/fumes. Use water SPRAY only to cool containers! Do not put water on leaked material.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Wear protective clothing as described in Section 8 of this safety data sheet.

6.2 Environmental precautions:

Do not allow ANY environmental contamination. Do not discharge into drains, water courses or onto the ground.

6.3 Methods and materials for containment and cleaning up:

Advice on how to clean up a spill

Collect with absorbent, non-combustible material into suitable containers.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Avoid spilling, skin and eye contact.



Wear full protective clothing for prolonged exposure and/or high concentrations. Always remove oil with soap and water or skin cleaning agent, never use organic solvents. Do not use oil-contaminated clothing or shoes, and do not put rags moistened with oil into pockets. Ventilate well, avoid breathing vapours. Use approved respirator if air contamination is above accepted level.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in tightly closed original container in a dry, cool and well-ventilated place.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values: Not Classified

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:



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Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. **Hand protection**

Chemical resistant gloves required for prolonged or repeated contact.

- **Respiratory protection:** In case of inadequate ventilation use suitable respirator. Wear mask supplied with: Gas cartridge suitable for organic substances.



Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Viscous liquid
Colour	Bright and Clear
Odour	like: - Gasoline
Melting point/freezing point	Not Classified
Boiling point or initial boiling point and boiling range	N.A.
Flammability	Not Classified
Lower and upper explosion limit	Not Classified
Flash point	240 °C (ASTM D 92)
Pour point	-12 °C
Auto-ignition temperature	Not Classified
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	10 cSt to 12 cst @ 100 °C (ASTM D 445)



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Viscosity	105
<u>Solubility(ies)</u>	
Water solubility	Not Classified
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not Classified
Soil organic carbon/water (log KOC)	Not Classified
Vapour pressure	N.A.
<u>Density and/or relative density</u>	
Density	0.8811 @ 15°C
Relative vapour density	Not Classified
Particle characteristics	Not Classified
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Not Classified.

10.2 Chemical stability

Stable under normal temperature conditions and recommended use.

10.3 Possibility of hazardous reactions

Not Classified.

10.4 Conditions to avoid

Not Classified.

10.5 Incompatible materials

Strong oxidising substances.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If inhaled**



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Not relevant at normal room temperatures. When heated, irritating vapours may be formed.

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Oral	LD50	>5000 mg/kg	rat
Dermal	LD50	>5000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: No specific health warnings noted.

11.1.3 Serious Eye Damage/ Eye Irritation: No specific health warnings noted.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC 50	>1000 mg/l	fish	96 h
EC 50	>1000 mg/l	aquatic invertebrates	48 h
IC 50	>1000 mg/l	ALGAE	72 h

12.2 Persistence and Degradability: The product is not readily biodegradable.

12.3 Bio Accumulative Potential: Will not bio-accumulate.

12.4 Mobility in Soil: The product is insoluble in water and will spread on the water surface.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method:

Dispose of waste and residues in accordance with local authority requirements.



13.1.1 Contaminated Packaging:

Incinerate with provision for removal of effluent gases by scrubber.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information



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The product is not covered by international regulation on the transport of dangerous goods (IMDG, IATA, ADR/RID).

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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6.18 MSDS of BS

Section 1: Chemical Product and Company Identification	
1.1 Chemical Product Identifiers:	
1.1.1 Chemical Name	Lubricating oils
1.1.1.1 Trade Name	Bright Stock
1.1.2 CAS No.	74869-22-0
1.2 Relevant identified uses of the chemical product and uses advised against	
Relevant identified uses	Manufacture of substances Intermediate Formulation [mixing] of preparations and/or re-packaging Coatings Cleaning agent Use in Oil and Gas field drilling and production operations Metal working fluids Release agent. Agrochemicals Building and construction work. Road work Manufacture of rubber products. Polymer preparations and compounds Fuels Lubricant Use as laboratory reagent. Mining (including offshore industries). Water treatment chemicals. Functional fluids Binding agent Explosive
Uses advised against	No additional information available
1.3 Details of the supplier of the safety data sheet	
1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product: Not classified

2.2 Label Elements



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Labelling according to Regulation (EC) No 1272/2008 (CLP)

No labelling applicable

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

When exposed to heat, may decompose liberating hazardous gases.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Bright Stock

3.1.1 Systematic Chemical Name or Trivial Name: Lubricating oils

3.1.2 CAS No.: 74869-22-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Remove victim to fresh air and keep at rest in a position comfortable for breathing. In case of breathing difficulties administer oxygen. In case of irregular breathing or respiratory arrest provide artificial respiration. If medical advice is needed, have product container or label at hand.

4.1.2 Skin Contact

After contact with skin, take off immediately all contaminated clothing, and wash immediately with plenty of water. Rinse thoroughly with plenty of water for at least 20 minutes and take medical advice. If medical advice is needed, have product container or label at hand.

4.1.3 Eye Contact

Rinse immediately and plentifully with water, also under the eyelids, for at least 20 minutes. Seek immediate medical advice. If medical advice is needed, have product container or label at hand.

4.1.4 Ingestion

If swallowed, do not induce vomiting: seek medical advice immediately and show this container label. If swallowed, rinse mouth with water (only if the person is conscious). Drink plenty of water.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis, skin defatting or dermatitis. Slightly irritating to respiratory system. Slightly irritating to eyes. Nausea.

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.



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Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Foam. Carbon dioxide (CO₂). Water fog. Powder.
- **Unsuitable extinguishing media**
Water spray. Do not use a solid water stream as it may scatter and spread fire.

5.2 Special hazards arising from the chemical product:

Combustion results toxic gases. Hazardous combustion products are carbon oxides, sulphur Oxides, nitrogen oxides, carbon monoxide.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon dioxide (CO₂). Sulphur dioxide (SO₂). Nitrogen oxides (NO_x). Carbon monoxide.

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool tanks/drums with water spray/remove them into safety.

5.4.2 Special protective equipment for fire-fighters: In case of fire: Wear self-contained breathing apparatus. Refer to section 8.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Wear suitable protective clothing, gloves and eye/face protection. Refer to section 8. Stop leak if safe to do so. Remove all sources of ignition.

Advice for emergency responders: Wear suitable protective clothing, gloves and eye/face protection. In case of fire: Wear self-contained breathing apparatus. Refer to section 8. Evacuate unnecessary personnel. Remove all sources of ignition. Stop leak if safe to do so. Eliminate leaks immediately.

6.2 Environmental precautions:



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Stop leak if safe to do so. Avoid release to the environment. Contaminated fire-fighting water must be collected separately. Prevent spreading over great surfaces (e.g. by damming or installing oil booms). Do not discharge into drains or the environment.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Collect in closed containers for disposal.

Advice on how to clean up a spill

Absorb with liquid-binding material (e.g. sand, diatomaceous earth, acid- or universal binding agents).

Other information relating to spills and releases

Relevant water authorities should be notified of any large spillage to water course or drain. Do not allow to enter into soil/subsoil. Eliminate all ignition sources if safe to do so. Keep away from heat/sparks/open flames/hot surfaces. - No smoking.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling: Handle in accordance with good industrial hygiene and safety procedures. Use only in well ventilated areas. Use personal protective equipment as required.

7.2 Conditions for Safe Storage, including requirement of hazardous area

Incompatible substances or mixtures

Oxidizing agents, strong.

Consideration of other advice:

Floors should be impenetrable, resistant to liquids and easy to clean. The floor should be leak tight, joint less and not absorbent.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Portable Tanks/vessels.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/mS]	STEL [ppm]	STEL [mg/mS]	Ceiling-C [ppm]	Ceiling-C [mg/mS]	Notation
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USA	BS	74869-22-0	ACGIH	0.0005	5	0.001	10			
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Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	2.7 mg/m ³ /day	human, inhalatory	worker (industry)	Long-term - systemic effects
DNEL	1 mg/kg bw/day	human, dermal	worker (industry)	Long-term - systemic effects
DNEL	0.74 mg/kg bodyweight	human, oral	General Population	Acute - systemic effects
DNEL	1.2 mg/m ³ /day	human, inhalatory	General Population	Long-term - systemic effects

DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Wear protective gloves/protective clothing/eye protection/face protection. The entire protective clothing must be washed after use. Used working clothes should not be used outside the work area. Do not eat, drink or smoke when using this product.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Wear eye protection/face protection.

- Skin protection





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a. Hand protection

Wear protective gloves.

b. Type of material

PVC (Polyvinyl chloride).

c. Other protection measures

Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Provide adequate ventilation.

- **Respiratory protection:** Wear respiratory protection. The filter class must be suitable for the maximum contaminant concentration (gas/vapour/aerosol/particulates) that may arise when handling the product. If the concentration is exceeded, closed-circuit breathing apparatus must



be used. In case of fire: Wear self-contained breathing apparatus.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	light yellow
Odour	Odourless
Melting point/freezing point	No data available
Boiling point or initial boiling point and boiling range	389-720 °C
Flammability	No data available
Lower and upper explosion limit	No data available
Flash point	288 °C ASTM D 92
Pour point	-6 °C
Auto-ignition temperature	360 °C (Combustion point/Fire point: >320°C)
Decomposition temperature	> 400 °C
pH (value)	No data available
Kinematic viscosity	29 cSt to 35 cst @ 100 °C
Viscosity Index	95
<u>Solubility(ies)</u>	
Water solubility	not significant
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	3.9-6
Vapour pressure	< 5 mmHg @ 25°C
<u>Density and/or relative density</u>	
Density	0.9 g/cm ³ @ 20°C



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Relative vapour density	No data available
Particle characteristics	No data available
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

On combustion, forms. Carbon dioxide (CO₂). Sulphur dioxide (SO₂). Nitrogen oxides (NO_x). Carbon monoxide.

10.2 Chemical stability

Stable at normal conditions. When exposed to heat, may decompose liberating hazardous gases.

10.3 Possibility of hazardous reactions

No data available.

10.4 Conditions to avoid

Keep away from heat/sparks/open flames/hot surfaces. - No smoking.

10.5 Incompatible materials

Oxidizing agents, strong.

10.6 Hazardous decomposition products

No data available.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not classified
- **If in eyes**
Not classified
- **If inhaled**
Not classified
- **If on skin**
Not classified
- **Other information**
None



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11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	> 5.53 mg/l/4h	rat
Oral	LD50	> 5000 mg/kg	rat
Dermal	LD50	> 2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Not classified

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified

11.1.4 Respiratory or Skin Sensitization: Not classified

11.1.5 Mutagenicity: Not classified

11.1.6 Carcinogenicity: Not classified

11.1.7 Reproductive Toxicity: Not classified.

11.1.8 Specific target organ toxicity-single exposure: Not classified

11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: Not classified

11.1.11 Additional Information: Not classified

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	> 100 mg/l	fish	96 h
LC50	> 1000 mg/l	aquatic invertebrates	21 days

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: The substance fulfils the very bio accumulative criterion.

n-octanol/water (log KOW)	3.9-6
BCF	132 estimated

12.4 Mobility in Soil: Low mobility (soil).

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations



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13.1 Waste Treatment Method: Consult the local waste disposal expert about waste disposal. Do not empty into drains or the aquatic environment. Dispose of this material and its container to hazardous or special waste collection point. Disposal must be done according to official regulations.



13.1.1 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified

14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.19 MSDS of IO-100

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	IO-100
1.1.1.1 Trade Name	IO-100
1.1.2 CAS No.	64742-65-0

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Base oil.
Uses advised against	No Data Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

The product is not classified as dangerous according to Regulation (EC) No. 1272/2008

2.2 Label Elements

Not classified/No labelling required

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Contaminated surfaces will be extremely slippery. Should not be released into the environment.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: IO-100

3.1.1 Systematic Chemical Name or Trivial Name: IO-100

3.1.2 CAS No.: 64742-65-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air.

4.1.2 Skin Contact



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Remove contaminated clothing and shoes, wash off with soap and water, Wash contaminated clothing before reuse

4.1.3 Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids.

4.1.4 Ingestion

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Call a physician or poison control centre immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed
gastrointestinal irritation, nausea, vomiting, Irritation, diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Carbon dioxide (CO₂). ABC powder. Foam. Water spray or fog.
- **Unsuitable extinguishing media**
Do not use a solid water stream as it may scatter and spread fire

5.2 Special hazards arising from the chemical product:

Incomplete combustion and thermolysis may produce gases of varying toxicity such as carbon monoxide, carbon dioxide, various hydrocarbons, aldehydes and soot. These may be highly dangerous if inhaled in confined spaces or at high concentration.

5.3 Hazardous Combustion Products:

Not Classified

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool containers / tanks with water spray, Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

5.4.2 Special protective equipment for fire-fighters: Wear self-contained breathing apparatus and protective suit.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Do not touch or walk through spilled material. Contaminated surfaces will be extremely slippery. Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition.

6.2 Environmental precautions:

Do not allow material to contaminate ground water system. Try to prevent the material from entering drains or water courses. Local authorities should be advised if significant spillages cannot be contained.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Collect with non-combustible absorbent material (sand, diatomaceous earth, earth, vermiculite).

Other information relating to spills and releases

Place in appropriate containers for disposal.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Use only in well-ventilated areas.

Measures to prevent fire as well as aerosol and dust generation

When using, do not eat, drink or smoke.



Take precautionary measures against static discharges. Ground/bond containers, tanks and transfer/receiving equipment.

Measures to protect the environment

Avoid release to the environment.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values: Contains no substances with occupational exposure limit values.



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8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Safety glasses with side-shields.

- **Skin protection**



- a. Hand protection**

Hydrocarbon-proof gloves. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the EC approved gloves.

- b. Type of material**

butyl-rubber

- c. Material thickness**

0.5mm

- d. Breakthrough times of the glove material**

4 h

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment.



8.2.3 Environmental Exposure Controls:

The product should not be allowed to enter drains, water courses or the soil.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties



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Physical state	liquid
Colour	Bright and Clear
Odour	like: - characteristic
Melting point/freezing point	No information available
Boiling point or initial boiling point and boiling range	No information available
Flammability	No information available
Lower and upper explosion limit	Not explosive
Flash point	160 °C
Pour point	-1°C
Auto-ignition temperature	> 320 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	18-23 cst; 40 °C
<u>Solubility(ies)</u>	
Water solubility	immiscible
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 hPa at 20 °C
<u>Density and/or relative density</u>	
Density	0.90 g/cm ³ ; 20 °C
Relative vapour density	No information available
Particle characteristics	No information available
<u>Other safety parameters</u>	
Oxidising properties	not applicable

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Chemical stability

Stable under recommended storage conditions.

10.2 Possibility of hazardous reactions

None under normal processing

10.3 Conditions to avoid

Heat (temperatures above flash point), sparks, ignition points, flames, static electricity.



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10.4 Incompatible materials

Strong oxidising agents.

10.5 Hazardous decomposition products

None under normal use

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not classified.
- **If in eyes**
Not classified.
- **If inhaled**
Not classified.
- **If on skin**
Not classified.
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	5 mg/l/4h	rat	OECD 403
Oral	LD50	> 5000 mg/kg bw	rat	OECD 420
Dermal	LD50	> 5000 mg/kg bw	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Not classified.

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified.

11.1.4 Respiratory or Skin Sensitization: None known.

11.1.5 Mutagenicity: This product is not classified as mutagenic.

11.1.6 Carcinogenicity: This product is not classified carcinogenic.

11.1.7 Reproductive Toxicity: This product does not contain any known or suspected reproductive hazards.

11.1.8 Specific target organ toxicity-single exposure: No information available.

11.1.9 Specific target organ toxicity - repeated exposure: No information available.

11.1.10 Aspiration Hazard: No information available.

11.1.11 Additional Information: Characteristic skin lesions (oil blisters) may develop following prolonged and repeated exposures (contact with contaminated clothing).



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Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 100 mg/l	fish	OECD 203	96 h
EL50	> 10000 mg/l	aquatic invertebrates	OECD 202	48 h

Aquatic toxicity (Chronic)

NOEL	10 mg/l	fish	QSAR Petrotox	14/28 d
NOEL	> 1000 mg/l	aquatic invertebrates	OECD 211	21 d

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: No information available.

n-octanol/water (log KOW)

> 6 @ 20 °C

12.4 Mobility in Soil: Given its physical and chemical characteristics, the product generally shows low soil mobility.

12.5 Other adverse effects: No information available.

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of in accordance with the European Directives on waste and hazardous waste. Where possible recycling is preferred to disposal or incineration.



13.1.1 Contaminated Packaging: Empty containers should be taken to an approved waste handling site for recycling or disposal

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified



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14.3.1 Hazard Labels: Not Classified

14.3.2 Symbols: Not Classified

14.3.3 Special Provisions: - Not Classified

14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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6.20 MSDS of IO-1600

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	IO-1600
1.1.1.1 Trade Name	IO-1600
1.1.2 CAS No.	64742-65-0

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Base oil.
Uses advised against	No Data Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product:

The product is not classified as dangerous according to Regulation (EC) No. 1272/2008

2.2 Label Elements

Not classified/No labelling required

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Contaminated surfaces will be extremely slippery. Should not be released into the environment.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: IO-1600

3.1.1 Systematic Chemical Name or Trivial Name: IO-1600

3.1.2 CAS No.: 64742-65-0

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air.

4.1.2 Skin Contact



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Remove contaminated clothing and shoes, wash off with soap and water, Wash contaminated clothing before reuse

4.1.3 Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids.

4.1.4 Ingestion

Do not induce vomiting. Never give anything by mouth to an unconscious person. Call a physician or poison control centre immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed
gastrointestinal irritation, nausea, vomiting, Irritation, diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

Treat symptomatically.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Carbon dioxide (CO₂). ABC powder. Foam. Water spray or fog.
- **Unsuitable extinguishing media**
Do not use a solid water stream as it may scatter and spread fire

5.2 Special hazards arising from the chemical product:

Incomplete combustion and thermolysis may produce gases of varying toxicity such as carbon monoxide, carbon dioxide, various hydrocarbons, aldehydes and soot. These may be highly dangerous if inhaled in confined spaces or at high concentration.

5.3 Hazardous Combustion Products: Not Classified

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Cool containers / tanks with water spray, Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

5.4.2 Special protective equipment for fire-fighters: Wear self-contained breathing apparatus and protective suit.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:





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Do not touch or walk through spilled material. Contaminated surfaces will be extremely slippery. Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition.

6.2 Environmental precautions:

Do not allow material to contaminate ground water system. Try to prevent the material from entering drains or water courses. Local authorities should be advised if significant spillages cannot be contained.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Collect with non-combustible absorbent material (sand, diatomaceous earth, earth, vermiculite).

Other information relating to spills and releases

Place in appropriate containers for disposal.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Use only in well-ventilated areas.

Measures to prevent fire as well as aerosol and dust generation

When using, do not eat, drink or smoke.



Take precautionary measures against static discharges. Ground/bond containers, tanks and transfer/receiving equipment.

Measures to protect the environment

Avoid release to the environment.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values: Contains no substances with occupational exposure limit values.

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:



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Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Safety glasses with side-shields.

- **Skin protection**



- a. Hand protection**

Hydrocarbon-proof gloves. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the EC approved gloves.

- b. Type of material**

butyl-rubber

- c. Material thickness**

0.5mm

- d. Breakthrough times of the glove material**

4 h

- **Respiratory protection:** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Respirator with combination filter for vapour/particulate (EN 141). The use of breathing apparatus must comply strictly



with the manufacturer's instructions and the regulations governing their choices and uses.

8.2.3 Environmental Exposure Controls:

The product should not be allowed to enter drains, water courses or the soil.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties



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Physical state	liquid
Colour	Bright and Clear
Odour	like: - characteristic
Melting point/freezing point	No information available
Boiling point or initial boiling point and boiling range	No information available
Flammability	No information available
Lower and upper explosion limit	Not explosive
Flash point	242 °C
Pour point	-1 °C
Auto-ignition temperature	> 320 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic Viscosity	289.2 cst @ 40 °C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	4 (pH value: 7, 20 °C) (ECHA)
Soil organic carbon/water (log KOC)	3.34 (ECHA)
Vapour pressure	160 hPa at 20 °C
<u>Density and/or relative density</u>	
Density	0.90 g/cm ³ ; 20 °C
Relative vapour density	No information available
Particle characteristics	No information available
<u>Other safety parameters</u>	
Oxidising properties	not applicable

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Chemical stability

Stable under recommended storage conditions.

10.2 Possibility of hazardous reactions

None under normal processing

10.3 Conditions to avoid

Heat (temperatures above flash point), sparks, ignition points, flames, static electricity.



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10.4 Incompatible materials

Strong oxidising agents.

10.5 Hazardous decomposition products

None under normal use

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not classified.
- **If in eyes**
Not classified.
- **If inhaled**
Not classified.
- **If on skin**
Not classified.
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	5 mg/l/4h	rat	OECD 403
Oral	LD50	> 5000 mg/kg bw	rat	OECD 420
Dermal	LD50	> 5000 mg/kg bw	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Not classified.

11.1.3 Serious Eye Damage/ Eye Irritation: Not classified.

11.1.4 Respiratory or Skin Sensitization: None known.

11.1.5 Mutagenicity: This product is not classified as mutagenic.

11.1.6 Carcinogenicity: This product is not classified carcinogenic.

11.1.7 Reproductive Toxicity: This product does not contain any known or suspected reproductive hazards.

11.1.8 Specific target organ toxicity-single exposure: No information available.

11.1.9 Specific target organ toxicity - repeated exposure: No information available.

11.1.10 Aspiration Hazard: No information available.

11.1.11 Additional Information: Characteristic skin lesions (oil blisters) may develop following prolonged and repeated exposures (contact with contaminated clothing).



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Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 100 mg/l	fish	OECD 203	96 h
EL50	> 10000 mg/l	aquatic invertebrates	OECD 202	48 h

Aquatic toxicity (Chronic)

NOEL	10 mg/l	fish	QSAR Petrotox	14/28 d
NOEL	> 1000 mg/l	aquatic invertebrates	OECD 211	21 d

12.2 Persistence and Degradability: Inherently biodegradable.

12.3 Bio Accumulative Potential: No information available.

n-octanol/water (log KOW)

> 6 @ 20 °C

12.4 Mobility in Soil: Given its physical and chemical characteristics, the product generally shows low soil mobility.

12.5 Other adverse effects: No information available.

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of in accordance with the European Directives on waste and hazardous waste. Where possible recycling is preferred to disposal or incineration.



13.1.1 Contaminated Packaging: Empty containers should be taken to an approved waste handling site for recycling or disposal

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Classified

14.2 UN Proper Shipping Name: Not Classified

14.3 Transport Class(es): Not Classified



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14.4 Packaging Group: Not Classified

14.5 Environmental Hazards: Not Classified

14.6 Special Precautions for User: Not Classified

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

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Edition: 1

6.21 MSDS of Diana Process Oil SR 28

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Diana Process Oil SR28
---------------------	------------------------

1.1.1.1 Trade Name	Diana Process Oil SR28
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1.2 Relevant identified uses of the chemical product and uses advised against

Uses advised against	No data available
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1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
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1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
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1.3.3 Contact No.	022-25076605
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1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
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1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product: Not Available

2.2 Label Elements: Not Available

2.3 Other hazards except those mentioned in DGCPL Rules, 201: Not Available

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Diana Process Oil SR28

3.1.1 Systematic Chemical Name or Trivial Name: Diana Process Oil SR28

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention.

4.1.2 Skin Contact

In case of contact, immediately flush skin with plenty of water. Remove contaminated clothing and shoes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.

4.1.3 Eye Contact

Check for and remove any contact lenses. In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Get medical attention.



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4.1.4 Ingestion

Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Loosen tight clothing such as a collar, tie, belt or waistband. Get medical attention if symptoms appear.

4.2 Most important symptoms and anticipated effects both acute and delayed: Not Available

4.3 Advice for immediate medical attention and special treatment needed

Notes for the doctor None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
SMALL FIRE: Use DRY chemical powder.
LARGE FIRE: Use water spray, fog or foam. Do not use water jet.

5.2 Special hazards arising from the chemical product:

Fire fighters should wear positive pressure self-contained breathing apparatus (SCBA) and full turnout gear.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Not Available

5.4 Specific Extinguishing Methods: Not Available

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Splash goggles. Full suit. Boots. Gloves. Suggested protective clothing might not be sufficient; consult a specialist before handling this product.

6.2 Environmental precautions: Absorb with an inert material and put the spilled material in an appropriate waste disposal.

6.3 Methods and materials for containment and cleaning up: Not Available

6.4 Reference to other sections: For personal protection, see Section 8.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:



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Keep away from heat. Keep away from sources of ignition. Empty containers pose a fire risk, evaporate the residue under a fume hood. Ground all equipment containing material. Do not ingest. Do not breathe gas/fumes/ vapor/spray.
If ingested, seek medical advice immediately and show the container or the label..



7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed. Keep container in a cool, well-ventilated area.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling - C [ppm]	Ceiling - C [mg/m ³]	Notation	Source
US	Diana Process Oil SR28				5						

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

Engineering measures: Provide exhaust ventilation or other engineering controls to keep the airborne concentrations of vapours below their respective occupational exposure limits. Ensure that eyewash stations and safety showers are proximal to the work-station location
Hygiene measures: Wash hands after handling compounds and before eating, smoking, using lavatory, and at the end of day.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:



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- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. **Hand protection**

Impervious gloves

8.2.3 Environmental Exposure Controls:

Store in a demarcated bunded area to prevent release to drains and/or watercourses.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Brown
Odour	Characteristic
Melting point/freezing point	Not Available
Boiling point or initial boiling point and boiling range	Not Available
Flammability	Not Available
Lower and upper explosion limit	Not Available
Flash point	Open cup: 296°C
Auto-ignition temperature	
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	Kinetic: 28.4 cSt (@100°C)
<u>Solubility(ies)</u>	
Water solubility	Insoluble in cold water.
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not Available
Vapour pressure	Not Available
<u>Density and/or relative density</u>	
Density	0.94 g/cm ³ (15°C / 59°F)



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Relative vapour density	Not Available
Other safety parameters	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

Chemical stability

The product is stable.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If in eyes**
Not Classified
- **If on skin**
Not Classified

11.1.1 Acute Toxicity:

Composition	Exposure route	Endpoint	Value	Species
Diana Process Oil SR28	oral	LD50	> 5000 mg/kg	rat

Chronic toxicity:

Repeated or prolonged exposure is not known to aggravate medical condition.

11.1.2 Skin Corrosion/ Irritation: Not Classified

11.1.3 Serious Eye Damage/ Eye Irritation: Not Classified

11.1.4 Respiratory or Skin Sensitization: Not Classified

11.1.5 Mutagenicity: Not Classified.

11.1.6 Carcinogenicity: Not Classified.

11.1.7 Reproductive Toxicity: Not Classified.

11.1.8 Specific target organ toxicity-single exposure: Not Classified

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified.

11.1.10 Aspiration Hazard: Not Classified

11.1.11 Additional Information: There is no additional information.



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Section 12: Ecological Information

12.1 Toxicity: Not Classified

12.2 Persistence and Degradability: Not Classified

12.3 Bio Accumulative Potential: Not Classified

12.4 Mobility in Soil: Not Classified

12.5 Other adverse effects: No data available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Waste must be disposed of in accordance with federal, state and local environmental control regulations.



13.1.1 Contaminated Packaging: Not Available

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: Not Available

14.2 UN Proper Shipping Name: Not Available

14.3 Transport Class(es): Not Available

14.4 Packaging Group: Not Available

14.5 Environmental Hazards: Not Available

14.6 Special Precautions for User: Not Available

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely



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6.22 MSDS of ELASTO SUPREME (TRAE)

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	ELASTO SUPREME
1.1.1.1 Trade Name	ELASTO SUPREME
1.1.2 CAS No.	64742-52-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Distillates (petroleum), hydrotreated heavy Naphthenic, Rubber processing oil – Naphthenic Type
Uses advised against	No information available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.: 022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P331 Do not induce vomiting.

P102 Keep out of reach of children

Precautionary statements – response

P301+P310 If swallowed immediately call a poison centre/doctor

P405 Store locked up.

P501 Dispose of contents/container to authorized chemical landfill or if organic to high temperature incineration.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: ELASTO SUPREME

3.1.1 Systematic Chemical Name or Trivial Name: ELASTO SUPREME

3.1.2 CAS No.: 64742-52-5

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation



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If fumes, aerosol or combustion products are inhaled remove from contaminated area. Other measures are usually unnecessary

4.1.2 Skin Contact

Remove contaminated clothing and footwear, and dispose of safely. Seek medical attention if skin irritation, swelling or redness develops and persists

4.1.3 Eye Contact

Wash out immediately with water. If irritation continues, seek medical attention. Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

4.1.4 Ingestion

Immediately give a glass of water. Do not give anything by mouth to an unconscious person. Do not induce vomiting as there is a risk of aspiration. Call a doctor immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Nausea, Vomiting

4.3 Advice for immediate medical attention and special treatment needed

None.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Dry chemical powder, foam, Carbon dioxide (CO₂), Water fog.

- **Unsuitable extinguishing media**

Does not use direct water jets on the burning product; they could cause splattering and spread the fire.

5.2 Special hazards arising from the chemical product:

Incomplete combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide and unidentified organic and inorganic compounds.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: None

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Work gloves providing adequate chemical resistance, specifically to hydrocarbons. Wear approved respiratory protection such as toxic dust, mist and fume respirator. Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations.

6.2 Environmental precautions:

Prevent product from entering sewers, rivers, waterways or other bodies of water.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Clean up all spills immediately. Avoid breathing vapours and contact with skin and eyes.

Advice on how to clean up a spill

Cleaning, inspection and maintenance of internal structure of storage tanks must be done only by properly equipped and qualified personnel as defined by national, local or company regulations

Other information relating to spills and releases

Empty containers may contain combustible product residues. Do not weld, solder, drill, cut or perform similar operations unless they have been properly cleaned.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Contaminated materials should not be allowed to accumulate in the workplaces and should never be kept inside the pockets. Keep away from food and beverages.

Measures to prevent fire as well as aerosol and dust generation

For fires involving this material, do not enter any enclosed or confined space without self contained breathing apparatus to protect against the hazardous effects of combustion



products or oxygen deficiency.

Measures to protect the environment

No particular measures.

7.2 Conditions for Safe Storage, including requirement of hazardous area



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Incompatible substances or mixtures

Avoid reaction with oxidizing agents.

Consideration of other advice:

For containers, or container linings use mild steel, stainless steel.

Ventilation requirements

Use local and general ventilation

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
Germany	Elasto Supreme	64742-52-5	ACGIH TLV		5		20				

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations or spill clean-up.



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- **Skin protection**



- a. **Hand protection**

Wear general protective gloves, e.g. light weight rubber gloves. Exposed employees should exercise reasonable personal cleanliness; this includes cleansing exposed skin several times daily with soap and water and laundering or dry cleaning soiled work clothing at least daily.

- b. **Other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

- **Respiratory protection:** If operating conditions create airborne concentrations which exceed the recommended exposure standard(s), the use of an approved respirator is recommended. Wear approved respiratory protection such as toxic dust, mist and fume



respirator.

8.2.3 Environmental Exposure Controls:

This material may present environmental risks common to oil spills.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Dark Homogeneous
Odour	Odourless
Melting point/freezing point	Not Applicable
Boiling point or initial boiling point and boiling range	Not Applicable
Flammability	Not Applicable
Lower and upper explosion limit	No data available
Flash point	250-350 °C, ASTM D 92
Pour point	18-38 °C
Aniline point	84-94 °C
Auto-ignition temperature	Not Applicable
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	700-1206 cst at 40 °C



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	25-31 cst at 100 °C
Refractive Index	1.5312 @ 20 °C
<u>Solubility(ies)</u>	
Water solubility	Negligible
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	not determined
Soil organic carbon/water (log KOC)	not determined
Vapour pressure	<0.013 kPa
<u>Density and/or relative density</u>	
Density	0.9 to 0.96 (g/cc at 15 °C)
Relative vapour density	not determined
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable.

10.2 Chemical stability

Product is considered stable and hazardous polymerisation will not occur.

10.3 Possibility of hazardous reactions

Avoid temperatures exceeding the flash point.

10.4 Conditions to avoid

No specific data

10.5 Incompatible materials

No specific data

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics



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- **If swallowed**
Not Classified
- **If in eyes**
Not Classified
- **If inhaled**
Not Classified
- **If on skin**
Not Classified
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	>5000 mg/kg (4h)	rat	ECHA
Oral	LD50	>5000 mg/kg bw	rat	ECHA
Dermal	LD50	>2000 mg/kg bw	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Shall not be classified as a corrosive/irritant to skin

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as irritation on eye contact

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as irritation on eye contact

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as Reproductive Toxicity

11.1.8 Specific target organ toxicity-single exposure: Shall not be classified as a specific target organ toxicant.

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified as a specific target organ toxicant

11.1.10 Aspiration Hazard: Shall not be classified as presenting an aspiration hazard.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	$\geq 10,000$ mg/L	Fish	96 h
EC50	≤ 100 mg/L	Fish	72 h
LC50	> 100 mg/L	Daphnia magna	48 h



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12.2 Persistence and Degradability: No information available.

12.3 Bio Accumulative Potential: No information available.

12.4 Mobility in Soil: No information available.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Place contained materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local environmental or health authorities for approved disposal of this material.



- **Sewage disposal-relevant information**

Prevent product from entering sewers, rivers, waterways or other bodies of water

Section 14: Transport Information

14.1 UN Number: Not Applicable

14.2 UN Proper Shipping Name Not Applicable

14.3 Transport Class(es): Not Applicable

14.4 Packaging Group: Not Applicable

14.5 Environmental Hazards: None (non-environmentally hazardous acc. to the dangerous goods regulations).

14.6 Special Precautions for User: Not Applicable

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage
- Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all



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6.23 MSDS of ELASTO 710 R

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	ELASTO 710 R
1.1.1.1 Trade Name	ELASTO 710 R
1.1.2 CAS No.	64742-52-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Distillates (petroleum), hydrotreated heavy Naphthenic, Rubber processing oil – Naphthenic Type
Uses advised against	No information available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P331 Do not induce vomiting.

P102 Keep out of reach of children

Precautionary statements – response

P301+P310 If swallowed immediately call a poison centre/doctor

P405 Store locked up.

P501 Dispose of contents/container to authorized chemical landfill or if organic to high temperature incineration.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: ELASTO 710 R

3.1.1 Systematic Chemical Name or Trivial Name: ELASTO 710 R

3.1.2 CAS No.: 64742-52-5

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation



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If fumes, aerosol or combustion products are inhaled remove from contaminated area. Other measures are usually unnecessary

4.1.2 Skin Contact

Remove contaminated clothing and footwear, and dispose of safely. Seek medical attention if skin irritation, swelling or redness develops and persists

4.1.3 Eye Contact

Wash out immediately with water. If irritation continues, seek medical attention. Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

4.1.4 Ingestion

Immediately give a glass of water. Do not give anything by mouth to an unconscious person. Do not induce vomiting as there is a risk of aspiration. Call a doctor immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Nausea, Vomiting

4.3 Advice for immediate medical attention and special treatment needed

None.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Dry chemical powder, foam, Carbon dioxide (CO₂), Water fog.

- **Unsuitable extinguishing media**

Does not use direct water jets on the burning product; they could cause splattering and spread the fire.

5.2 Special hazards arising from the chemical product:

Incomplete combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide and unidentified organic and inorganic compounds.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: None

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Work gloves providing adequate chemical resistance, specifically to hydrocarbons. Wear approved respiratory protection such as toxic dust, mist and fume respirator. Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations.

6.2 Environmental precautions:

Prevent product from entering sewers, rivers, waterways or other bodies of water.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Clean up all spills immediately. Avoid breathing vapours and contact with skin and eyes.

Advice on how to clean up a spill

Cleaning, inspection and maintenance of internal structure of storage tanks must be done only by properly equipped and qualified personnel as defined by national, local or company regulations

Other information relating to spills and releases

Empty containers may contain combustible product residues. Do not weld, solder, drill, cut or perform similar operations unless they have been properly cleaned.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Contaminated materials should not be allowed to accumulate in the workplaces and should never be kept inside the pockets. Keep away from food and beverages.

Measures to prevent fire as well as aerosol and dust generation

For fires involving this material, do not enter any enclosed or confined space without self contained breathing apparatus to protect against the hazardous effects of combustion



products or oxygen deficiency.

Measures to protect the environment

No particular measures.

7.2 Conditions for Safe Storage, including requirement of hazardous area

Incompatible substances or mixtures

Avoid reaction with oxidizing agents.



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Consideration of other advice:

For containers, or container linings use mild steel, stainless steel.

Ventilation requirements

Use local and general ventilation

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
Germany	Elasto 710 R	64742-52-5	ACGIH TLV		5		20				

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations or spill clean-up.



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- **Skin protection**



- a. Hand protection**

Wear general protective gloves, e.g. light weight rubber gloves. Exposed employees should exercise reasonable personal cleanliness; this includes cleansing exposed skin several times daily with soap and water and laundering or dry cleaning soiled work clothing at least daily.

- b. Other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

- **Respiratory protection:** If operating conditions create airborne concentrations which exceed the recommended exposure standard(s), the use of an approved respirator is recommended. Wear approved respiratory protection such as toxic dust, mist and fume



respirator.

8.2.3 Environmental Exposure Controls:

This material may present environmental risks common to oil spills.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Dark Homogeneous
Odour	Odourless
Melting point/freezing point	Not Applicable
Boiling point or initial boiling point and boiling range	Not Applicable
Flammability	Not Applicable
Lower and upper explosion limit	No data available
Flash point	218 °C, ASTM D 92
Pour point	18-38 °C
Aniline point	39.3-52 °C
Auto-ignition temperature	Not Applicable
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	20-28.5 cst at 100 °C



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Refractive Index	1.5650-1.5850 @ 6=0 °C
<u>Solubility(ies)</u>	
Water solubility	Negligible
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	not determined
Soil organic carbon/water (log KOC)	not determined
Vapour pressure	<0.013 kPa
<u>Density and/or relative density</u>	
Density	0.9 to 0.96 (g/cc at 15 °C)
Relative vapour density	not determined
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable.

10.2 Chemical stability

Product is considered stable and hazardous polymerisation will not occur.

10.3 Possibility of hazardous reactions

Avoid temperatures exceeding the flash point.

10.4 Conditions to avoid

No specific data

10.5 Incompatible materials

No specific data

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**



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- Not Classified
- **If in eyes**
Not Classified
 - **If inhaled**
Not Classified
 - **If on skin**
Not Classified
 - **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	>5000 mg/kg (4h)	rat	ECHA
Oral	LD50	>5000 mg/kg bw	rat	ECHA
Dermal	LD50	>2000 mg/kg bw	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Shall not be classified as a corrosive/irritant to skin

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as irritation on eye contact

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as irritation on eye contact

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as Reproductive Toxicity

11.1.8 Specific target organ toxicity-single exposure: Shall not be classified as a specific target organ toxicant.

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified as a specific target organ toxicant

11.1.10 Aspiration Hazard: Shall not be classified as presenting an aspiration hazard.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	$\geq 10,000$ mg/L	Fish	96 h
EC50	≤ 100 mg/L	Fish	72 h
LC50	> 100 mg/L	Daphnia magna	48 h

12.2 Persistence and Degradability: No information available.



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12.3 Bio Accumulative Potential: No information available.

12.4 Mobility in Soil: No information available.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Place contained materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local environmental or health authorities for approved disposal of this material.



- **Sewage disposal-relevant information**

Prevent product from entering sewers, rivers, waterways or other bodies of water

Section 14: Transport Information

14.1 UN Number: Not Applicable

14.2 UN Proper Shipping Name Not Applicable

14.3 Transport Class(es): Not Applicable

14.4 Packaging Group: Not Applicable

14.5 Environmental Hazards: None (non-environmentally hazardous acc. to the dangerous goods regulations).

14.6 Special Precautions for User: Not Applicable

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage
- Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet



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6.24 MSDS of ELASTO SUPREME H (RAE)

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	ELASTO SUPREME H
1.1.1.1 Trade Name	ELASTO SUPREME H
1.1.2 CAS No.	64742-52-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Distillates (petroleum), hydrotreated heavy Naphthenic, Rubber processing oil – Naphthenic Type
Uses advised against	No information available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.:	022-25076606
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Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P331 Do not induce vomiting.

P102 Keep out of reach of children

Precautionary statements – response

P301+P310 If swallowed immediately call a poison centre/doctor

P405 Store locked up.

P501 Dispose of contents/container to authorized chemical landfill or if organic to high temperature incineration.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: ELASTO SUPREME H

3.1.1 Systematic Chemical Name or Trivial Name: ELASTO SUPREME H

3.1.2 CAS No.: 64742-52-5

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation



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If fumes, aerosol or combustion products are inhaled remove from contaminated area. Other measures are usually unnecessary

4.1.2 Skin Contact

Remove contaminated clothing and footwear, and dispose of safely. Seek medical attention if skin irritation, swelling or redness develops and persists

4.1.3 Eye Contact

Wash out immediately with water. If irritation continues, seek medical attention. Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

4.1.4 Ingestion

Immediately give a glass of water. Do not give anything by mouth to an unconscious person. Do not induce vomiting as there is a risk of aspiration. Call a doctor immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Nausea, Vomiting

4.3 Advice for immediate medical attention and special treatment needed

None.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Dry chemical powder, foam, Carbon dioxide (CO₂), Water fog.

- **Unsuitable extinguishing media**

Does not use direct water jets on the burning product; they could cause splattering and spread the fire.

5.2 Special hazards arising from the chemical product:

Incomplete combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide and unidentified organic and inorganic compounds.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: None

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Work gloves providing adequate chemical resistance, specifically to hydrocarbons. Wear approved respiratory protection such as toxic dust, mist and fume respirator. Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations.

6.2 Environmental precautions:

Prevent product from entering sewers, rivers, waterways or other bodies of water.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Clean up all spills immediately. Avoid breathing vapours and contact with skin and eyes.

Advice on how to clean up a spill

Cleaning, inspection and maintenance of internal structure of storage tanks must be done only by properly equipped and qualified personnel as defined by national, local or company regulations

Other information relating to spills and releases

Empty containers may contain combustible product residues. Do not weld, solder, drill, cut or perform similar operations unless they have been properly cleaned.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Contaminated materials should not be allowed to accumulate in the workplaces and should never be kept inside the pockets. Keep away from food and beverages.

Measures to prevent fire as well as aerosol and dust generation

For fires involving this material, do not enter any enclosed or confined space without self contained breathing apparatus to protect against the hazardous effects of combustion



products or oxygen deficiency.

Measures to protect the environment

No particular measures.

7.2 Conditions for Safe Storage, including requirement of hazardous area

Incompatible substances or mixtures

Avoid reaction with oxidizing agents.



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Consideration of other advice:

For containers, or container linings use mild steel, stainless steel.

Ventilation requirements

Use local and general ventilation

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling g-C [ppm]	Ceiling g-C [mg/m ³]	Notation	Source
Germany	Elasto Supreme H	6474-2-52-5	ACGIH TLV		5		20				

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations or spill clean-up.



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- **Skin protection**



- a. **Hand protection**

Wear general protective gloves, e.g. light weight rubber gloves. Exposed employees should exercise reasonable personal cleanliness; this includes cleansing exposed skin several times daily with soap and water and laundering or dry cleaning soiled work clothing at least daily.

- b. **Other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

- **Respiratory protection:** If operating conditions create airborne concentrations which exceed the recommended exposure standard(s), the use of an approved respirator is recommended. Wear approved respiratory protection such as toxic dust, mist and fume



respirator.

8.2.3 Environmental Exposure Controls:

This material may present environmental risks common to oil spills.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Dark Homogeneous
Odour	Odourless
Melting point/freezing point	Not Applicable
Boiling point or initial boiling point and boiling range	Not Applicable
Flammability	Not Applicable
Lower and upper explosion limit	No data available
Flash point	270 °C, ASTM D 92
Pour point	27 °C
Aniline point	68 °C
Auto-ignition temperature	Not Applicable
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	60 cst at 100 °C



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Refractive Index	1.55 @ 60 °C
<u>Solubility(ies)</u>	
Water solubility	Negligible
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	not determined
Soil organic carbon/water (log KOC)	not determined
Vapour pressure	<0.013 kPa
<u>Density and/or relative density</u>	
Density	0.9 to 0.96 (g/cc at 15 °C)
Relative vapour density	not determined
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable.

10.2 Chemical stability

Product is considered stable and hazardous polymerisation will not occur.

10.3 Possibility of hazardous reactions

Avoid temperatures exceeding the flash point.

10.4 Conditions to avoid

No specific data

10.5 Incompatible materials

No specific data

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**



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- Not Classified
- **If in eyes**
Not Classified
 - **If inhaled**
Not Classified
 - **If on skin**
Not Classified
 - **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	>5000 mg/kg (4h)	rat	ECHA
Oral	LD50	>5000 mg/kg bw	rat	ECHA
Dermal	LD50	>2000 mg/kg bw	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Shall not be classified as a corrosive/irritant to skin

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as irritation on eye contact

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as irritation on eye contact

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as Reproductive Toxicity

11.1.8 Specific target organ toxicity-single exposure: Shall not be classified as a specific target organ toxicant.

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified as a specific target organ toxicant

11.1.10 Aspiration Hazard: Shall not be classified as presenting an aspiration hazard.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	$\geq 10,000$ mg/L	Fish	96 h
EC50	≤ 100 mg/L	Fish	72 h
LC50	> 100 mg/L	Daphnia magna	48 h

12.2 Persistence and Degradability: No information available.



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12.3 Bio Accumulative Potential: No information available.

12.4 Mobility in Soil: No information available.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Place contained materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local environmental or health authorities for approved disposal of this material.



- **Sewage disposal-relevant information**
Prevent product from entering sewers, rivers, waterways or other bodies of water

Section 14: Transport Information

14.1 UN Number: Not Applicable

14.2 UN Proper Shipping Name Not Applicable

14.3 Transport Class(es): Not Applicable

14.4 Packaging Group: Not Applicable

14.5 Environmental Hazards: None (non-environmentally hazardous acc. to the dangerous goods regulations).

14.6 Special Precautions for User: Not Applicable

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage
- Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet



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6.25 MSDS of RPO-M

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	RPO-M
1.1.1.1 Trade Name	RPO-M
1.1.2 CAS No.	64742-52-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Distillates (petroleum), hydrotreated heavy Naphthenic, Rubber processing oil – Naphthenic Type
Uses advised against	No information available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.: 022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P331 Do not induce vomiting.

P102 Keep out of reach of children

Precautionary statements – response

P301+P310 If swallowed immediately call a poison centre/doctor

P405 Store locked up.

P501 Dispose of contents/container to authorized chemical landfill or if organic to high temperature incineration.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: RPO-M

3.1.1 Systematic Chemical Name or Trivial Name: RPO-M

3.1.2 CAS No.: 64742-52-5

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation



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If fumes, aerosol or combustion products are inhaled remove from contaminated area. Other measures are usually unnecessary

4.1.2 Skin Contact

Remove contaminated clothing and footwear, and dispose of safely. Seek medical attention if skin irritation, swelling or redness develops and persists

4.1.3 Eye Contact

Wash out immediately with water. If irritation continues, seek medical attention. Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

4.1.4 Ingestion

Immediately give a glass of water. Do not give anything by mouth to an unconscious person. Do not induce vomiting as there is a risk of aspiration. Call a doctor immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Nausea, Vomiting

4.3 Advice for immediate medical attention and special treatment needed

None.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Dry chemical powder, foam, Carbon dioxide (CO₂), Water fog.

- **Unsuitable extinguishing media**

Does not use direct water jets on the burning product; they could cause splattering and spread the fire.

5.2 Special hazards arising from the chemical product:

Incomplete combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide and unidentified organic and inorganic compounds.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: None

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Work gloves providing adequate chemical resistance, specifically to hydrocarbons. Wear approved respiratory protection such as toxic dust, mist and fume respirator. Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations.

6.2 Environmental precautions:

Prevent product from entering sewers, rivers, waterways or other bodies of water.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Clean up all spills immediately. Avoid breathing vapours and contact with skin and eyes.

Advice on how to clean up a spill

Cleaning, inspection and maintenance of internal structure of storage tanks must be done only by properly equipped and qualified personnel as defined by national, local or company regulations

Other information relating to spills and releases

Empty containers may contain combustible product residues. Do not weld, solder, drill, cut or perform similar operations unless they have been properly cleaned.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Contaminated materials should not be allowed to accumulate in the workplaces and should never be kept inside the pockets. Keep away from food and beverages.

Measures to prevent fire as well as aerosol and dust generation

For fires involving this material, do not enter any enclosed or confined space without self contained breathing apparatus to protect against the hazardous effects of combustion



products or oxygen deficiency.

Measures to protect the environment

No particular measures.

7.2 Conditions for Safe Storage, including requirement of hazardous area

Incompatible substances or mixtures

Avoid reaction with oxidizing agents.



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Consideration of other advice:

For containers, or container linings use mild steel, stainless steel.

Ventilation requirements

Use local and general ventilation

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
Germany	RP O-M	6474-2-52-5	ACGIH TLV		5		20				

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations or spill clean-up.

- **Skin protection**





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a. Hand protection

Wear general protective gloves, e.g. light weight rubber gloves. Exposed employees should exercise reasonable personal cleanliness; this includes cleansing exposed skin several times daily with soap and water and laundering or dry cleaning soiled work clothing at least daily.

b. Other protection measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

- **Respiratory protection:** If operating conditions create airborne concentrations which exceed the recommended exposure standard(s), the use of an approved respirator is recommended. Wear approved respiratory protection such as toxic dust, mist and fume



respirator.

8.2.3 Environmental Exposure Controls:

This material may present environmental risks common to oil spills.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Dark Homogeneous
Odour	Odourless
Melting point/freezing point	Not Applicable
Boiling point or initial boiling point and boiling range	Not Applicable
Flammability	Not Applicable
Lower and upper explosion limit	No data available
Flash point	218 °C, ASTM D 92
Pour point	+38 °C (Apr. to Sept.), +32 °C (Oct. to Mar.)
Aniline point	40 °C to 52 °C
Auto-ignition temperature	Not Applicable
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	20 cst to 28.5 cst at 100 °C
Refractive Index	1.54 to 1.56 @ 60 °C
<u>Solubility(ies)</u>	
Water solubility	Negligible
<u>Partition coefficient</u>	



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Partition coefficient n-octanol/water (log value):	not determined
Soil organic carbon/water (log KOC)	not determined
Vapour pressure	<0.013 kPa
<u>Density and/or relative density</u>	
Density	0.85 to 0.95 (kg/l at 25 °C)
Relative vapour density	not determined
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable.

10.2 Chemical stability

Product is considered stable and hazardous polymerisation will not occur.

10.3 Possibility of hazardous reactions

Avoid temperatures exceeding the flash point.

10.4 Conditions to avoid

No specific data

10.5 Incompatible materials

No specific data

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not Classified
- **If in eyes**
Not Classified
- **If inhaled**
Not Classified



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- **If on skin**
Not Classified
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	>5000 mg/kg (4h)	rat	ECHA
Oral	LD50	>5000 mg/kg bw	rat	ECHA
Dermal	LD50	>2000 mg/kg bw	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Shall not be classified as a corrosive/irritant to skin

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as irritation on eye contact

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as irritation on eye contact

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as Reproductive Toxicity

11.1.8 Specific target organ toxicity-single exposure: Shall not be classified as a specific target organ toxicant.

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified as a specific target organ toxicant

11.1.10 Aspiration Hazard: Shall not be classified as presenting an aspiration hazard.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	$\geq 10,000$ mg/L	Fish	96 h
EC50	≤ 100 mg/L	Fish	72 h
LC50	> 100 mg/L	Daphnia magna	48 h

12.2 Persistence and Degradability: No information available.

12.3 Bio Accumulative Potential: No information available.

12.4 Mobility in Soil: No information available.

12.5 Other adverse effects: Data are not available



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Section 13: Disposal considerations

13.1 Waste Treatment Method: Place contained materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local environmental or health authorities for approved disposal of this material.



- **Sewage disposal-relevant information**
Prevent product from entering sewers, rivers, waterways or other bodies of water

Section 14: Transport Information

14.1 UN Number: Not Applicable

14.2 UN Proper Shipping Name Not Applicable

14.3 Transport Class(es): Not Applicable

14.4 Packaging Group: Not Applicable

14.5 Environmental Hazards: None (non-environmentally hazardous acc. to the dangerous goods regulations).

14.6 Special Precautions for User: Not Applicable

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage
- Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.26 MSDS of RPO-H

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	RPO-H
1.1.1.1 Trade Name	RPO-H
1.1.2 CAS No.	64742-52-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Distillates (petroleum), hydrotreated heavy Naphthenic, Rubber processing oil – Naphthenic Type
Uses advised against	No information available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor

1.4 Emergency Telephone No.: 022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Aspiration hazard	1	Asp. Tox. 1	H304

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)



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2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H304 May be fatal if swallowed and enters airways

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P331 Do not induce vomiting.

P102 Keep out of reach of children

Precautionary statements – response

P301+P310 If swallowed immediately call a poison centre/doctor

P405 Store locked up.

P501 Dispose of contents/container to authorized chemical landfill or if organic to high temperature incineration.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: RPO-H

3.1.1 Systematic Chemical Name or Trivial Name: RPO-H

3.1.2 CAS No.: 64742-52-5

Section 4: First-Aid Measures

4.1 Description of first aid measures





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4.1.1 Inhalation

If fumes, aerosol or combustion products are inhaled remove from contaminated area. Other measures are usually unnecessary

4.1.2 Skin Contact

Remove contaminated clothing and footwear, and dispose of safely. Seek medical attention if skin irritation, swelling or redness develops and persists

4.1.3 Eye Contact

Wash out immediately with water. If irritation continues, seek medical attention. Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

4.1.4 Ingestion

Immediately give a glass of water. Do not give anything by mouth to an unconscious person. Do not induce vomiting as there is a risk of aspiration. Call a doctor immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Nausea, Vomiting

4.3 Advice for immediate medical attention and special treatment needed

None.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Dry chemical powder, foam, Carbon dioxide (CO₂), Water fog.

- **Unsuitable extinguishing media**

Does not use direct water jets on the burning product; they could cause splattering and spread the fire.

5.2 Special hazards arising from the chemical product:

Incomplete combustion is likely to give rise to a complex mixture of airborne solid and liquid particulates and gases, including carbon monoxide and unidentified organic and inorganic compounds.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: None

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given.

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



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Work gloves providing adequate chemical resistance, specifically to hydrocarbons. Wear approved respiratory protection such as toxic dust, mist and fume respirator. Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations.

6.2 Environmental precautions:

Prevent product from entering sewers, rivers, waterways or other bodies of water.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Clean up all spills immediately. Avoid breathing vapours and contact with skin and eyes.

Advice on how to clean up a spill

Cleaning, inspection and maintenance of internal structure of storage tanks must be done only by properly equipped and qualified personnel as defined by national, local or company regulations

Other information relating to spills and releases

Empty containers may contain combustible product residues. Do not weld, solder, drill, cut or perform similar operations unless they have been properly cleaned.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Contaminated materials should not be allowed to accumulate in the workplaces and should never be kept inside the pockets. Keep away from food and beverages.

Measures to prevent fire as well as aerosol and dust generation

For fires involving this material, do not enter any enclosed or confined space without self contained breathing apparatus to protect against the hazardous effects of combustion



products or oxygen deficiency.

Measures to protect the environment

No particular measures.

7.2 Conditions for Safe Storage, including requirement of hazardous area

Incompatible substances or mixtures

Avoid reaction with oxidizing agents.



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Consideration of other advice:

For containers, or container linings use mild steel, stainless steel.

Ventilation requirements

Use local and general ventilation

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
Germany	RP O-H	64742-52-5	ACGIH TLV		5		20				

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- Eye/face protection



Safety glasses, chemical type goggles, or face shield appropriate where splashing or misting is expected during routine operations or spill clean-up.



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- **Skin protection**



- a. **Hand protection**

Wear general protective gloves, e.g. light weight rubber gloves. Exposed employees should exercise reasonable personal cleanliness; this includes cleansing exposed skin several times daily with soap and water and laundering or dry cleaning soiled work clothing at least daily.

- b. **Other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

- **Respiratory protection:** If operating conditions create airborne concentrations which exceed the recommended exposure standard(s), the use of an approved respirator is recommended. Wear approved respiratory protection such as toxic dust, mist and fume



respirator.

8.2.3 Environmental Exposure Controls:

This material may present environmental risks common to oil spills.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Dark Homogeneous
Odour	Odourless
Melting point/freezing point	Not Applicable
Boiling point or initial boiling point and boiling range	Not Applicable
Flammability	Not Applicable
Lower and upper explosion limit	No data available
Flash point	288 °C, ASTM D 92
Pour point	40 °C
Aniline point	70 °C
Auto-ignition temperature	Not Applicable
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	32 cst to 65 cst at 100 °C



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Refractive Index	1.54 to 1.56 @ 60 °C
<u>Solubility(ies)</u>	
Water solubility	Negligible
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	not determined
Soil organic carbon/water (log KOC)	not determined
Vapour pressure	<0.013 kPa
<u>Density and/or relative density</u>	
Density	0.85 to 0.95 (kg/l at 25 °C)
Relative vapour density	not determined
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable.

10.2 Chemical stability

Product is considered stable and hazardous polymerisation will not occur.

10.3 Possibility of hazardous reactions

Avoid temperatures exceeding the flash point.

10.4 Conditions to avoid

No specific data

10.5 Incompatible materials

No specific data

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**



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- Not Classified
- **If in eyes**
Not Classified
 - **If inhaled**
Not Classified
 - **If on skin**
Not Classified
 - **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	>5000 mg/kg (4h)	rat	ECHA
Oral	LD50	>5000 mg/kg bw	rat	ECHA
Dermal	LD50	>2000 mg/kg bw	rabbit	ECHA

11.1.2 Skin Corrosion/ Irritation: Shall not be classified as a corrosive/irritant to skin

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as irritation on eye contact

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as irritation on eye contact

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Shall not be classified as Reproductive Toxicity

11.1.8 Specific target organ toxicity-single exposure: Shall not be classified as a specific target organ toxicant.

11.1.9 Specific target organ toxicity - repeated exposure: Shall not be classified as a specific target organ toxicant

11.1.10 Aspiration Hazard: Shall not be classified as presenting an aspiration hazard.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LL50	$\geq 10,000$ mg/L	Fish	96 h
EC50	≤ 100 mg/L	Fish	72 h
LC50	> 100 mg/L	Daphnia magna	48 h

12.2 Persistence and Degradability: No information available.



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12.3 Bio Accumulative Potential: No information available.

12.4 Mobility in Soil: No information available.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Place contained materials in disposable containers and dispose of in a manner consistent with applicable regulations. Contact local environmental or health authorities for approved disposal of this material.



- **Sewage disposal-relevant information**
Prevent product from entering sewers, rivers, waterways or other bodies of water

Section 14: Transport Information

14.1 UN Number: Not Applicable

14.2 UN Proper Shipping Name Not Applicable

14.3 Transport Class(es): Not Applicable

14.4 Packaging Group: Not Applicable

14.5 Environmental Hazards: None (non-environmentally hazardous acc. to the dangerous goods regulations).

14.6 Special Precautions for User: Not Applicable

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage
- Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet



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6.27 MSDS of LSHS

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Low Sulphur Heavy Stock
1.1.1.1 Trade Name	LSHS
1.1.2 CAS No.	97862-78-7

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Industrial use
Uses advised against	Not Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable liquid	4	Flam. Liq. 4	H227
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Reproductive toxicity	2	Repr. 2	H361f
Carcinogenicity	1	Carc. 1	H350
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Hazardous to the aquatic environment - chronic hazard	1	Aquatic Chronic 1	H410
Acute toxicity	4	Acute Tox. 4	H332

For full text of abbreviations: see SECTION 16

The most important adverse physicochemical, human health and environmental effects

Delayed or immediate effects can be expected after short or long-term exposure. The product is combustible and can be ignited by potential ignition sources. Spillage and fire water can cause pollution of watercourses.



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2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H227	Combustible liquid
H315	Causes skin irritation
H332	Harmful if inhaled
H350	May cause cancer
H361	Suspected of damaging fertility or the unborn child
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H410	Very toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention	
P201	Obtain special instructions before use
P202	Do not handle until all safety precautions have been read and understood
P264	Wash hands thoroughly after handling
P210	Keep away from heat/sparks/open flames/hot surfaces. No smoking
P233	Keep container tightly closed
P240	Ground and bond container and receiving equipment
P241	Use explosion-proof electrical/ ventilating/ lighting equipment
P242	Use non-sparking tools
P243	Take action to prevent static discharges
P280	Wear protective gloves/eye protection
P271	Use only outdoors or in a well-ventilated area.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray..
P273	Avoid release to the environment



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Precautionary statements – response	
P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]
P370 + P378	In case of fire: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide to extinguish
P308+P313	IF exposed or concerned: Get medical advice/attention
P332+P313	If skin irritation occurs: Get medical advice/attention.

2.3 Other hazards except those mentioned in DGCPL Rules, 201:
Results of PBT and vPvB assessment
According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: LSHS

3.1.1 Systematic Chemical Name or Trivial Name: Low Sulphur Heavy Stock

3.1.2 CAS No.: 97862-78-7

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air in case of accidental inhalation of vapours. If breathing is irregular or stopped, administer artificial respiration. If symptoms persist, call a physician.

4.1.2 Skin Contact

Wash off immediately with soap and plenty of water. If skin irritation persists, call a physician. For thermal burns, cool affected areas as quickly as possible by immersing in water. Consult a physician.

4.1.3 Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Call a physician immediately. If burns occur, treat as thermal burns.

4.1.4 Ingestion

If swallowed, seek medical advice immediately and show this container or label. Do not induce vomiting. Prevent vomiting if possible. If a person vomits when lying on his back, place him in the recovery position. Never give anything by mouth to an unconscious person.

4.2 Most important symptoms and anticipated effects both acute and delayed



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Refer to SECTION 11

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray. Foam. Dry powder. Carbon dioxide (CO₂).

5.2 Special hazards arising from the chemical product:

Do not allow run-off from firefighting to enter drains or water courses.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides. Hydrocarbons

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In the event of fire, wear self-contained breathing apparatus. Use personal protective equipment.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Keep people away from and upwind of spill/leak. Remove all sources of ignition. Do not breathe vapours or spray mist.

6.2 Environmental precautions:

Prevent product from entering drains. Prevent further leakage or spillage if safe to do so.

6.3 Methods and materials for containment and cleaning up:

Keep in suitable, closed containers for disposal.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.



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Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Ensure adequate ventilation. Do not breathe vapours or spray mist. Barrier creams may help to protect the exposed areas of skin, they should however not be applied once exposure has occurred. Wear personal protective equipment.

Measures to prevent fire as well as aerosol and dust generation

Keep away from open flames, hot surfaces and sources of ignition.



7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep containers tightly closed in a cool, well-ventilated place.

Consideration of other advice:

Keep away from oxidizing agents, and acidic or alkaline products.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

Engineering measures: Provide sufficient air exchange and/or exhaust in work rooms.

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



a. Hand protection

Impervious gloves



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Hygiene measures: Wash hands before breaks and immediately after handling the product.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Black dark green Dark brown
Odour	Petroleum
Melting point/freezing point	No data available
Boiling point or initial boiling point and boiling range	> 450 °C
Flammability	No data available
Lower and upper explosion limit	No data available
Flash point	90 °C
Auto-ignition temperature	No data available
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	30 - 70 mm ² /s; 50 °C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	No data available
<u>Density and/or relative density</u>	
Density	0.94 g/cm ³ ; 20 °C
Relative vapour density	No data available
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Stable under normal conditions.



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10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

None known

10.4 Conditions to avoid

Heat, flames and sparks.

10.5 Incompatible materials

Strong acids and oxidizing agents

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
No data available
- **If in eyes**
No data available
- **If inhaled**
No data available
- **If on skin**
No data available
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	4.8 mg/l/4h	rat
Oral	LD50	> 2000 mg/kg	rat
Dermal	LD50	>2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Moderately irritating

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer



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- 11.1.5 Mutagenicity:** Shall not be classified as germ cell mutagenic.
- 11.1.6 Carcinogenicity:** Shall not be classified as carcinogenic.
- 11.1.7 Reproductive Toxicity:** No data available
- 11.1.8 Specific target organ toxicity-single exposure:** No data available.
- 11.1.9 Specific target organ toxicity - repeated exposure:** No data available
- 11.1.10 Aspiration Hazard:** No data available
- 11.1.11 Additional Information:** There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (acute)

Endpoint	Value	Species	Source	Exposure time
LL50	12.51 mg/l	fish	ECHA	96 h
EL50	21.85 mg/l	aquatic invertebrates	ECHA	48 h

12.2 Persistence and Degradability: Expected to be biodegradable

12.3 Bio Accumulative Potential: No data available

12.4 Mobility in Soil: No data available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of in accordance with local regulations.



Section 14: Transport Information

14.1 UN Number: UN 3082

14.2 UN Proper Shipping Name: LSHS

14.3 Transport Class(es): Not Classified

14.4 Packaging Group: III

14.5 Environmental Hazards: Yes

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.



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Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.28 MSDS of IFO-Burn Grade

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	IFO-Burn Grade
1.1.1.1 Trade Name	Fuel Oil
1.1.2 CAS No.	68476-33-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Industrial use, Professional use
Uses advised against	Not Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Acute Toxicity	4	Acute Tox. 4	H332
Carcinogenicity	1B	Carc. 1B	H350
Reproductive toxicity	2	Repr. 2	H361d
Hazardous to the aquatic environment	1	Aquatic Acute 1	H400
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	1	Aquatic Chronic 1	H410

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS07

GHS08

GHS09



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2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H332	Harmful if inhaled
H304	May be fatal if swallowed and enters airways
H350	May cause cancer
H400	Very toxic to aquatic life
H361d	Suspected of damaging the unborn child.
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H410	Very toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P201	Obtain special instructions before use.
P260	Do not breathe dust/fume/gas/mist/vapours/spray.
P273	Avoid release to the environment
P281	Use personal protective equipment as required.

Precautionary statements – response

P301+P310	IF SWALLOWED: Immediately call a POISON CENTER/doctor/.
P331	Do NOT induce vomiting.
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Fuel Oil

3.1.1 Systematic Chemical Name or Trivial Name: IFO-Burn Grade

3.1.2 CAS No.: 68476-33-5

3.1.3 Other Identifiers

Molar mass 221-238 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



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4.1.1 Inhalation

Remove person to fresh air and keep comfortable for breathing. When in doubt or if symptoms are observed, get medical advice. If breathing is irregular or stopped, administer artificial respiration. Get immediate medical advice/attention.

4.1.2 Skin Contact

Take off contaminated clothing. Gently wash with plenty of soap and water. When in doubt or if symptoms are observed, get medical advice. In the event of a high pressure injection injury, worker should obtain immediate medical assistance. Contact with hot product will cause thermal burns. Immerse in cool water/wrap in wet bandages. Get medical advice/attention.

4.1.3 Eye Contact

Rinse immediately carefully and thoroughly with eye-bath or water. When in doubt or if symptoms are observed, get medical advice. Get medical advice/attention.

4.1.4 Ingestion

Rinse mouth thoroughly with water. Do NOT induce vomiting. Get immediate medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Dizziness, Drowsiness, Narcosis, Erythema, Nausea, Vomiting and Diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, alcohol resistant foam, Dry extinguishing powder, Carbon dioxide, inert gas, Sand, Earth.

- **Unsuitable extinguishing media**

Strong water jet

5.2 Special hazards arising from the chemical product:

Combustible. Heating causes rise in pressure with risk of bursting.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides, Organic compounds, (As appropriate: Sulphur oxides, Hydrogen sulphide (H₂S), Sulphuric acid)



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5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire: Wear self-contained breathing apparatus. Use water spray jet to protect personnel and to cool endangered containers. Use foam on spills to minimise vapours. Evacuate area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Do not allow run-off from fire-fighting to enter drains or water courses. Dispose according to legislation.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Personal protection equipment: see section 8 Do not breathe vapour/spray. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Ensure that the equipment is adequately grounded. Avoid contact with skin, eyes and clothes. Use explosion-proof machinery, apparatus, ventilation facilities, tools etc. Use only non-sparking tools. As appropriate: Product may release Hydrogen Sulphide: A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

Advice for emergency personnel: Ensure procedures and training for emergency decontamination and disposal are in place.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Absorb with liquid-binding material (e.g.sand, diatomaceous earth, acid- or universal binding agents). Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of waste product or used containers according to local regulations.

6.4 Reference to other sections:



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Personal protection equipment: see section 8 Disposal: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provide adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.



Take any precaution to avoid mixing with incompatible materials.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep in a dry, cool and well-ventilated place.

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

Bund storage facilities to prevent soil and water pollution in the event of spillage.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters: Not applicable

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.



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- **Skin protection**



- a. **Hand protection**

- Wear chemically resistant gloves (tested to EN374)

- b. **Type of material**

- NBR (Nitrile rubber)

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Half-face mask (EN 140) Full face mask (EN 136) Filter type: ABEK / P (EN 141) The filter class must be suitable for the maximum contaminant concentration



(gas/vapour/aerosol/particulates) that may arise when handling the product. If the concentration is exceeded, self-contained breathing apparatus must be used. (EN 137)

8.2.3 Environmental Exposure Controls:

Comply with applicable Community environmental protection legislation. Do not allow contact with soil, surface or ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Brown to black colour
Odour	Characteristic smell (Diesel like)
Melting point/freezing point	-1 - 13 °C
Boiling point or initial boiling point and boiling range	185 – 500 °C
Flammability	Flammable
Lower and upper explosion limit	1 vol% (LEL) – 5 vol% (UEL)
Flash point	66 °C
Pour point	30 °C
Auto-ignition temperature	263-407° C
Decomposition temperature	No data available
pH (value)	Neutral
Kinematic viscosity	55-115 cSt @ 50°C
Solubility(ies)	
Water solubility	< 0.1 g/l (at 20 °C)
Partition coefficient	



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Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	> 5 hPa (at 20 °C)
<u>Density and/or relative density</u>	
Density	982 kg/m ³ (at 15 °C)
Relative vapour density	2.79 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Combustible

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

None under normal processing.

10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Oxidising substances

10.6 Hazardous decomposition products

Hazardous combustion products: Burning produces noxious and toxic fumes. Reference to other sections: 5.2

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not Classified
- **If in eyes**
Not Classified



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- **If inhaled**
Not Classified
- **If on skin**
Not Classified
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Oral	LD50	>2000 mg/kg	rat
Dermal	LD50	>2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Not classified

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: May cause cancer

11.1.7 Reproductive Toxicity: Suspected of damaging the unborn child.

11.1.8 Specific target organ toxicity-single exposure: Not classified

11.1.9 Specific target organ toxicity - repeated exposure: May cause damage to organs through prolonged or repeated exposure.

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Very toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	35 mg/l	Fish 1(Species: Pimephales promelas [flow-through])	96h
LC50	48 mg/l	Fish 2(Species: Brachydanio rerio [semi-static])	96h

12.2 Persistence and Degradability: No data available

12.3 Bio Accumulative Potential: No data available

12.4 Mobility in Soil: No data available

12.5 Other adverse effects: Data are not available



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Section 13: Disposal considerations

13.1 Waste Treatment Method: Handle with care. Do not allow contact with soil, surface or ground water. Dispose of empty containers and wastes safely.



- **Sewage disposal-relevant information**

Recycling is preferred to disposal or incineration. If recycling is not possible, eliminate in accordance with local valid waste disposal regulations.

13.1.1 Contaminated Packaging: Never use pressure to empty container. Do not pierce or burn, even after use. Handle contaminated packages in the same way as the substance itself. Dispose according to legislation.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3082

14.2 UN Proper Shipping Name: IFO

14.3 Transport Class(es): 9

14.3.1 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: No data available.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- Chemical hazard awareness
- Safety control measures including PPE
- Housekeeping, hygiene and chemical storage



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iv. Responding to any kind of emergency

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6.29 MSDS of IFO-180

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	IFO-180
1.1.1.1 Trade Name	Fuel Oil
1.1.2 CAS No.	68476-33-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Industrial use, Professional use
Uses advised against	Not Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Acute Toxicity	4	Acute Tox. 4	H332
Carcinogenicity	1B	Carc. 1B	H350
Reproductive toxicity	2	Repr. 2	H361d
Hazardous to the aquatic environment	1	Aquatic Acute 1	H400
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	1	Aquatic Chronic 1	H410

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS07

GHS08

GHS09



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2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H332	Harmful if inhaled
H304	May be fatal if swallowed and enters airways
H350	May cause cancer
H400	Very toxic to aquatic life
H361d	Suspected of damaging the unborn child.
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H410	Very toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P201	Obtain special instructions before use.
P260	Do not breathe dust/fume/gas/mist/vapours/spray.
P273	Avoid release to the environment
P281	Use personal protective equipment as required.

Precautionary statements – response

P301+P310	IF SWALLOWED: Immediately call a POISON CENTER/doctor/.
P331	Do NOT induce vomiting.
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Fuel Oil

3.1.1 Systematic Chemical Name or Trivial Name: IFO-180

3.1.2 CAS No.: 68476-33-5

3.1.3 Other Identifiers

Molar mass 221-238 g/mol



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Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Remove person to fresh air and keep comfortable for breathing. When in doubt or if symptoms are observed, get medical advice. If breathing is irregular or stopped, administer artificial respiration. Get immediate medical advice/attention.

4.1.2 Skin Contact

Take off contaminated clothing. Gently wash with plenty of soap and water. When in doubt or if symptoms are observed, get medical advice. In the event of a high pressure injection injury, worker should obtain immediate medical assistance. Contact with hot product will cause thermal burns. Immerse in cool water/wrap in wet bandages. Get medical advice/attention.

4.1.3 Eye Contact

Rinse immediately carefully and thoroughly with eye-bath or water. When in doubt or if symptoms are observed, get medical advice. Get medical advice/attention.

4.1.4 Ingestion

Rinse mouth thoroughly with water. Do NOT induce vomiting. Get immediate medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Dizziness, Drowsiness, Narcosis, Erythema, Nausea, Vomiting and Diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, alcohol resistant foam, Dry extinguishing powder, Carbon dioxide, inert gas, Sand, Earth.

- **Unsuitable extinguishing media**

Strong water jet

5.2 Special hazards arising from the chemical product:

Combustible. Heating causes rise in pressure with risk of bursting.



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5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides, Organic compounds, (As appropriate : Sulphur oxides, Hydrogen sulphide (H₂S), Sulphuric acid)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire: Wear self-contained breathing apparatus. Use water spray jet to protect personnel and to cool endangered containers. Use foam on spills to minimise vapours. Evacuate area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Do not allow run-off from fire-fighting to enter drains or water courses. Dispose according to legislation.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Personal protection equipment: see section 8 Do not breathe vapour/spray. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Ensure that the equipment is adequately grounded. Avoid contact with skin, eyes and clothes. Use explosion-proof machinery, apparatus, ventilation facilities, tools etc. Use only non-sparking tools. As appropriate: Product may release Hydrogen Sulphide: A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

Advice for emergency personnel: Ensure procedures and training for emergency decontamination and disposal are in place.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Absorb with liquid-binding material (e.g. sand, diatomaceous earth, acid- or universal binding agents). Collect in closed and suitable containers for disposal.



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Other information relating to spills and releases

Dispose of waste product or used containers according to local regulations.

6.4 Reference to other sections:

Personal protection equipment: see section 8 Disposal: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provide adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.



Take any precaution to avoid mixing with incompatible materials.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep in a dry, cool and well-ventilated place.

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

Bund storage facilities to prevent soil and water pollution in the event of spillage.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters: Not applicable

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:



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- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. **Hand protection**

Wear chemically resistant gloves (tested to EN374)

- b. **Type of material**

NBR (Nitrile rubber)

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Half-face mask (EN 140) Full face mask (EN 136) Filter type: ABEK / P (EN 141) The filter class must be suitable for the maximum contaminant concentration



(gas/vapour/aerosol/particulates) that may arise when handling the product. If the concentration is exceeded, self-contained breathing apparatus must be used. (EN 137)

8.2.3 Environmental Exposure Controls:

Comply with applicable Community environmental protection legislation. Do not allow contact with soil, surface or ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Brown to black colour
Odour	Characteristic smell (Diesel like)
Melting point/freezing point	-1 - 13 °C
Boiling point or initial boiling point and boiling range	185 – 500 °C
Flammability	Flammable
Lower and upper explosion limit	1 vol% (LEL) – 5 vol% (UEL)
Flash point	66 °C
Pour point	27 °C
Auto-ignition temperature	263-407° C



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Decomposition temperature	No data available
pH (value)	Neutral
Kinematic viscosity	125-180 cSt @ 50°C
<u>Solubility(ies)</u>	
Water solubility	< 0.1 g/l (at 20 °C)
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	> 5 hPa (at 20 °C)
<u>Density and/or relative density</u>	
Density	984.2 Kg/m ³ (at 15 °C)
Relative vapour density	2.79 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Combustible

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

None under normal processing.

10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Oxidising substances

10.6 Hazardous decomposition products

Hazardous combustion products: Burning produces noxious and toxic fumes. Reference to other sections: 5.2



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Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not Classified
- **If in eyes**
Not Classified
- **If inhaled**
Not Classified
- **If on skin**
Not Classified
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Oral	LD50	>2000 mg/kg	rat
Dermal	LD50	>2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Not classified

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: May cause cancer

11.1.7 Reproductive Toxicity: Suspected of damaging the unborn child.

11.1.8 Specific target organ toxicity-single exposure: Not classified

11.1.9 Specific target organ toxicity - repeated exposure: May cause damage to organs through prolonged or repeated exposure.

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Very toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	35 mg/l	Fish 1(Species: Pimephales promelas [flow-through])	96h



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LC50	48 mg/l	Fish 2(Species: Brachydanio rerio [semi-static])	96h
12.2 Persistence and Degradability: No data available			
12.3 Bio Accumulative Potential: No data available			
12.4 Mobility in Soil: No data available			
12.5 Other adverse effects: Data are not available			

Section 13: Disposal considerations

13.1 Waste Treatment Method: Handle with care. Do not allow contact with soil, surface or ground water. Dispose of empty containers and wastes safely.



- **Sewage disposal-relevant information**

Recycling is preferred to disposal or incineration. If recycling is not possible, eliminate in accordance with local valid waste disposal regulations.

13.1.1 Contaminated Packaging: Never use pressure to empty container. Do not pierce or burn, even after use. Handle contaminated packages in the same way as the substance itself. Dispose according to legislation.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3082

14.2 UN Proper Shipping Name: IFO

14.3 Transport Class(es): 9

14.3.1 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: No data available.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable



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Section 16: Other Information

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16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.30 MSDS of IFO-380

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	IFO-380
1.1.1.1 Trade Name	Fuel Oil
1.1.2 CAS No.	68476-33-5

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Industrial use, Professional use
Uses advised against	Not Available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Acute Toxicity	4	Acute Tox. 4	H332
Carcinogenicity	1B	Carc. 1B	H350
Reproductive toxicity	2	Repr. 2	H361d
Hazardous to the aquatic environment	1	Aquatic Acute 1	H400
Specific target organ toxicity - repeated exposure	2	STOT RE 2	H373
Aspiration hazard	1	Asp. Tox. 1	H304
Hazardous to the aquatic environment - chronic hazard	1	Aquatic Chronic 1	H410

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS07

GHS08

GHS09



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2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H332	Harmful if inhaled
H304	May be fatal if swallowed and enters airways
H350	May cause cancer
H400	Very toxic to aquatic life
H361d	Suspected of damaging the unborn child.
H373	May cause damage to organs (nervous system) through prolonged or repeated exposure (if inhaled)
H410	Very toxic to aquatic life with long lasting effects

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P201	Obtain special instructions before use.
P260	Do not breathe dust/fume/gas/mist/vapours/spray.
P273	Avoid release to the environment
P281	Use personal protective equipment as required.

Precautionary statements – response

P301+P310	IF SWALLOWED: Immediately call a POISON CENTER/doctor/.
P331	Do NOT induce vomiting.
P308+P313	IF exposed or concerned: Get medical advice/attention

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Fuel Oil

3.1.1 Systematic Chemical Name or Trivial Name: IFO-380

3.1.2 CAS No.: 68476-33-5

3.1.3 Other Identifiers

Molar mass 221-238 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



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4.1.1 Inhalation

Remove person to fresh air and keep comfortable for breathing. When in doubt or if symptoms are observed, get medical advice. If breathing is irregular or stopped, administer artificial respiration. Get immediate medical advice/attention.

4.1.2 Skin Contact

Take off contaminated clothing. Gently wash with plenty of soap and water. When in doubt or if symptoms are observed, get medical advice. In the event of a high pressure injection injury, worker should obtain immediate medical assistance. Contact with hot product will cause thermal burns. Immerse in cool water/wrap in wet bandages. Get medical advice/attention.

4.1.3 Eye Contact

Rinse immediately carefully and thoroughly with eye-bath or water. When in doubt or if symptoms are observed, get medical advice. Get medical advice/attention.

4.1.4 Ingestion

Rinse mouth thoroughly with water. Do NOT induce vomiting. Get immediate medical advice/attention.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Dizziness, Drowsiness, Narcosis, Erythema, Nausea, Vomiting and Diarrhoea

4.3 Advice for immediate medical attention and special treatment needed

None

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**

Water spray, alcohol resistant foam, Dry extinguishing powder, Carbon dioxide, inert gas, Sand, Earth.

- **Unsuitable extinguishing media**

Strong water jet

5.2 Special hazards arising from the chemical product:

Combustible. Heating causes rise in pressure with risk of bursting.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon oxides, Organic compounds, (As appropriate: Sulphur oxides, Hydrogen sulphide (H₂S), Sulphuric acid)



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5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire: Wear self-contained breathing apparatus. Use water spray jet to protect personnel and to cool endangered containers. Use foam on spills to minimise vapours. Evacuate area. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Do not allow run-off from fire-fighting to enter drains or water courses. Dispose according to legislation.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Stay upwind/keep distance from source. Provide adequate ventilation. Use personal protective equipment as required. Personal protection equipment: see section 8 Do not breathe vapour/spray. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Ensure that the equipment is adequately grounded. Avoid contact with skin, eyes and clothes. Use explosion-proof machinery, apparatus, ventilation facilities, tools etc. Use only non-sparking tools. As appropriate: Product may release Hydrogen Sulphide: A specific assessment of inhalation risks from the presence of hydrogen sulphide in tank headspaces, confined spaces, product residue, tank waste and waste water, and unintentional releases should be made to help determine controls appropriate to local circumstances.

Advice for emergency personnel: Ensure procedures and training for emergency decontamination and disposal are in place.

6.2 Environmental precautions:

Do not allow to enter into ground-water, surface water or drains. If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Dam up.

Advice on how to clean up a spill

Absorb with liquid-binding material (e.g. sand, diatomaceous earth, acid- or universal binding agents). Collect in closed and suitable containers for disposal.

Other information relating to spills and releases

Dispose of waste product or used containers according to local regulations.

6.4 Reference to other sections:



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Personal protection equipment: see section 8 Disposal: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Provide adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.



Take any precaution to avoid mixing with incompatible materials.

Measures to protect the environment

Avoid release to the environment.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep in a dry, cool and well-ventilated place.

Incompatible substances or mixtures

Oxidising substances

Consideration of other advice:

Bund storage facilities to prevent soil and water pollution in the event of spillage.

Ventilation requirements

Use local and general ventilation.

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters: Not applicable

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.



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- **Skin protection**



- a. **Hand protection**

- Wear chemically resistant gloves (tested to EN374)

- b. **Type of material**

- NBR (Nitrile rubber)

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment. Half-face mask (EN 140) Full face mask (EN 136) Filter type: ABEK / P (EN 141) The filter class must be suitable for the maximum contaminant concentration



(gas/vapour/aerosol/particulates) that may arise when handling the product. If the concentration is exceeded, self-contained breathing apparatus must be used. (EN 137)

8.2.3 Environmental Exposure Controls:

Comply with applicable Community environmental protection legislation. Do not allow contact with soil, surface or ground water.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	Brown to black colour
Odour	Characteristic smell (Diesel like)
Melting point/freezing point	-1 - 13 °C
Boiling point or initial boiling point and boiling range	185 – 500 °C
Flammability	Flammable
Lower and upper explosion limit	1 vol% (LEL) – 5 vol% (UEL)
Flash point	60 °C
Pour point	30°C
Auto-ignition temperature	263-407° C
Decomposition temperature	No data available
pH (value)	Neutral
Kinematic viscosity	380 mm ² /s @ 50 °C
<u>Solubility(ies)</u>	
Water solubility	< 0.1 g/l (at 20 °C)



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<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	> 5 hPa (at 20 °C)
<u>Density and/or relative density</u>	
Density	991 kg/m ³ (at 15 °C)
Relative vapour density	2.79 (air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:
Information with regard to physical hazard classes:
There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Combustible

10.2 Chemical stability

The product is stable under storage at normal ambient temperatures.

10.3 Possibility of hazardous reactions

None under normal processing.

10.4 Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials

Oxidising substances

10.6 Hazardous decomposition products

Hazardous combustion products: Burning produces noxious and toxic fumes. Reference to other sections: 5.2

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
Not Classified



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- **If in eyes**
Not Classified
- **If inhaled**
Not Classified
- **If on skin**
Not Classified
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Oral	LD50	>2000 mg/kg	rat
Dermal	LD50	>2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Not classified

11.1.3 Serious Eye Damage/ Eye Irritation: Shall not be classified as seriously damaging to the eye or eye irritant.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: May cause cancer

11.1.7 Reproductive Toxicity: Suspected of damaging the unborn child.

11.1.8 Specific target organ toxicity-single exposure: Not classified

11.1.9 Specific target organ toxicity - repeated exposure: May cause damage to organs through prolonged or repeated exposure.

11.1.10 Aspiration Hazard: May be fatal if swallowed and enters airways.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Very toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	35 mg/l	Fish 1(Species: Pimephales promelas [flow-through])	96h
LC50	48 mg/l	Fish 2(Species: Brachydanio rerio [semi-static])	96h

12.2 Persistence and Degradability: No data available

12.3 Bio Accumulative Potential: No data available



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12.4 Mobility in Soil: No data available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Handle with care. Do not allow contact with soil, surface or ground water. Dispose of empty containers and wastes safely.



- **Sewage disposal-relevant information**

Recycling is preferred to disposal or incineration. If recycling is not possible, eliminate in accordance with local valid waste disposal regulations.

13.1.1 Contaminated Packaging: Never use pressure to empty container. Do not pierce or burn, even after use. Handle contaminated packages in the same way as the substance itself. Dispose according to legislation.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3082

14.2 UN Proper Shipping Name: IFO

14.3 Transport Class(es): 9

14.3.1 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: No data available.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023



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16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.31 MSDS of ASPHALT VG 10

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Bitumen VG 10
1.1.1.1 Trade Name	Asphalt
1.1.2 CAS No.	8052-42-4

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	road and highway paving applications waterproofing and sealing applications coatings fuel Lubricants
Uses advised against	None

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Carcinogenicity	2	Carc. 2	H351

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H351 Suspected of causing cancer



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2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P202	Do not handle until all safety precautions have been read and understood
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P280	Wear protective gloves/eye protection
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Precautionary statements – response

P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
------------------	---

P308+P313	IF exposed or concerned: Get medical advice/attention
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2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Asphalt

3.1.1 Systematic Chemical Name or Trivial Name: Bitumen VG 10

3.1.2 CAS No.: 8052-42-4

3.1.3 Other Identifiers

Molar mass	500-900 g/mol
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Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Get medical attention.

4.1.2 Skin Contact

In case of contact with hot or molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily. Remove contaminated clothing and shoes. Get medical attention if irritation develops or persists. Wash clothing separately before reuse. Destroy or thoroughly clean contaminated shoes. If high pressure injection under the skin occurs, always seek medical attention.

4.1.3 Eye Contact



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Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.

4.1.4 Ingestion

Rinse mouth thoroughly. Do not induce vomiting without advice from poison control center. Do not give mouth-to-mouth resuscitation. If vomiting occurs, keep head low so that stomach content does not get into the lungs. Never give anything by mouth to a victim who is unconscious or is having convulsions. Get medical attention immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis, Edema, Conjunctivitis, Proteinuria

4.3 Advice for immediate medical attention and special treatment needed

In case of shortness of breath, give oxygen. Keep victim warm. Keep victim under observation. Symptoms may be delayed.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray. Water fog. Dry chemical powder. Carbon dioxide (CO₂).
- **Unsuitable extinguishing media**
Water jet

5.2 Special hazards arising from the chemical product:

Thermal decomposition or combustion may liberate toxic gases or fumes.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Wear full protective clothing, including helmet, self-contained positive pressure or pressure demand breathing apparatus, protective clothing and face mask. Withdraw immediately in case of rising sound from venting safety devices or any discoloration of tanks due to fire. Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Move containers from fire area if you can do it without risk. In the event of fire, cool tanks with water spray. Cool containers exposed to flames with water until well after the fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn. Water runoff can cause environmental damage. Use compatible foam to minimize vapor generation as needed.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Keep unnecessary personnel away. Local authorities should be advised if significant spills cannot be contained. Keep upwind. Keep out of low areas. Ventilate closed spaces before entering. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.

6.2 Environmental precautions:

If facility or operation has an "oil or hazardous substance contingency plan", activate its procedures. Stay upwind and away from spill. Wear appropriate protective equipment including respiratory protection as conditions warrant. Do not enter or stay in area unless monitoring indicates that it is safe to do so. Isolate hazard area and restrict entry to emergency crew.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible.

Advice on how to clean up a spill

Do not allow material to contaminate ground water system. Clean surface thoroughly to remove residual contamination. Wipe up with absorbent material (e.g. cloth, fleece).

Other information relating to spills and releases

Keep combustibles (wood, paper, oil, etc.) away from spilled material.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Avoid prolonged exposure. Use only with adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Do not handle, store or open near an open flame or sources of ignition.



Take precautionary measures against static discharge.

Measures to protect the environment

Avoid release to the environment.



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7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed in a cool, well-ventilated place.

Incompatible substances or mixtures

Strong oxidizing agents

Consideration of other advice:

All equipment used when handling the product must be grounded. Use only non-sparking tools.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 120 – 190 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
US	Asphalt	8052-42-4	ACGIH	0.025	0.5			0.025	0.5		

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
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DNEL	2.9 mg/m ³	human, inhalatory	worker (industry)	Long term systemic effects
DNEL: Derived No Effect Level				
8.2 Exposure Controls				
8.2.1 Personal Protective Equipment: Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.				
8.2.2 Suggestions for protection of eye, skin and body and respiratory system: • Eye/face protection  Use safety goggle with side protection. • Skin protection  a. Hand protection Thick, thermally insulated protective gloves. Change protective gloves regularly. Protective gloves according to standards EN 374 and EN 407. b. Type of material Polyvinyl alcohol (PVA), Fluoro-elastomer c. Other protection measures Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. • Respiratory protection: Respiratory protection necessary at: Bitumen fumes. Type: A1/P2 				
8.2.3 Environmental Exposure Controls: Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation.				

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties



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Physical state	Liquid
Colour	Brown/black
Odour	like: - Strong petroleum
Melting point/freezing point	37.78 - 65.56 °C
Boiling point or initial boiling point and boiling range	371.11 - 593.39 °C
Flammability	Not available
Lower and upper explosion limit	Not considered to be explosive
Flash point	220°C
Softening point	40°C
Auto-ignition temperature	> 315.61 °C
Decomposition temperature	not relevant
pH (value)	not determined
Absolute viscosity	800-1200 Poises @ 60°C
Kinematic viscosity	250 cst @ 135°C
Viscosity ratio	4 @ 60°C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not available
Soil organic carbon/water (log KOC)	Not available
Vapour pressure	< 0.01 kPa @ 20 °C
<u>Density and/or relative density</u>	
Density	~ 0.93 – 1.10 @ 15°C
Relative vapour density	> 1.6 (Air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

There are no known reactivity hazards associated with this product

10.2 Chemical stability

Stable at normal ambient temperatures and when used as recommended.



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10.3 Possibility of hazardous reactions

No potentially hazardous reactions known.

10.4 Conditions to avoid

Excessive heating above the maximum recommended handling and storage temperature may cause degradation of the substance and evolution of irritant vapours and fumes.

10.5 Incompatible materials

Oxidising agents.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
nausea, vomiting, aspiration hazard
- **If in eyes**
slightly irritant but not relevant for classification
- **If inhaled**
irritant effects, headache, vertigo, fatigue, dizziness, narcosis
- **If on skin**
causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	> 94.4 mg/m ³	rat	OECD 403
Oral	LD50	> 5000 mg/kg	rat	OECD 401
Dermal	LD50	>2000 mg/kg	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: May cause eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Based on available data, the classification criteria are not met.



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11.1.8 Specific target organ toxicity-single exposure: Based on available data, the classification criteria are not met.

11.1.9 Specific target organ toxicity - repeated exposure

Based on available data, the classification criteria are not met.

11.1.10 Aspiration Hazard: Based on available data, the classification criteria are not met.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (Acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 1000 mg/l	fish	QSAR	96 h
LL50	> 1000 mg/l	aquatic invertebrates	QSAR	48 h
EL50	> 1000 mg/l	aquatic plants	QSAR	72 hours
LL50	> 1000 mg/l	Microorganisms	QSAR	40 hours

Aquatic toxicity (Chronic)

LL50	> 1000 mg/l	fish early life stage	QSAR	28 days
NOEL	≥ 1000 mg/l	aquatic invertebrates	QSAR	21 days

12.2 Persistence and Degradability: Not available.

12.3 Bio Accumulative Potential: Not available.

12.4 Mobility in Soil: Not available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- Sewage disposal-relevant information**

Do not contaminate ponds, waterways or ditches with chemical or used container.



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13.1.1 Contaminated Packaging: Offer rinsed packaging material to local recycling facilities.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3257

14.2 UN Proper Shipping Name: Elevated temperature liquid

14.3 Transport Class(es): 9

14.3.1 Hazard Labels: 9

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.32 MSDS of ASPHALT VG 30

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Bitumen VG 30
1.1.1.1 Trade Name	Asphalt
1.1.2 CAS No.	8052-42-4

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	road and highway paving applications waterproofing and sealing applications coatings fuel Lubricants
Uses advised against	None

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Carcinogenicity	2	Carc. 2	H351

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H351 Suspected of causing cancer



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2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P202	Do not handle until all safety precautions have been read and understood
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P280	Wear protective gloves/eye protection
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Precautionary statements – response

P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
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P308+P313	IF exposed or concerned: Get medical advice/attention
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2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Asphalt

3.1.1 Systematic Chemical Name or Trivial Name: Bitumen VG 30

3.1.2 CAS No.: 8052-42-4

3.1.3 Other Identifiers

Molar mass	500-900 g/mol
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Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Get medical attention.

4.1.2 Skin Contact

In case of contact with hot or molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily. Remove contaminated clothing and shoes. Get medical attention if irritation develops or persists. Wash clothing separately before reuse. Destroy or thoroughly clean contaminated shoes. If high pressure injection under the skin occurs, always seek medical attention.

4.1.3 Eye Contact



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Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.

4.1.4 Ingestion

Rinse mouth thoroughly. Do not induce vomiting without advice from poison control center. Do not give mouth-to-mouth resuscitation. If vomiting occurs, keep head low so that stomach content does not get into the lungs. Never give anything by mouth to a victim who is unconscious or is having convulsions. Get medical attention immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis, Edema, Conjunctivitis, Proteinuria

4.3 Advice for immediate medical attention and special treatment needed

In case of shortness of breath, give oxygen. Keep victim warm. Keep victim under observation. Symptoms may be delayed.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray. Water fog. Dry chemical powder. Carbon dioxide (CO₂).
- **Unsuitable extinguishing media**
Water jet

5.2 Special hazards arising from the chemical product:

Thermal decomposition or combustion may liberate toxic gases or fumes.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Wear full protective clothing, including helmet, self-contained positive pressure or pressure demand breathing apparatus, protective clothing and face mask. Withdraw immediately in case of rising sound from venting safety devices or any discoloration of tanks due to fire. Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Move containers from fire area if you can do it without risk. In the event of fire, cool tanks with water spray. Cool containers exposed to flames with water until well after the fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn. Water runoff can cause environmental damage. Use compatible foam to minimize vapor generation as needed.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Keep unnecessary personnel away. Local authorities should be advised if significant spills cannot be contained. Keep upwind. Keep out of low areas. Ventilate closed spaces before entering. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.

6.2 Environmental precautions:

If facility or operation has an "oil or hazardous substance contingency plan", activate its procedures. Stay upwind and away from spill. Wear appropriate protective equipment including respiratory protection as conditions warrant. Do not enter or stay in area unless monitoring indicates that it is safe to do so. Isolate hazard area and restrict entry to emergency crew.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible.

Advice on how to clean up a spill

Do not allow material to contaminate ground water system. Clean surface thoroughly to remove residual contamination. Wipe up with absorbent material (e.g. cloth, fleece).

Other information relating to spills and releases

Keep combustibles (wood, paper, oil, etc.) away from spilled material.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Avoid prolonged exposure. Use only with adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Do not handle, store or open near an open flame or sources of ignition.



Take precautionary measures against static discharge.

Measures to protect the environment

Avoid release to the environment.



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7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed in a cool, well-ventilated place.

Incompatible substances or mixtures

Strong oxidizing agents

Consideration of other advice:

All equipment used when handling the product must be grounded. Use only non-sparking tools.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 120 – 190 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling-C [ppm]	Ceiling-C [mg/m ³]	Notation	Source
US	Asphalt	8052-42-4	ACGIH	0.025	0.5			0.025	0.5		

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
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DNEL	2.9 mg/m ³	human, inhalatory	worker (industry)	Long term systemic effects
DNEL: Derived No Effect Level				
8.2 Exposure Controls				
8.2.1 Personal Protective Equipment:				
Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.				
8.2.2 Suggestions for protection of eye, skin and body and respiratory system:				
• Eye/face protection				
				
Use safety goggle with side protection.				
• Skin protection				
				
a. Hand protection				
Thick, thermally insulated protective gloves. Change protective gloves regularly. Protective gloves according to standards EN 374 and EN 407.				
b. Type of material				
Polyvinyl alcohol (PVA), Fluoro-elastomer				
c. Other protection measures				
Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.				
• Respiratory protection: Respiratory protection necessary at: Bitumen fumes. Type: A1/P2				
				
8.2.3 Environmental Exposure Controls:				
Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation.				

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties



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Physical state	Liquid
Colour	Brown/black
Odour	like: - Strong petroleum
Melting point/freezing point	37.78 - 65.56 °C
Boiling point or initial boiling point and boiling range	371.11 - 593.39 °C
Flammability	Not available
Lower and upper explosion limit	Not considered to be explosive
Flash point	220°C
Softening point	47°C
Auto-ignition temperature	> 315.61 °C
Decomposition temperature	not relevant
pH (value)	not determined
Absolute viscosity	2400-3600 Poises @ 60°C
Kinematic viscosity	350 cst @ 135°C
Viscosity ratio	4 @ 60°C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not available
Soil organic carbon/water (log KOC)	Not available
Vapour pressure	< 0.01 kPa @ 20 °C
<u>Density and/or relative density</u>	
Density	~ 0.93 – 1.10 @ 15°C
Relative vapour density	> 1.6 (Air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

There are no known reactivity hazards associated with this product

10.2 Chemical stability

Stable at normal ambient temperatures and when used as recommended.



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10.3 Possibility of hazardous reactions

No potentially hazardous reactions known.

10.4 Conditions to avoid

Excessive heating above the maximum recommended handling and storage temperature may cause degradation of the substance and evolution of irritant vapours and fumes.

10.5 Incompatible materials

Oxidising agents.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
nausea, vomiting, aspiration hazard
- **If in eyes**
slightly irritant but not relevant for classification
- **If inhaled**
irritant effects, headache, vertigo, fatigue, dizziness, narcosis
- **If on skin**
causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	> 94.4 mg/m ³	rat	OECD 403
Oral	LD50	> 5000 mg/kg	rat	OECD 401
Dermal	LD50	>2000 mg/kg	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: May cause eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.



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11.1.7 Reproductive Toxicity: Based on available data, the classification criteria are not met.

11.1.8 Specific target organ toxicity-single exposure: Based on available data, the classification criteria are not met.

11.1.9 Specific target organ toxicity - repeated exposure
Based on available data, the classification criteria are not met.

11.1.10 Aspiration Hazard: Based on available data, the classification criteria are not met.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (Acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 1000 mg/l	fish	QSAR	96 h
LL50	> 1000 mg/l	aquatic invertebrates	QSAR	48 h
EL50	> 1000 mg/l	aquatic plants	QSAR	72 h
LL50	> 1000 mg/l	Microorganisms	QSAR	40 h

Aquatic toxicity (Chronic)

LL50	> 1000 mg/l	fish early life stage	QSAR	28 days
NOEL	≥ 1000 mg/l	aquatic invertebrates	QSAR	21 days

12.2 Persistence and Degradability: Not available.

12.3 Bio Accumulative Potential: Not available.

12.4 Mobility in Soil: Not available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- **Sewage disposal-relevant information**



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Do not contaminate ponds, waterways or ditches with chemical or used container.

13.1.1 Contaminated Packaging: Offer rinsed packaging material to local recycling facilities.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 3257

14.2 UN Proper Shipping Name: Elevated temperature liquid

14.3 Transport Class(es): 9

14.3.1 Hazard Labels: 9

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.33 MSDS of ASPHALT VG 40

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Bitumen VG 40
1.1.1.1 Trade Name	Asphalt
1.1.2 CAS No.	8052-42-4

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	road and highway paving applications waterproofing and sealing applications coatings fuel Lubricants
Uses advised against	None

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Carcinogenicity	2	Carc. 2	H351

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS08

2.2.2 Signal Word(s): Danger

2.2.3 Hazard Statements(s):

H351 Suspected of causing cancer



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2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P202	Do not handle until all safety precautions have been read and understood
-------------	--

P280	Wear protective gloves/eye protection
-------------	---------------------------------------

Precautionary statements – response

P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
------------------	---

P308+P313	IF exposed or concerned: Get medical advice/attention
------------------	---

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Asphalt

3.1.1 Systematic Chemical Name or Trivial Name: Bitumen VG 40

3.1.2 CAS No.: 8052-42-4

3.1.3 Other Identifiers

Molar mass	500-900 g/mol
-------------------	---------------

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

Move to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Get medical attention.

4.1.2 Skin Contact

In case of contact with hot or molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily. Remove contaminated clothing and shoes. Get medical attention if irritation develops or persists. Wash clothing separately before reuse. Destroy or thoroughly clean contaminated shoes. If high pressure injection under the skin occurs, always seek medical attention.

4.1.3 Eye Contact



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Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.

4.1.4 Ingestion

Rinse mouth thoroughly. Do not induce vomiting without advice from poison control center. Do not give mouth-to-mouth resuscitation. If vomiting occurs, keep head low so that stomach content does not get into the lungs. Never give anything by mouth to a victim who is unconscious or is having convulsions. Get medical attention immediately.

4.2 Most important symptoms and anticipated effects both acute and delayed

Aspiration hazard, Irritation, Dizziness, Drowsiness, Narcosis, Edema, Conjunctivitis, Proteinuria

4.3 Advice for immediate medical attention and special treatment needed

In case of shortness of breath, give oxygen. Keep victim warm. Keep victim under observation. Symptoms may be delayed.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- **Suitable extinguishing media**
Water spray. Water fog. Dry chemical powder. Carbon dioxide (CO₂).
- **Unsuitable extinguishing media**
Water jet

5.2 Special hazards arising from the chemical product:

Thermal decomposition or combustion may liberate toxic gases or fumes.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Carbon dioxide (CO₂)

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

Wear full protective clothing, including helmet, self-contained positive pressure or pressure demand breathing apparatus, protective clothing and face mask. Withdraw immediately in case of rising sound from venting safety devices or any discoloration of tanks due to fire. Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Move containers from fire area if you can do it without risk. In the event of fire, cool tanks with water spray. Cool containers exposed to flames with water until well after the fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn. Water runoff can cause environmental damage. Use compatible foam to minimize vapor generation as needed.

5.4.2 Special protective equipment for fire-fighters: None



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Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Advice for non-emergency personnel: Keep unnecessary personnel away. Local authorities should be advised if significant spills cannot be contained. Keep upwind. Keep out of low areas. Ventilate closed spaces before entering. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.

6.2 Environmental precautions:

If facility or operation has an "oil or hazardous substance contingency plan", activate its procedures. Stay upwind and away from spill. Wear appropriate protective equipment including respiratory protection as conditions warrant. Do not enter or stay in area unless monitoring indicates that it is safe to do so. Isolate hazard area and restrict entry to emergency crew.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible.

Advice on how to clean up a spill

Do not allow material to contaminate ground water system. Clean surface thoroughly to remove residual contamination. Wipe up with absorbent material (e.g. cloth, fleece).

Other information relating to spills and releases

Keep combustibles (wood, paper, oil, etc.) away from spilled material.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Avoid prolonged exposure. Use only with adequate ventilation.

Measures to prevent fire as well as aerosol and dust generation

Do not handle, store or open near an open flame or sources of ignition.



Take precautionary measures against static discharge.

Measures to protect the environment

Avoid release to the environment.



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7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Keep container tightly closed in a cool, well-ventilated place.

Incompatible substances or mixtures

Strong oxidizing agents

Consideration of other advice:

All equipment used when handling the product must be grounded. Use only non-sparking tools.

Ventilation requirements

Use local and general ventilation.

Specific designs for storage rooms or vessels

Recommended storage temperature: 120 – 190 °C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m ³]	STEL [ppm]	STEL [mg/m ³]	Ceiling - C [ppm]	Ceiling - C [mg/m ³]	Notation	Source
US	Asphalt	8052-42-4	ACGIH	0.025	0.5			0.025	0.5		

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

Human health values

Relevant DNELs and other threshold levels

Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
DNEL	2.9 mg/m ³	human, inhalatory	worker (industry)	Long term systemic effects



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DNEL: Derived No Effect Level

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

Use of PPE to be ensured by working crew for exposure control. Different PPEs for exposure control of different body parts is described in the next section.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



- a. Hand protection**

Thick, thermally insulated protective gloves. Change protective gloves regularly. Protective gloves according to standards EN 374 and EN 407.

- b. Type of material**

Polyvinyl alcohol (PVA), Fluoro-elastomer

- c. Other protection measures**

Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

- **Respiratory protection:** Respiratory protection necessary at: Bitumen fumes. Type: A1/P2



8.2.3 Environmental Exposure Controls:

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation.

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid
Colour	Brown/black



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Odour	like: - Strong petroleum
Melting point/freezing point	37.78 - 65.56 °C
Boiling point or initial boiling point and boiling range	371.11 - 593.39 °C
Flammability	Not available
Lower and upper explosion limit	Not considered to be explosive
Flash point	220°C
Softening point	50°C
Auto-ignition temperature	> 315.61 °C
Decomposition temperature	not relevant
pH (value)	not determined
Absolute viscosity	3200-4800 Poises @ 60°C
Kinematic viscosity	400 cst @ 135°C
Viscosity ratio	4 @ 60°C
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	Not available
Soil organic carbon/water (log KOC)	Not available
Vapour pressure	< 0.01 kPa @ 20 °C
<u>Density and/or relative density</u>	
Density	~ 0.93 – 1.10 @ 15°C
Relative vapour density	> 1.6 (Air = 1)
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:

There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

There are no known reactivity hazards associated with this product

10.2 Chemical stability

Stable at normal ambient temperatures and when used as recommended.

10.3 Possibility of hazardous reactions

No potentially hazardous reactions known.



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10.4 Conditions to avoid

Excessive heating above the maximum recommended handling and storage temperature may cause degradation of the substance and evolution of irritant vapours and fumes.

10.5 Incompatible materials

Oxidising agents.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed**
nausea, vomiting, aspiration hazard
- **If in eyes**
slightly irritant but not relevant for classification
- **If inhaled**
irritant effects, headache, vertigo, fatigue, dizziness, narcosis
- **If on skin**
causes skin irritation
- **Other information**
None

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species	Source
Inhalation	LC50	> 94.4 mg/m ³	rat	OECD 403
Oral	LD50	> 5000 mg/kg	rat	OECD 401
Dermal	LD50	>2000 mg/kg	rabbit	OECD 402

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: May cause eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Shall not be classified as germ cell mutagenic.

11.1.6 Carcinogenicity: Shall not be classified as carcinogenic.

11.1.7 Reproductive Toxicity: Based on available data, the classification criteria are not met.

11.1.8 Specific target organ toxicity-single exposure: Based on available data, the classification criteria are not met.

11.1.9 Specific target organ toxicity - repeated exposure



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Based on available data, the classification criteria are not met.

11.1.10 Aspiration Hazard: Based on available data, the classification criteria are not met.

11.1.11 Additional Information: There is no additional information.

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Aquatic toxicity (Acute)

Endpoint	Value	Species	Source	Exposure time
LL50	> 1000 mg/l	fish	QSAR	96 h
LL50	> 1000 mg/l	aquatic invertebrates	QSAR	48 h
EL50	> 1000 mg/l	aquatic plants	QSAR	72 h
LL50	> 1000 mg/l	Microorganisms	QSAR	40 h

Aquatic toxicity (Chronic)

LL50	> 1000 mg/l	fish early life stage	QSAR	28 days
NOEL	≥ 1000 mg/l	aquatic invertebrates	QSAR	21 days

12.2 Persistence and Degradability: Not available.

12.3 Bio Accumulative Potential: Not available.

12.4 Mobility in Soil: Not available

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.



- Sewage disposal-relevant information**

Do not contaminate ponds, waterways or ditches with chemical or used container.

13.1.1 Contaminated Packaging: Offer rinsed packaging material to local recycling facilities.

13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**



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Section 14: Transport Information

14.1 UN Number: UN 3257

14.2 UN Proper Shipping Name: Elevated temperature liquid

14.3 Transport Class(es): 9

14.3.1 Hazard Labels: 9

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.



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6.34 MSDS of SULPHUR

Section 1: Chemical Product and Company Identification

1.1 Chemical Product Identifiers:

1.1.1 Chemical Name	Sulphur
1.1.1.1 Trade Name	Molten Sulphur
1.1.2 CAS No.	7704-34-9

1.2 Relevant identified uses of the chemical product and uses advised against

Relevant identified uses	Industrial use.
Uses advised against	No data available

1.3 Details of the supplier of the safety data sheet

1.3.1 Company Name	Hindustan Petroleum Corporation Limited
1.3.2 Address	HPCL Mumbai Refinery, B D Patil Marg Chembur, Mumbai-400074
1.3.3 Contact No.	022-25076605
1.3.4 Contact person in emergency	Fire & Safety Shift Supervisor
1.4 Emergency Telephone No.:	022-25076606

Section 2: Hazards Identification

2.1 Classification of the chemical product:

Hazard class	Category	Hazard class & category	Hazard statement
Flammable solid	2	Flam. Solid 2	H228
Skin corrosion/irritation	2	Skin Irrit. 2	H315
Acute toxicity, inhalation	3	Acute Tox. 3	H331
Hazardous to the aquatic environment, acute hazard	3	Aquatic Chronic 3	H412
Specific target organ toxicity, single exposure	3	STOT SE 3	H335

2.2 Label Elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

2.2.1 GHS pictograms:



GHS02 GHS06 GHS07

2.2.2 Signal Word(s): Warning

2.2.3 Hazard Statements(s):



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H228 Flammable solid
H315 Causes skin irritation
H331 Toxic if inhaled
H412 Harmful to aquatic life with long lasting effects
H335 May cause respiratory irritation

2.2.4 Precautionary Statement(s):

Precautionary statements - prevention

P240	Ground/bond container and receiving equipment
P241	Use explosion-proof electrical, ventilating, and lighting equipment
P210	Keep away from heat/sparks/open flames/hot surfaces. No smoking.
P273	Avoid release to the environment.
P264	Wash hands, forearms, and other exposed areas thoroughly after handling
P280	Wear protective gloves, protective clothing, and eye protection

Precautionary statements – response

P302+P352	IF ON SKIN: Wash with plenty of water
P321	Specific treatment
P332+P313	If skin irritation occurs: Get medical advice/attention
P362+P364	Take off contaminated clothing and wash it before reuse
P370+P378	In case of fire: Use appropriate media (see section 5) to extinguish
P501	Dispose of contents/container in accordance with local, regional, national, territorial, provincial, and international regulations

2.3 Other hazards except those mentioned in DGCPL Rules, 201:

Results of PBT and vPvB assessment

Unleaded Motor Gasoline is extremely flammable liquid which may generate explosive mixtures of hydrocarbon vapours and air at ambient temperatures.

Section 3: Composition/Information on Ingredients

3.1 Name of Product: Sulphur

3.1.1 Systematic Chemical Name or Trivial Name: Molten Sulphur

3.1.2 CAS No.: 7704-34-9



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3.1.3 Other Identifiers

Molar mass

32 g/mol

Section 4: First-Aid Measures

4.1 Description of first aid measures



4.1.1 Inhalation

When symptoms occur: go into open air and ventilate suspected area. Obtain medical attention if breathing difficulty persists.

4.1.2 Skin Contact

If skin irritation occurs: Get medical advice/attention. Remove contaminated clothing. Drench affected area with water for at least 15 minutes. In molten form: Cool skin rapidly with cold water after contact with molten product, Removal of solidified molten material from skin requires medical assistance. Wash contaminated clothing before reuse.

4.1.3 Eye Contact

Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Obtain medical attention. Protect skin and eyes from contact with molten material.

4.1.4 Ingestion

Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Obtain medical attention. Protect skin and eyes from contact with molten material.

4.2 Most important symptoms and anticipated effects both acute and delayed

Irritation, Dizziness, Drowsiness, Cough, Mental Confusion Headache, Erythema, Pain in the nose

4.3 Advice for immediate medical attention and special treatment needed

If exposed or concerned, get medical advice and attention. If medical advice is needed, have product container or label at hand.

Section 5: Firefighting Measures

5.1 Extinguishing Media



- Suitable extinguishing media



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Dry chemical, carbon dioxide, foam, water spray

- **Unsuitable extinguishing media**

Apply aqueous extinguishing media carefully to prevent frothing/steam explosion. Do not use a heavy water stream. Use of heavy stream of water may spread fire.

5.2 Special hazards arising from the chemical product:

Flammable solid. Sulfur burns with a pale blue flame that may be difficult to see in daylight. Flammable vapors can accumulate in head space of closed systems.

5.3 Hazardous Combustion Products:

In case of fire may be liberated: Sulfur dioxide. Sulfur oxides

5.4 Specific Extinguishing Methods

For this substance/mixture no limitations of extinguishing agents are given

5.4.1 Advice for Fire-Fighters

In case of fire: Use water spray or fog for cooling exposed containers. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.

5.4.2 Special protective equipment for fire-fighters: None

Section 6: Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:



Avoid all contact with skin, eyes, or clothing. Avoid breathing dust, or vapors. Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. No smoking.

6.2 Environmental precautions:

Prevent entry to sewers and public waters. Avoid release to the environment.

6.3 Methods and materials for containment and cleaning up:

Advice on how to contain a spill

Contain solid spills with appropriate barriers and prevent migration and entry into sewers or streams. As an immediate precautionary measure, isolate spill or leak area in all directions.

Advice on how to clean up a spill

Clean up spills immediately and dispose of waste safely. If melted: allow liquid to solidify before taking it up. In solid form: Eliminate all ignition sources. Avoid generation of dust during clean-up of spills. Recover the product by vacuuming, shoveling or sweeping. Transfer spilled material to a suitable container for disposal. Contact competent authorities after a spill.

Other information relating to spills and releases

Dispose of waste product or used containers according to local regulations.

6.4 Reference to other sections:

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.



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Section 7: Handling and Storage

7.1 Precautions for Safe Handling:

Risk of thermal burns on contact with molten product. Flammable vapors can accumulate in head space of closed systems. If stored under heat for extended periods or significantly agitated, this material might evolve or release hydrogen sulfide, a flammable gas, which can raise and widen this material's actual flammability limits and significantly lower its auto-ignition temperature.

Measures to prevent fire as well as aerosol and dust generation

Sources of ignition, such as smoking and open flames, are prohibited.



Measures to protect the environment

Do not allow to enter into surface water or drains.

7.2 Conditions for Safe Storage, including requirement of hazardous area classification and any incompatibilities:

Store in a well-ventilated place. Keep container tightly closed. Store in a dry, cool place.

Incompatible substances or mixtures

Strong oxidizers. Under certain conditions: Sodium, tin, nickel, zinc

Consideration of other advice:

Use explosion-proof electrical, ventilating, and lighting equipment.

Ventilation requirements

Keep/Store away from direct sunlight, extremely high or low temperatures, ignition sources, and incompatible materials. Store in a well-ventilated place.

Specific designs for storage rooms or vessels

Recommended storage temperature: 127°C to 149°C

Section 8: Exposure Controls and Personal Protection

8.1 Control Parameters

National Limit Values

Occupational exposure limit values (Workplace Exposure Limits)

Country	Name of agent	CAS No	Identifier	TWA	TWA [mg/m ³]	STEL	STEL [mg/m ³]	Ceiling g-C	Ceiling g-C [ppm]	Notation	Source
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				[ppm l]		[ppm l]			[mg/ mS]		
Alber a	Sulf ur	7704 -34- 9	OEL	1		5					

Notation

- Ceiling-C: Ceiling value is a limit value above which exposure should not occur
- STEL: Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15- minute period (unless otherwise specified)
- TWA: Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours' time-weighted average (unless otherwise specified)

8.2 Exposure Controls

8.2.1 Personal Protective Equipment:

The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

8.2.2 Suggestions for protection of eye, skin and body and respiratory system:

- **Eye/face protection**



Use safety goggle with side protection.

- **Skin protection**



a. Hand protection

Wear chemically resistant protective gloves. If material is hot, wear thermally resistant protective gloves.

b. Type of material

Neoprene, PVC

c. Other protection measures

When using, do not eat, drink or smoke



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- **Respiratory protection:** If exposure limits are exceeded or irritation is experienced, approved respiratory protection should be worn. In case of inadequate ventilation, oxygen deficient atmosphere, or where exposure levels are not known wear approved respiratory



protection.

8.2.3 Environmental Exposure Controls: No data available

Section 9: Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Physical state	Solid
Colour	Bright whitish yellow
Odour	like: - Rotten eggs
Melting point/freezing point	114 - 119 °C
Boiling point or initial boiling point and boiling range	444.6 °C
Flammability	Flammable solid
Lower and upper explosion limit	4 vol% (LEL) – 44 vol% (UEL)
Flash point	207 °C
Auto-ignition temperature	232 °C
Decomposition temperature	not relevant
pH (value)	not determined
Kinematic viscosity	Not available
<u>Solubility(ies)</u>	
Water solubility	Insoluble
<u>Partition coefficient</u>	
Partition coefficient n-octanol/water (log value):	No data available
Soil organic carbon/water (log KOC)	No data available
Vapour pressure	0.015 kPa
<u>Density and/or relative density</u>	
Density	Not available
Relative vapour density	3.64 [Air = 1]
Particle characteristics	not relevant (liquid)
<u>Other safety parameters</u>	
Oxidising properties	none

9.2 Other Information:

Information with regard to physical hazard classes:



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There is no additional information.

Section 10: Stability and Reactivity

10.1 Reactivity

Reacts violently with strong oxidizers. Increased risk of fire or explosion.

10.2 Chemical stability

Stable under recommended handling and storage conditions (see section 7).

10.3 Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4 Conditions to avoid

Direct sunlight, extremely high or low temperatures, heat, hot surfaces, sparks, open flames, incompatible materials, and other ignition sources. Dust accumulation (to minimize explosion hazard).

10.5 Incompatible materials

Strong oxidizers. Under certain conditions: Sodium, tin, nickel, zinc.

10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

Section 11: Toxicological Information

11.1 Information on Toxicological Effects: Symptoms related to the physical, chemical and toxicological characteristics

11.1.1 Acute Toxicity:

Exposure route	Endpoint	Value	Species
Inhalation	LC50	> 9.23 mg/l/4h	rat
Oral	LD50	> 3000 mg/kg	rat
Dermal	LD50	> 2000 mg/kg	rabbit

11.1.2 Skin Corrosion/ Irritation: Causes skin irritation.

11.1.3 Serious Eye Damage/ Eye Irritation: Causes eye irritation.

11.1.4 Respiratory or Skin Sensitization: Shall not be classified as a respiratory or skin sensitizer

11.1.5 Mutagenicity: Not classified

11.1.6 Carcinogenicity: Not classified

11.1.7 Reproductive Toxicity: Not classified

11.1.8 Specific target organ toxicity-single exposure: Not classified



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11.1.9 Specific target organ toxicity - repeated exposure: Not classified

11.1.10 Aspiration Hazard: Not classified

11.1.11 Additional Information: Symptoms related to the physical, chemical and toxicological characteristics, For further information see section 4

Section 12: Ecological Information

12.1 Toxicity: Toxic to aquatic life with long lasting effects.

Endpoint	Value	Species	Exposure time
LC50	0.0448 mg/l	fish	96 h
EC50	0.022 mg/l	Daphnia	48 h

12.2 Persistence and Degradability: Expected to be biodegradable.

12.3 Bio Accumulative Potential: Heavier hydrocarbons of the product are possibly accumulative in aquatic organisms.

n-octanol/water (log KOW)	0.45 (at 25 °C)
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12.4 Mobility in Soil: Spillage, depending on the ambient temperatures, may evaporate in significant quantities, and the rest of the spilled product may penetrate the soil and contaminate ground waters.

12.5 Other adverse effects: Data are not available

Section 13: Disposal considerations

13.1 Waste Treatment Method: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations



- Sewage disposal-relevant information**

Do not pressurize, cut, or weld containers. Handle empty containers with care because residual product is flammable. Empty containers should be taken for recycle, recovery or waste in accordance with local regulation.

13.1.1 Contaminated Packaging: Used plastic packaging shall be placed in suitable waste containers and disposed of in accordance with applicable legal regulations.



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13.1.2 Local Regulations (if any) on Disposal: Waste material must be disposed of in accordance with the **Hazardous Waste Management Rules-2016**

Section 14: Transport Information

14.1 UN Number: UN 2448

14.2 UN Proper Shipping Name: MOLTEN SULFUR

14.3 Transport Class(es): 4.1

14.3.1 Hazard Labels: 4.1

14.3.2 Symbols:



14.4 Packaging Group: III

14.5 Environmental Hazards: No

14.6 Special Precautions for User: Provisions for dangerous goods (ADR) should be complied within the premises.

Section 15: Regulatory Information

15.1 Safety, health and Environmental: Regulations/legislation specific for the substance or mixture relevant provisions of **Non - Toxic/Flammable Substance** are applicable

Section 16: Other Information

16.1 Date of Issue: March 2023

16.2 Necessity of Specific Training: Working and handling personnel to be trained on following aspects

- i. Chemical hazard awareness
- ii. Safety control measures including PPE
- iii. Housekeeping, hygiene and chemical storage
- iv. Responding to any kind of emergency

16.3 Disclaimer: This M.S.D.S and the information it contains is offered to you in good faith as accurate. We believe that information to be correct but cannot guarantee its accuracy or completeness. Health and safety precautions in this data sheet may not be adequate for all individuals and/or situations. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. No statement made in this data sheet shall be construed as a permission or recommendation for the use of any product in a manner that might infringe existing patents.